Essays on the Labour Market Decisions of Older Workers

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Abstract

This dissertation comprises three essays examining the labour market decision-making of workers aged 50 and older in Canada and the United States. Older workers make up a large share of the labour force, so understanding their labour market decisions can inform public policy on public pension plans and public health care expenditure.

In the first essay, I compare the labour market attachment of workers aged 50 to 69 in the United States and Canada from 1997 to 2019. I utilise the panel structure of the Canadian Labour Force Survey and the US Current Population Survey to create six-month and four-month panels for Canada and the United States, respectively. I use the survey questions on reasons for (1) working part-time and (2) being non-employed (unemployed or not-in-the-labour force) to determine how the reasons provided differ based on age and gender. I also examine the different types of labour force transitions within the panel and how it changes with age and gender. The descriptive results show that workers become less attached to the labour market as they age and have a preference for more short-term, flexible work arrangements. The increase in part-time employment is driven primarily by supply-side factors. Personal preference is the main reason in Canada, while retirement and/or social security earning limits is the main reason in the United States. For the workers marginally attached or not attached to the labour market, the dominant factor seems to be retirement or the ending of short-term jobs.

The second essay studies the short-term impact of an involuntary job separation on the probability of re-employment and labour market exit for workers in Canada aged 50 to 69

years. Like the first essay, I use the Canadian Labour Force survey from 1997 to 2019 to create six-month panels. I use the panels to estimate linear probability models for returning to work and exiting the labour force. My results show that a prior involuntary separation increases the likelihood of labour market exit by approximately 4.7 percentage point and decreases the chance of re-employment by 5.3 percentage points. The findings suggest that an involuntary job separation is a pathway through which older workers exit the labour market.

The third essay uses the US Health and Retirement Survey from 1998 to 2014 to determine whether physical activity protects the cognitive function of retired older workers aged 51 to 75 in the United States. In this essay, I first estimate causal relationships between (1) retirement and cognitive function; (2) retirement and physical activity using the eligibility ages for early and normal social security benefit claiming as instruments for retirement. I also establish the association between physical activity and cognitive function. I then analyse whether engaging in physical activity is a mechanism through which retirement impacts the cognitive outcomes of older workers. My results show that the combined effect of retiring and meeting the physical activity guidelines increases cognitive scores by approximately 0.5 points. This corresponds to a 4.5% increase in the cognitive score compared to the sample average.

Abrégé

Cette thèse comprend trois essais examinant la prise de décision sur le marché du travail des travailleurs âgés de 50 ans et plus au Canada et aux États-Unis. Les travailleurs âgés représentent une part importante de la population active, donc comprendre leurs décisions sur le marché du travail peut éclairer les politiques publiques sur les régimes de retraite publics et les dépenses publiques de santé.

Dans le premier essai, je compare l'attachement au marché du travail des travailleurs âgés de 50 à 69 ans aux États-Unis et au Canada de 1997 à 2019. J'utilise la structure de panel de l'Enquête sur la population active du Canada et de l'Enquête sur la population actuelle des États-Unis pour créer des données sur six mois. et des panels de quatre mois pour le Canada et les États-Unis, respectivement. J'utilise les questions de l'enquête sur les raisons pour (1) travailler à temps partiel et (2) être sans emploi (sans emploi ou inactif) pour déterminer comment les raisons fournies diffèrent en fonction de l'âge et du sexe. J'examine également les différents types de transitions de la main-d'œuvre au sein du panel et comment cela change avec l'âge et le sexe. Les résultats descriptifs montrent que les travailleurs deviennent moins attachés au marché du travail à mesure qu'ils vieillissent et ont une préférence pour des modalités de travail plus flexibles et à plus court terme. L'augmentation de l'emploi à temps partiel est principalement due à des facteurs liés à l'offre. La préférence personnelle est la principale raison au Canada, tandis que la retraite et/ou les limites de gains de la sécurité sociale sont la principale raison aux États-Unis. Pour les travailleurs marginalement attachés ou non au marché du travail, le facteur dominant semble être la retraite ou la fin des emplois

de courte durée.

Le deuxième essai étudie l'impact à court terme d'une cessation d'emploi involontaire sur la probabilité de réemploi et de sortie du marché du travail pour les travailleurs au Canada âgés de 50 à 69 ans. Comme le premier essai, j'utilise l'Enquête sur la population active du Canada de 1997 à 2019 pour créer des panels de six mois. J'utilise les panels pour estimer des modèles de probabilité linéaire de retour au travail et de sortie de la population active. Mes résultats montrent qu'une séparation involontaire antérieure augmente la probabilité de sortie du marché du travail d'environ 4,7 points de pourcentage et diminue les chances de réemploi de 5,3 points de pourcentage. Les résultats suggèrent qu'une cessation d'emploi involontaire est une voie par laquelle les travailleurs âgés quittent le marché du travail.

Le troisième essai utilise l'enquête américaine sur la santé et la retraite de 1998 à 2014 pour déterminer si l'activité physique protège la fonction cognitive des travailleurs âgés à la retraite âgés de 51 à 75 ans aux États-Unis. Dans cet essai, j'estime d'abord les relations causales entre (1) la retraite et la fonction cognitive ; (2) la retraite et l'activité physique en utilisant les âges d'éligibilité aux prestations de sécurité sociale anticipées et normales comme instruments de retraite. J'établis également l'association entre l'activité physique et la fonction cognitive. J'analyse ensuite si la pratique d'une activité physique est un mécanisme par lequel la retraite affecte les résultats cognitifs des travailleurs âgés. Mes résultats montrent que l'effet combiné de la retraite et du respect des directives en matière d'activité physique augmente les scores cognitifs d'environ 0,5 point. Cela correspond à une augmentation de 4,5% du score cognitif par rapport à la moyenne de l'échantillon.

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I want to dedicate this thesis to my daughter, Kalea Campbell, my grandmother, Carmen Fairclough, and the loving memory of my late grandparents, Naomi McCalla and Sydney Fairclough. I want to thank them for their love and encouragement and for always believing in me even when I doubted myself.

Contribution to Original Knowledge

Research on older workers needs to be more extensive in the literature, given that workers 55 and older comprise a large share of countries' labour force worldwide. Additionally, understanding the labour market outcomes of these workers is essential to formulating public policies concerning the labour supply, pension benefits and public health care services. This thesis adds to the body of literature on the decision of older workers to remain attached or to exit the labour market. In my first essay, in addition to examining the main reasons for part-time employment and non-employment by older workers, I challenge the view that discrimination and lack of job opportunities are the main reasons for older workers remaining unattached from the labour market. Instead, I argue that workers are unemployed, not-in-labour force, or choose part-time employment due to supply-side rather than demand-side factors.

The second essay contributes to the literature on the impact of involuntary job separation on older workers. Currently, the existing literature focuses on the duration of unemployment and the earnings loss of workers younger than 55 years who experience an involuntary separation (Morissette et al. (2007); Morissette and Bonikowska (2012); Merkurieva (2019); Chen and Morissette (2010)). The main contribution of this chapter is an investigation of involuntary job separation as a pathway through which older workers exit the labour market. The paper closest to my analysis is the work done by Chan and Stevens (1999, 2001, 2004) using the Health and Retirement Survey in the United States. However, while Chan and Stevens (2001) estimate the hazard of exiting the labour market after returning to employment, I focus on the impact of involuntary separation on the probability of labour market exit. To my knowledge, this is the first paper to evaluate this probability.

The third essay contributes to the literature on the health outcomes of retirement. While numerous studies examine the impact of retirement on physical and mental health outcomes and health behaviours, there is little empirical evidence on the mechanism through which retirement impacts these outcomes (Neuman (2008); Johnston and Lee (2009); Coe and Zamarro (2011); Insler (2014); Eibich (2015); Kämpfen and Maurer (2016); Shai (2018); Rohwedder and Willis (2010); Mazzonna and Peracchi (2012); Bonsang et al. (2012); Bingley and Martinello (2013); Celidoni et al. (2017); Heller-Sahlgren (2017)). The main contribution of my paper is to determine whether engaging in physical activity is a mechanism through which retirement impacts cognitive function. In the literature, there is an established negative relationship between retirement and cognitive function (Bonsang et al. (2012); Rohwedder and Willis (2010); Mazzonna and Peracchi (2012)). The literature also shows that retirement positively impacts the individual's health behaviours and lifestyle choices (Eibich (2015); Kämpfen and Maurer (2016); Atalay et al. (2019); Celidoni and Rebba (2017); Kesavayuth et al. (2018); Zhao et al. (2017)). Additionally, given the positive association that exists between physical activity and cognitive function, including physical activity as a mechanism could be protective of cognitive function and help to slow its deterioration (Frith and Loprinzi (2017); Weuve et al. (2004); Cadar (2018)).

The analysis in this essay is similar to Eibich (2015), which examines the impact of health behaviours on various self-reported health outcomes, including mental health. However, my study utilizes a panel data analysis with nine waves of data rather than the single cross-sectional analysis used by Eibich (2015). In addition, I use an objective measure for the health outcome (cognitive function) in this study rather than the self-reported health measure used by Eibich (2015). As a result, the health outcome is not susceptible to biases and measurement errors from using self-reported health measures.

Contribution of Authors

This thesis contains three essays, and I am the sole author of the three essays.

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Acronyms

2SLS Two-Stage Least Squares.

CPP Canada Pension Plan.

CPS Current Population Survey.

EIP Employability Improvement Project.

GIS Guaranteed Income Supplement.

HRS Health and Retirement Survey.

ILO International Labor Organization.

IRA Individual Retirement Accounts.

LFPR Labour Force Participation Rate.

LFS Labour Force Survey.

LPM Linear Probability Models.

MRA Minimum Retirement Age.

NILF Not-in-the-Labour Force.

NRA Normal Retirement Age for Benefit Claiming.

OAS Old Age Security.

OASDI Old Age, Survivors, and Disability Insurance.

OECD The Organization for Economic Cooperation and Development.

OLS Ordinary Least Squares.

QPP Quebec Pension Plan.

RET Retirement Earnings Test.

RRSP Registered Retirement Savings Plans.

SSI Supplemental Security Income.

TFSA Tax-Free Savings Accounts.

Chapter 1

General Introduction

Given their proximity to the early and normal eligibility ages for public pension benefits, older workers face a unique decision concerning their timing of labour market exits. Individual characteristics, as well as their earnings, work industry, occupation type and any unexpected events, influence the exit decision. This thesis investigates how the labour market attachment of employed and non-employed older workers changes with age (chapter 2) and their decision whether to remain attached or to exit the labour market following an involuntary job separation (chapter 3). The final chapter of the thesis focuses on retired older workers and explores how the decision to retire impacts their cognitive scores (chapter 4).

In chapter two, “The Labour Market Attachment of Older Workers: Comparison of Canada and the United States”, I use the Canadian Labour Force Survey (LFS) and the US Current Population Survey (CPS) from 1997 to 2019 to carry out a descriptive analysis of the labour market attachment of employed and non-employed older workers. In the chapter, I use the responses to the questions: (1) What is the main reason for working part-time? and (2) What is the main reason for non-employment? to investigate whether supply-side or demand-side factors influence the older workers' labour market decisions. My findings show an increased preference for part-time employment amongst older workers age 60 to 69 years, especially older women. Further, the primary reasons for part-time and non-employment are personal preference, retirement and/or social security limits.

In chapter three, "Older Workers and Involuntary Job Loss in Canada", I estimate the probability of re-employment and labour market exit for older workers who experience an involuntary job separation. Previous papers focused on the effect of an involuntary job separation on the duration of unemployment and the wages earned upon returning to work for older workers (Morissette et al. (2007); Merkurieva (2019); Chen and Morissette (2010)). In this chapter, I use six-month panels from the Canadian LFS from 1997 to 2019 to estimate linear probability models for returning to work and exiting the labour force. My results show that a prior job separation increases the probability of labour market exit by 4.7 percentage points and decreases the likelihood of re-employment by 5.3 percentage points. Additionally, male workers are less likely to return to employment than their female counterparts. Male workers that return to work are more likely to return full-time than part-time.

In chapter four, "Cognitive Functioning in Retirement: Does physical activity improve the impact?" I use longitudinal data from the U.S. Health and Retirement Survey (HRS) for 1998 - 2014 to determine if engaging in physical activity is a possible mechanism through which retirement affects cognition. Existing literature established a negative causal link between retirement, retirement duration and health outcomes of older individuals (Bonsang et al. (2012); Rohwedder and Willis (2010); Atalay et al. (2019)). In this chapter, I estimate a Two-Stage Least Squares (2SLS) using the retirement benefit claiming age thresholds as instruments for retirement. My results show that a retirement duration of at least one year increases the likelihood of meeting the U.S. 2018 Physical Activity Guidelines by 12.6 percentage points but decreases the cognitive score of respondents by 5.7% compared to the sample average. Additionally, the combined effect of retiring for at least one year and meeting the physical activity guidelines increases the cognitive scores by 4.5%.

Chapter 2

The Labour Market Attachment of Older Workers: Comparison of Canada and the United States

2.1 Introduction

In December 2019, the labour force participation of workers aged 55 and over in Canada was 38.2%. This rate is comparable to the labour force participation rate of 40.3% for the same age group in the United States. Numerous studies examine how age impacts labour force participation rates as well as factors that contribute to the increase in the participation rates for older workers (OECD (2019a,c); Fields (2017); Canada (2018)). Given the increasing share of older workers that make up the labour force worldwide, public policies in many countries focus on (1) keeping older workers employed longer, (2) getting unemployed workers back to employment, and (3) getting inactive older workers to re-establish an attachment to the labour market (OECD (2019b,c); Canada (2016)). As workers get closer to the early and normal age of retirement benefit claiming (NRA), their attachment to the labour market weakens, and they start planning their retirement transitions.

Individuals are classified as attached to the labour market if they are working for pay on a full-time, part-time, seasonal or temporary basis or if they are actively looking for a job. They are not attached to the labour market if they cannot work, do not want a job (not actively looking) or are classified as retired (Jones and Riddell (2019)). In the literature, socioeconomic factors impact an individual's level of labour market attachment. For example, individuals with low levels of education, persistent health problems, "caregiving" responsibilities or individuals with a long duration of unemployment typically have weaker labour market attachments (OECD (2019b,c); Canada (2018); Chan and Stevens (1999, 2001, 2004); Farber (2005)).

In this chapter, I investigate the labour market attachment of employed and non-employed older workers in Canada and the United States. Despite differences in levels, labour market trends among older workers in Canada and the United States are remarkably similar (Figure 2.1 and Figure 2.2). This chapter describes these trends in light of differences in institutional contexts (e.g. pensions and health care systems), underlying demographic patterns, and differences in the measurement of the labour force. I use the panel structure of the Canadian Labour Force Survey (LFS) and the US Current Population Survey (CPS) from 1997 to 2019 to:

1. Discuss how the work arrangements of employed older workers change with age and gender.

2. Investigate whether age discrimination or demand-side conditions are the primary drivers of non-employment amongst older workers.

This chapter contributes to the literature on the labour market participation of older workers in two ways. Firstly, there has been an increase in the proportion of part-time workers. Workers' desire for more flexible work arrangements drives the growth in part-time employment (Gash (2008)). Though the proportion of older workers, especially women, opting for part-time jobs is increasing, there needs to be more research on the work arrangements of these workers. The work arrangement literature focuses mainly on analysing prime-aged workers. This paper uses the LFS and CPS to examine the pervasiveness of part-time employment among older workers and whether demand-side or supply-side factors are influencing the decision-making of these workers.

Secondly, in the existing literature, the reasons given for the older worker's decisions to exit the labour market include ageism, lack of transferable skills, or lack of job opportunities for older workers (Bélanger et al. (2016); OECD (2019b); Canada (2018)). These existing studies are primarily theoretical. Canada (2018) uses responses from the Survey of Older Workers in Canada for 2008 to determine the challenges that older workers face to re-enter or remain in the labour force. In this chapter, I use data from the LFS and CPS that allows me to delve deeper into the main reasons for non-employment in Canada and the United States and to analyse how these reasons change with age and gender. My descriptive analysis shows that older workers in Canada and the United States prefer part-time versus full-time employment. The main reason is personal preference in Canada and retirement and/or social security limit in the United States. This preference for part-time work increases at the eligibility age thresholds for public pension benefit claiming in Canada and the United States.

Additionally, the worker's gender is critical in the decision to work part-time. Older women in Canada are almost three times more likely than men to be employed part-time, while in the United States, women are two times more likely. Surprisingly, caregiving does not play a critical role in working part-time, but rather personal preference and retirement. However, workers aged 50 - 54 are more likely to state caregiving as the reason for working part-time compared to workers aged 55 and older. The chapter also shows that the main reasons given by older workers in Canada and the United States for non-employment are supply-side factors rather than discrimination or lack of available jobs. In the United States, an examination of the workers classified as Not-in-the-Labour-Force who want a job shows family responsibilities and their illness as the main reasons for not looking for a job in the last four weeks.

In the remainder of this chapter, Section 2 reviews the institutional framework in Canada and the United States. Section 3 discusses the data and sample construction and provides descriptive statistics for the main variables. Section 4 discusses how the work arrangements of the employed workers change with age. I examine the labour market attachment of the non-employed workers in Section 5; Section 6 investigates transitions between labour market states, and Section 7 concludes the chapter.

2.2 Institutional Framework

Access to health care, the pension systems and the measurements of the labour force used in a country influence the labour market attachment of older workers.

2.2.1 Conceptual Definitions

In the United States and Canada, monthly household surveys, the CPS and the LFS, respectively, provide information on labour market activity. Though both countries comply with the guidelines established by the International Labor Organization (ILO) for measuring employment, unemployment, and labour force participation, there are differences in the measurement of labour force activity. According to Jones and Riddell (2019), the principal differences include the following:

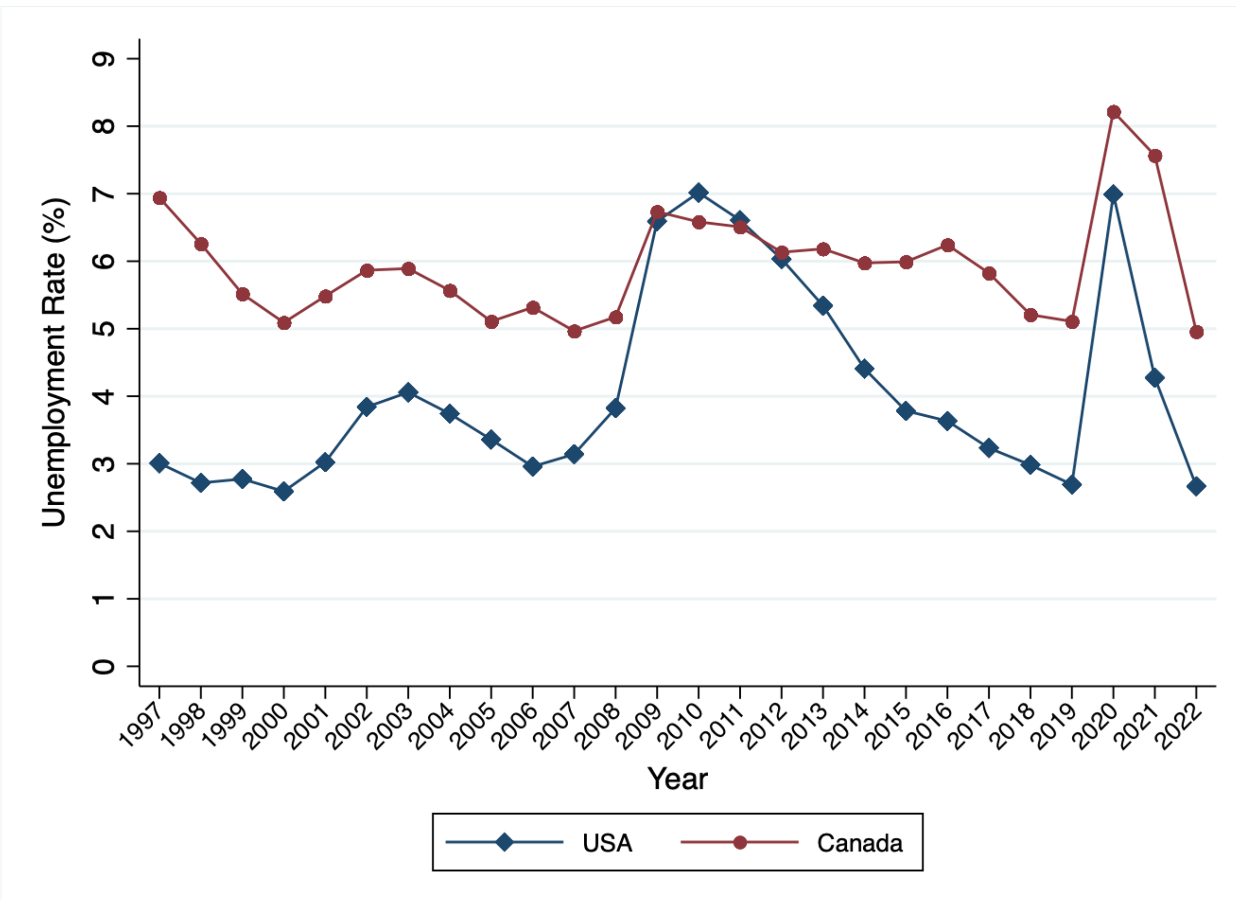
1. The definition of the adult population. In Canada, the adult population starts at the age of 15 and older, but 16 years of age and older in the United States.

2. In the United States, respondents are classified as out of the labour force if (i) used only "passive" search methods, such as "looked at job ads"; (ii) had a job to start within the next four weeks; and (iii) not available for work because of personal or family responsibilities. While for Canada, the LFS classifies (i) as unemployed and (ii) as employed. The LFS also classifies respondents (iii) as out of the labour force.

3. In the United States, the CPS classifies full-time students looking for full-time work as unemployed. In contrast, the LFS ranks them as Not-in-the-Labour Force (NILF).

Defining the adult population as 16 years and older and including full-time students actively looking for a job as unemployed, increases the labour force participation rate in the United States compared to Canada. However, the more restrictive definition of unemployment in item 2 would lower the unemployment and participation rate in the United States. Focusing on respondents 50 - 69 years eliminates the problem with principal differences (1) and (3).

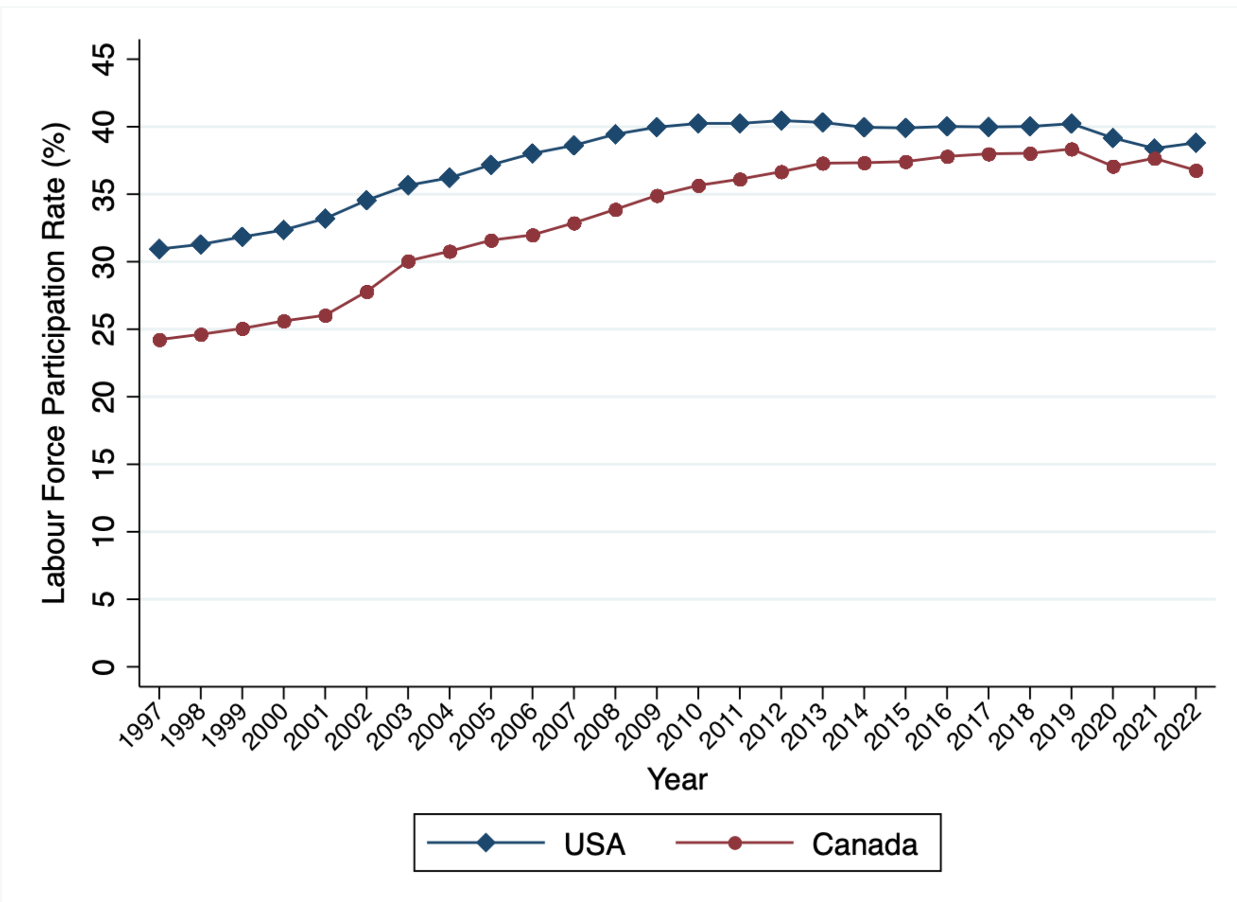
Figure 2.1: Unemployment Rate - 55+ (1997 -2022)



This figure shows the unemployment rate for workers 55 years an older in Canada and the United States. Source: FRED®, Federal Reserve Bank of St. Louis.

The requirement for using "active search" methods for unemployed workers results in an overstating of the unemployment rate in Canada compared to the United States. Figure 2.1 illustrates that while the unemployment rate for workers 55 years and over in Canada and the United States move together over time, Canada's unemployment rate is generally higher than the United States. Figure 2.1 shows that the gap between the unemployment rates remained fairly constant but decreased during the 2008 - 2010 and 2019 - 2020 financial crises. Chan and Stevens (2001, 2004) find that there is a positive association between the unemployment rate and the decision of older workers to exit the labour market. In addition to the loss of earnings from work, older workers typically have a longer duration of unemployment and earn lower wages once re-employed than younger workers.

Figure 2.2: Labour Force Participation Rate - 55+ (1997 -2022)



Source: FRED®, Federal Reserve Bank of St. Louis.

Figure 2.2 shows that the Labour Force Participation Rate (LFPR) for workers 55 and older increased from 1997 to 2019 and is higher in the United States than in Canada. The difference in the LFPR in 1997 was approximately 5%, which decreased to 2% by 2019.

2.2.2 Demographics

Population ageing occurs due to declining birthrates, improved living standards, healthcare advancements and healthier lifestyles. In 2018, the median age in Canada was 40.8 years, compared to 38.2 years in the United States (OECD (2019a)). Additionally, given the increase in the life expectancy of both countries, the percentage of the population 65 years and older increased between 1997 and 2019 (Table 2.1). The life expectancy and share of the population 65 and over was higher for both men and women in Canada relative to the United States. In 1997 the old-age dependency ratio in Canada and the United States was 18% and 19%, respectively. The ratio increased to 26% and 24% in 2019 (Table 2.1). The dependency ratio indicates that the cost per worker in Canada to raise benefits for retirees by $1 increased by $0.08 from 1997 to 2019 compared to $0.05 in the United States (Turner (1984)). In 2019, the old-age dependency ratio was almost the same in both countries though Canada has a longer life expectancy (+4 years in 2019) and a relatively older population. This suggests that the demographic differences between the two countries have little impact on the financing of social security benefits (Latulippe and Turner (2019)).

2.2.3 Health Care System

In the United States, health care is self-funded and tied to employment. With the exception of low income workers, active workers cannot access government-provided health care until age 65 when they become eligible for Medicare. Monthly premiums and taxes finance Medicare; the Medicare enrollee is also responsible for paying a deductible and co-payment for some services (Ridic et al. (2012)). In contrast, Canadian citizens and permanent residents have universal access to health care services the Canadian and provincial governments provide, financed through taxes. There are no co-payments required in the Canadian health-

Table 2.1: Population Demographics 1997 vs 2019

	1997			**2019**		
	Female	Male	Total	Female	Male	Total
Age dependency ratio			18%			26%
Life expectancy at birth (years)	81.3	75.7	78.4	84.2	80	82
Share of population (65+)	14%	10%	12%	19%	16%	18%

(a) Canada

	1997			**2019**		
	Female	Male	Total	Female	Male	Total
Age dependency ratio			19%			24%
Life expectancy at birth (years)	79.4	73.6	76.4	81.4	76.3	78.8
Share of population (65+)	14%	11%	13%	17%	14%	16%

(b) United States

Source: United Nations Population Division. World Population Prospects: 2022 Revision

care system for services provided by physicians and/or in-hospital. Further, active workers in Canada have supplemental private insurance for services not covered by the universal plan, such as prescription drugs and dental services (Feeny et al. (2010)). The premium costs for the supplemental private insurance is shared by the employer and the worker.

The differences in the health care systems contribute to the higher life expectancy in Canada. Moreover, older workers in Canada tend to be healthier than their counterparts in the United States (Feeny et al. (2010)). Furthermore, the health care systems' differences impact older workers' labour market attachment. In the United States, older workers under the Normal Retirement Age for Benefit Claiming (NRA) might remain employed full-time until they reach the Medicare age eligibility threshold (Mastrobuoni (2009)).

2.2.4 Public Pensions

Publicly funded pensions incentivise labour market exits for older workers and can result in weaker attachments as workers approach the early and NRA for retirement benefit claiming. The publicly provided pension benefit in the United States is the Old Age, Survivors, and

Disability Insurance (OASDI), commonly called Social Security. In Canada, the publicly funded pension consists of the Canada Pension Plan (CPP) (or Quebec Pension Plan (QPP) for residents of Quebec) in addition to the Old Age Security (OAS) programme (OECD (2017, 2021)). The OAS provides pension benefits based on meeting legal status and residence requirements.[1] Canada also provides the Guaranteed Income Supplement (GIS) to pensioners with little or no income other than an OAS pension. In the United States, Supplemental Security Income (SSI) benefits low-income Americans. The GIS and SSI benefits are income-tested and are reduced as income increases (Latulippe and Turner (2019); OECD (2017, 2021)).

The primary public pension plans in Canada and the United States are defined-benefit programs. Benefits under Social Security and CPP-QPP are based on earnings and cover employed and self-employed persons.[2] Eligibility for retirement benefits depends on the number of years contributions are made, with a minimum requirement of ten years.

For older workers, being close to the early and NRA for retirement benefit claiming influences the level of labour market attachment. In the United States, the early age for retirement benefit claiming is 62 years, compared to 60 years in Canada.[3] At the early age for retirement benefits claiming, workers have access to their retirement pensions but incur a penalty for claiming early. In the United States, the benefit is reduced by 6 2/3% for each year of retirement before the NRA (Gustman and Steinmeier (2000); Gruber and Wise (2010)). However, at age 65, the reduction falls to 5% (OECD (2019a); Fields (1984)). In Canada, the early pension adjustment for the CPP reduces the pension amount by 0.6% for each month the retirees claim pension before age 65. The maximum possible reduction is 36% for claiming at age 60 (OECD (2019a); Fields (1984)). If workers claim their pension early, the penalty for claiming early could result in the worker's remaining attached to the

[1]Workers residing in Canada for at least 40 years after age 18 receive a full basic OAS. Residence of at least ten years after age 18 results in partial pension payment. The partial OAS benefit is calculated as 1/40th of the full pension for each complete year of residence (OECD (2017, 2021)).

[2]In Canada the CPP-QPP is calculated based on your best 40 years of earnings, while in the United States the OASDI uses 35 years of earnings (OECD (2017, 2021)

[3]In Canada, early retirement is only possible under the CPP.

labour force to make up the shortfall in monthly earnings. The option to claim early also leads to spikes in retirement rates at those ages.

Workers can access their full retirement benefits at the NRA for claiming benefits. In Canada, the NRA is 65 years. In 1983, the United States amended the Social Security Act to increase the NRA from 65 to 67 years gradually. The changes in the Social Security Act took effect in 2000, and the NRA for benefit claiming now depends on the individual's birth year. For individuals born before 1937, the NRA is 65 years, while the NRA increases in 2-month increments for those born after 1937. A further increase in the NRA from 66 to 67 years is expected by 2027 (Mastrobuoni (2009); Latulippe and Turner (2019)).

In the US and Canadian systems, workers can defer their pension claims until age 70. Pension deferral increases pension by 8% for each year of deferral in the United States. In Canada, the OAS pension increases by 0.6% for each month of deferral, up to a maximum of 36% if the benefit is claimed at age 70. The CPP-QPP deferral increases benefits by 0.7% per month, up to a maximum of 36% if the benefit is claimed at age 70 (OECD (2019a)).

Workers are also able to combine work and pension receipts. In the United States, this is subject to a Retirement Earnings Test (RET) where the pension benefit is reduced by 50% of earnings over the RET-exempt amount (Latulippe and Turner (2019)).[4] In 1997, the RET-exempt amount was US$8,640 for individuals under 65 years and US$13,500 for individuals 65 to 69 years. In January 2000, the retirement earnings test (RET) was removed from individuals who reached the NRA. In 2019 the RET-exempt amount for individuals under 65 years was US$17,640 (IRS (2022b)).

In Canada, there is no retirement earnings test. However, before 2012, workers claiming pension benefits before the NRA was required to have a reduction in earnings in the first month and the month after the retiree claimed pension benefits. This means that workers would either have to stop working or work part-time to claim their pension at age 60 (Latulippe and Turner (2019)). The cessation test negatively impacted the LFPR and the labour

[4] Benefits in excess of the RET-exempt amount that are withheld are added to the monthly pension benefit once reaching the NRA.

force attachment of early pension claimers. Moreover, for retirees 65 years and older, there is an OAS pension recovery tax based on the retiree's net income in the previous calendar year. The OAS recovery tax threshold was CAN$77,580 for the 2019 tax year. The OAS pension benefit is reduced by 15 cents for every dollar above the threshold (OECD (2021)).

Additionally, CPP beneficiaries younger than 65 with earnings over the basic exemption must contribute to the CPP. After reaching the NRA, workers can continue contributing up to age 70. Under the QPP, beneficiaries must contribute to the plan if younger than 70 years. Retirees contributing to the CPP/QPP receive an annual pension supplement (Latulippe and Turner (2019)).

2.2.5 Private Pensions

Private pension plans in Canada and the United States include employee-employer-funded programmes and individual retirement savings plans. The Canadian Registered Retirement Savings Plans (RRSP) are similar to the traditional Individual Retirement Accounts (IRA) in the United States (OECD (2009, 2017, 2021)). RRSPs and traditional IRAs are tax-deferred with a maximum yearly contribution limit set by the government.[5] However, Canadian retirement accounts have a more generous contribution limit than US traditional IRAs. Individuals can contribute up to 18 per cent of their previous year's income. In 2023 the maximum contribution limit is CA$31,560 (CRA (2005)). The maximum contribution limit for IRAs in 2023 is US$6,500. A 10% tax penalty is also charged for funds withdrawn before age 59.5 (Schwab (2023)). Another difference between RRSPs and traditional IRAs is the maximum age for contributions. In Canada, the maximum age for contributing to an RRSP is 71 years, while in the United States, the maximum age, prior to 2019, is 70.5 years (OECD (2009, 2017, 2021)). In 2019, the United States eliminated the age limit for IRA contributions.

In the United States, traditional 401ks and Roth 401ks are defined contribution plans of-

[5]IRAs can be traditional or Roth IRA. Roth IRAs are after tax contributions. Withdrawals from Roth IRAs are tax-free after age 59.5 (Schwab (2023)).

fered through an employer. Traditional 401ks are tax-free in the contribution year, but funds are taxed on withdrawal. For Roth 401ks, contributions are made using after-tax income and withdrawals are tax-free (Royal (2023)). The contribution limit to traditional 401ks in the US is similar to that of Canadian RRSPs. In 2023, the 401k maximum contribution limit is US$22,500 for workers younger than 50 years and US$30,000 per year for workers 50 and older (IRS (2022a)).

The contribution limits for Roth 401ks are similar to Tax-Free Savings Accounts (TFSA) in Canada. In 2023, the contribution limit for a Roth 401k is US$7,500 with an additional US$1000 allowable for workers 55 years and older (Royal (2023)). The TFSA has a contribution limit of CA$7,500 for 2023, but the contribution limit is cumulative, so unused contribution space in a given year is rolled over. Additionally, a first-time contributor who turned 18 in 2009 can deposit CA$88,000 in their TFSA (CRA (2017)).

In summary, while the economies of Canada and the United States are very similar, there are vital differences in the demography, health care systems, private and public pensions and definitions of the labour market. These differences may factor into the labour market decisions of older workers as they get closer to the early and normal eligibility age thresholds for pension benefit claims in the United States and Canada. The institutional framework is summarized in the Appendix (Table 2.8).

2.3 Data

This paper uses monthly data from the Canadian Labour Force Survey (LFS) and the US Current Population Survey (CPS) from 1997 to 2019 for respondents aged 50 - 69.

2.3.1 Labour Force Survey

The LFS is a monthly household survey administered by Statistics Canada and is representative of Canada's civilian, non-institutionalised population. Each month, interviews are conducted with 56,000 households for all household members aged 15 and over (approximately 100,000 observations). The survey collects labour force and demographic information such as gender, age, and education. Respondents answer questions related to the reference week (i.e. the week containing the 15th of the month). The LFS relies on a rotating panel design. Each household remains in the sample for six consecutive months, and one-sixth of the sample is replaced monthly with a new cohort of respondents (Statistics Canada (2017)).

I use the restricted access microdata of the LFS to create six-month panels for respondents. To make the panel, I construct a unique individual identifier by concatenating dwelling, household, and individual-within-household codes. The final sample includes individuals aged 50 - 69 who are employed as paid workers during the 1st month of the panel and are present all six consecutive months.

2.3.2 Current Population Survey

The CPS is a monthly household survey administered jointly by the U.S. Census Bureau and the Bureau of Labor Statistics. The survey is representative of the civilian, non-institutionalized population in the United States. Each month, interviews are conducted with approximately 65,000 households. Respondents answer questions related to the reference week (i.e. the week containing the 12th of the month). In the CPS, a reference person within a household is interviewed for four months, and then they are not interviewed for the next eight months. After these eight months, they are included in the CPS again for the next four months (Flood et al. (2022)).

I use the monthly IPUMS-CPS data to construct four-month panels for workers aged 50 - 69 years over the period 1997 to 2019. To create the panel, I use the individual identifier and the month-in-sample variable in the IPUMS-CPS to match the individuals across consecutive months. I focus only on the first four-month panel for each respondent. Similar to the LFS, the CPS sample I use in this paper consists of all individuals employed as paid workers in the 1st month of the panel and present for all four consecutive months.

Table 2.7 provides descriptive information on the Canadian and the US sample of employed older workers in Month 1. The average age in the two samples is 55.8 and 53.5 years, respectively, with 50.1% and 50.8% being men. Table 2.2 shows how the sample's labour force status changes by month. By design, all workers are employed in month 1 of the panel. The percentage of workers unemployed and Not-in-the-Labour Force (NILF) increase each month, with the highest numbers observed in months 5 and 6.

Table 2.2: Labour Force Status by Month in Sample

	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Total
Employed	100%	97.7%	96.4%	95.5%	94.7%	94%	96.4%
Unemployed		1%	1.6%	2%	2.4%	2.6%	1.6%
NILF		1.3%	1.9%	2.5%	3%	3.5%	2%
No. Individuals	472,356						

(a) Canada

	Month 1	Month 2	Month 3	Month 4	Total
Employed	100%	96.6%	95.4%	94.4%	95.5%
Unemployed		0.9%	1.3%	1.6%	1.3%
NILF		2.5%	3.3%	4%	3.2%
No. Individuals	680,769				

(b) United States

The table shows the labour force status of workers in Canada and the United States for each month in the panel. All workers are employed in Month 1.
Source: Canada Labour Force Survey and US Current Population Survey (1997 - 2019).

The descriptive analysis in this paper focuses on the responses to the following questions:

1. *What is the main reason for working part-time?* Both the LFS and the CPS ask this question.

2. *What is the main reason you left your last job?* In Canada, the LFS asks this question of all persons classified as unemployed or NILF who last worked within the previous year. However, in the United States, respondents are asked: *What is the main reason you are unemployed?*[6] The CPS asked persons categorized as NILF who indicated

[6]Civilians age 15+ who were unemployed (looking for work) or on layoff during the previous week.

they wanted a job: *What is the main reason you were not looking for work during the previous four weeks?*

The LFS provides ten reasons for working part-time. I combine the responses to get six categories: Personal preference, own illness; personal and/or family responsibilities; caregiver for own children or elder relative (60+); could not find a full-time job and business conditions. The CPS gives seventeen reasons, and I combine responses to get five categories: demand-side, retired; own-illness; personal and/or family responsibilities and other. Table 2.10 and Table 2.9 in the Appendix provide more information on how the I combined the categories.

For the non-employed, the LFS provides eighteen reasons in response to the question: *What is the main reason you left your last job?* I combine the answers to get five categories: end seasonal, end casual job; end temporary; demand-side; quit and retired. In the CPS, I use two questions to determine why workers are non-employed in the United States: (1) *What is the main reason you are unemployed?* and (2) *What is the main reason you were not looking for work during the previous four weeks?* Question 1 provides six categories: layoff; other job loser, end temporary, quit, re-entrant and new entrant. Question 2 gives twelve categories, and I combine responses to get six categories: demand-side, lack of skill, own-illness, personal and/or family responsibilities, discrimination, and other. Table 2.11 and Table 2.12 in the Appendix provide more information on how I combined the categories.

2.4 Work Arrangements of the Employed

In this section, I examine how the work arrangement of employed older workers in the United States and Canada change with age. I focus on answering three questions:

1. Do older workers have a preference for part-time employment as they age?

2. Is the decision of older workers to work part-time driven by demand-side or supply-side factors?

3. Does gender affect the work arrangements of older workers?

Table 2.3 depicts the full-time versus part-time status of workers by month in sample. While the proportion full-time versus part-time is stable over the sample period, the table shows that a larger proportion of older workers in Canada are employed full-time (86%) compared to 79% in the United States. These findings suggest that full-time employment remains the dominant choice of work arrangement for older workers. Furthermore, Table 2.3 indicates that a larger proportion of workers aged 50 - 69 are employed part-time in the United States (21%) compared to Canada (14%). The US Census Bureau classifies workers as part-time if they work less than 35 hours weekly.

Table 2.3: Full-time versus Part-time Status by Month in Sample

	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Total
Full-time	85.7%	85.9%	85.9%	86%	86.1%	86.1%	86%
Part-time	14.3%	14.1%	14.1%	14%	13.9%	13.9%	14%
No. Individuals	472,356						

(a) Canada

	Month 1	Month 2	Month 3	Month 4	Total
Full-time	77.6%	78.8%	79.1%	79%	79%
Part-time	22.4%	21.2%	20.9%	21%	21%
No. Individuals	680,769				

(b) United States

The table shows the full-time versus part-time status of workers in Canada and the United States for each month in the panel. All workers are employed in Month 1.
Source: Canada Labour Force Survey and US Current Population Survey (1997 - 2019).

To address question 1, Figure 2.3 illustrates how full-time and part-time employment changes over the age profile in Canada and the United States from 1997 - 2019. For Canada, in Figure 2.3 (figure a), there is a slight increase in ages 50 to 54. The proportion of part-time continues to increase steadily in the age group 55 - 59 and increases rapidly in the age group 60 - 64. At age 64, there is a large increase in the proportion of workers engaged part-time, and the gap between full-time and part-time employment decreases by 20 percentage points by age 69. Figure 2.3 (figure b) shows a similar pattern for the United States, and by age

69, the gap between full-time and part-time employment is almost completely eliminated. However, in contrast to Canada, only a slight increase in the proportion of part-time is seen from age 50 to 59 years until age 60 onwards, when it increases quickly. The descriptive evidence shows that as workers age, there is an increased preference for part-time employment in Canada and the United States (Figure 2.3).

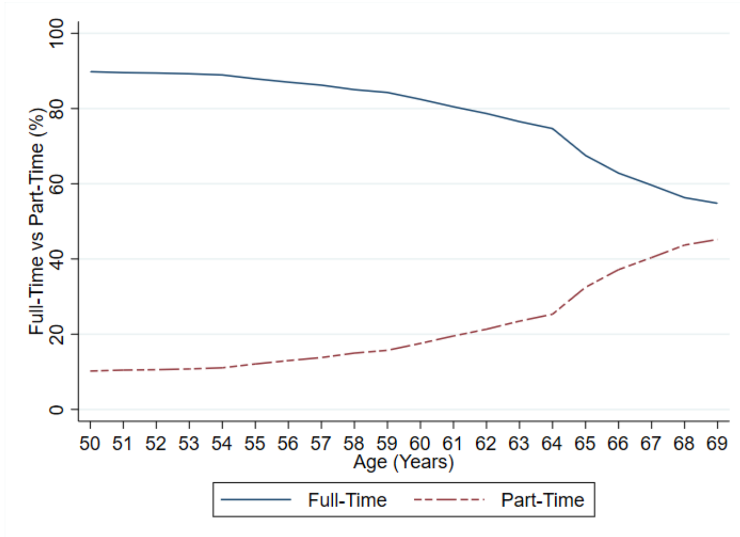
Though the proportion of workers engaged in part-time jobs increases for all ages, the share of part-time employment rises rapidly after age 64 for both countries. In the United States, the considerable increase in part-time employment after age 65 could be due to (1) Medicare eligibility at age 65 and (2) Amendments to the Social Security earnings test (Dunn (2018)). Reaching the Medicare eligibility age reduces the need for older workers to work full-time to maintain health insurance coverage.

Moreover, before 2000 the Social Security Act reduced social security benefits if the retiree's earnings exceeded the RET-exempt threshold. Although the CPP/QPP is not subject to an earnings test in Canada, combined earnings from work and pension benefits could put retirees into a higher tax bracket. According to Milligan and Schirle (2021), the increase in the marginal tax bracket could result in lower employment rates. It could also result in a preference for part-time over full-time employment. Further, eligibility for the OAS and GIS at age 65 could encourage older workers in Canada to work part-time. OAS recipients who qualify for the GIS can earn up to $3,500 per year before GIS benefits are reduced.

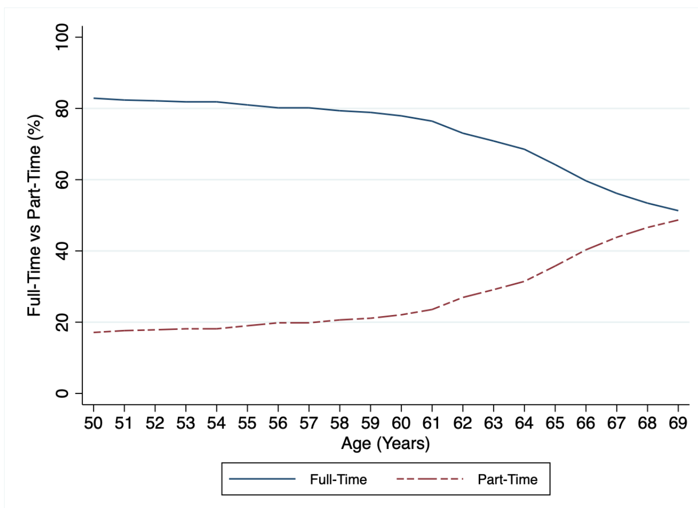
2.4.1 Reasons Part-Time

In the LFS and the CPS, workers are classified based on the reason for working part-time. To address question 2, I use the responses to the question: *What is the main reason for working part-time?* Table 2.4 (column 3) gives the sample in each category for Canada and the United States. I group these reasons into (1) demand-side and (2) supply-side factors. Demand-side accounts for 24.5% and 32.4% of total part-time employment in Canada and

Figure 2.3: Full-Time versus Part-Time Employment by Age



(a) Canada



(b) United States

Notes: The figure plots the proportion of workers employed full-time and part-time. Figure A plots the proportions for Canada and Figure B the proportions for the United States. **Source:** Canada Labour Force Survey and US Current Population Survey (1997 - 2019). No. Observations = 2,834,136 (Canada) and 2.723,076 (United States).

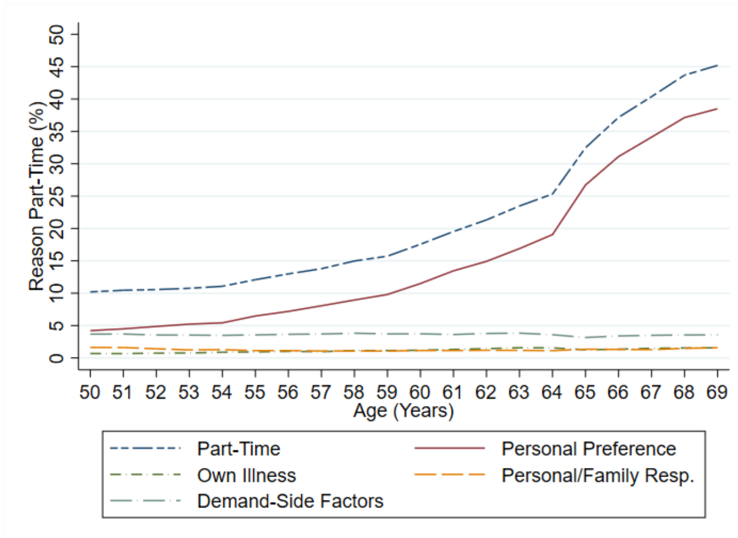
the United States, respectively. These workers' main reasons for part-time employment are business conditions, full-time work is less than 35 hours, or they could not find a full-time job. In contrast, supply-side reasons account for 75.5% and 67.6% of total part-time employment, respectively. These findings support the existing literature and suggest that it is the worker's personal choices about whether to work part-time that accounts for the prevalence of part-time employment rather than difficulties in finding full-time employment, or business conditions (Gash (2008); O'Sullivan et al. (2021); Hakim (2002, 2001)).

Table 2.4: Reasons Part-Time by Gender

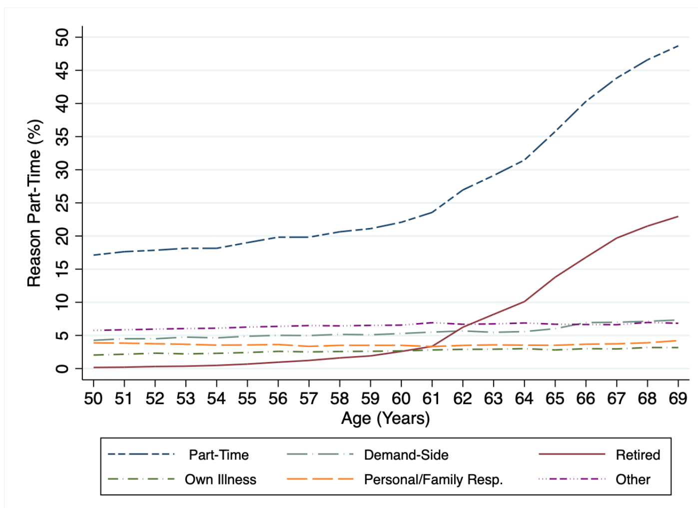
	Canada			**United States**			
Reasons Part-Time	Female	Male	Total	Reasons Part-time	Female	Male	Total
Supply-side factors				*Supply-side factors*			
Personal preference	59.5%	58.9%	59.4%	Retired	11.7%	18.8%	14.2%
Own-illness	6.2%	7.9%	6.5%	Own-illness	11.2%	13.7%	12.1%
Personal or Family Resp.	7.1%	5.9%	6.9%	Personal or Family Resp.	35.3%	27.3%	32.5%
Caregiver	3.1%	1.1%	2.7%	Caregiver	0.8%	0.2%	0.6%
Demand-side factors				*Demand-side factors*			
Business conditions	15.8%	17.2%	16.1%	Other	9.5%	6%	8.3%
Could not find job	8.3%	9%	8.4%	Demand-side	31.5%	34%	32.4%
Share of sample	77.3%	22.7%	100%	Share of sample	65%	35%	100%
No. Observations	2,834,136			No. Observations	2,723,076		
No. Individuals	472,356			No. Observations	680,769		
No. Waves (months)	6			No. Waves (months)	4		

The table shows the average responses over the sample for men and women to the question: *What is the main reason you are part-time?* classified into six categories. All respondents were aged between 50 and 69 years. **Source:** Canada Labour Force Survey and US Current Population Survey (1997 - 2019)

Figure 2.4: Reasons Employed Part-Time by Age



(a) Canada



(b) United States

Notes: The figure plots the proportion of workers employed part-time by reason. Figure A plots the proportion for Canada and Figure B the proportion for the United States. **Source:** Canada Labour Force Survey and US Current Population Survey (1997 - 2019). No. Observations = 2,834,136 (Canada) and 2.723,076 (United States).

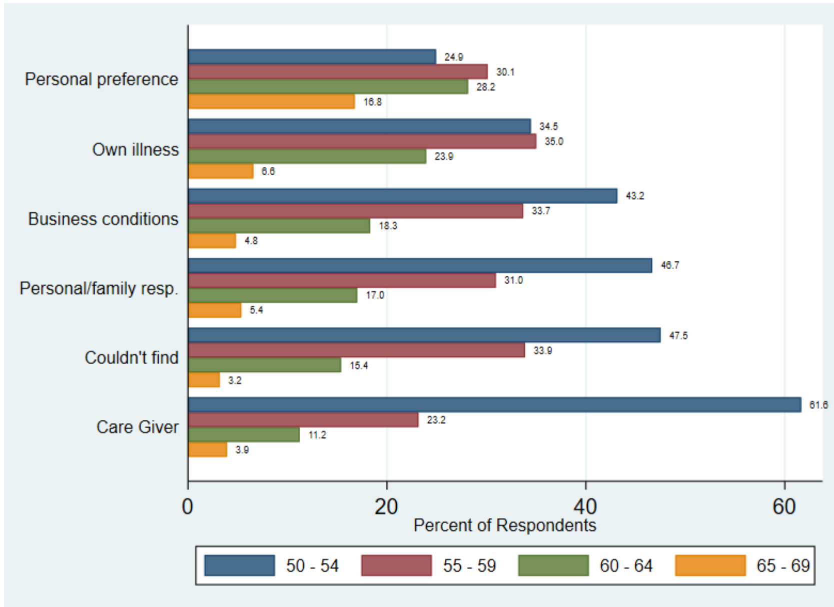
In Figure 2.4 (panel a), personal preference is the primary reason for working part-time in Canada. The proportion employed part-time for personal reasons increases consistently from ages 50 to 69. Personal preference accounts for approximately 50% of the workers employed part-time in Canada from ages 50 - 54; however, the share increases to about

70% by age 64 and increases further to approximately 78% by age 65, the NRA for benefit claiming in Canada. In Figure 2.4 (panel b), the primary reason given by older workers for working part-time in the United States is retiring from full-time employment and/or the Social Security earnings limit. This proportion increases considerably after age 60, the early retirement age for benefit claiming in the US. By age 65, the percentage of part-time due to retirement accounts for nearly 50% of the proportion of workers employed part-time. This share remains relatively constant over the ages 60 to 69 years.

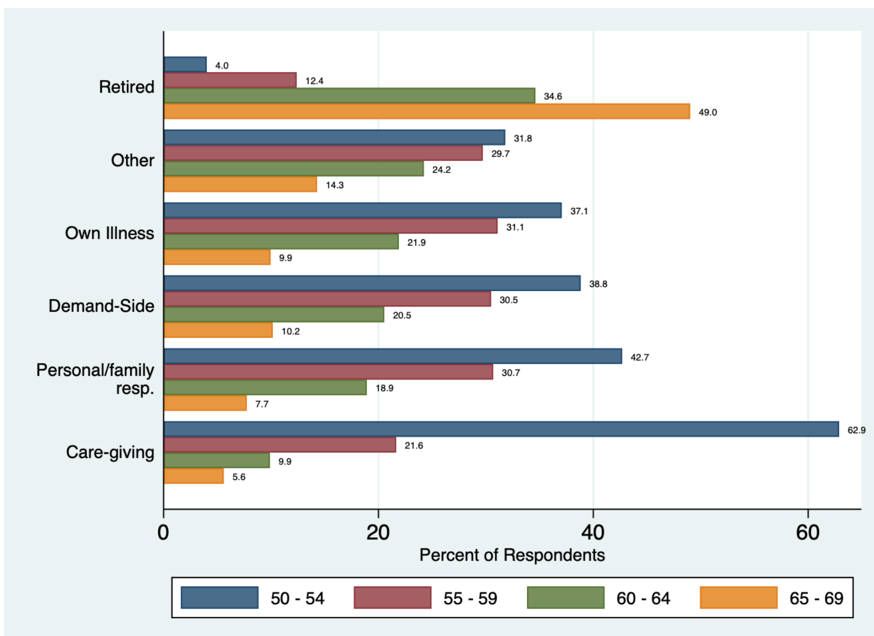
Examining the age groups in Figure 2.5 reveals that the main reason for part-time employment differs considerably. In Canada, the main reason given in the group 50 - 54 years is caregiving, the 55 - 59 age group cite being unable to find a job, while age groups 60 - 64 and 65 - 69 years give personal preference. In the United States, the age group 50 - 54 shows caregiving is the main reason for being part-time. In the 55 - 59 age group, own illness, demand-side factors, and personal and or family responsibilities are almost equal. In the 60 - 64 and 65 - 69 age groups, the retired and/or social security earnings limit is the main reason for working part-time.

Personal preference and retirement as the main reason for choosing part-time are due to workers reaching the early and NRA eligibility thresholds for claiming public pension benefits in Canada and the United States. In the United States, the earnings test impacted the hours worked by older workers, as it restricted the amount they could earn without incurring a reduction in their social security benefits. In 2000, the United States removed the earnings test for beneficiaries who exceed their NRA.

Figure 2.5: Reasons Employed Part-Time by Age Group



(a) Canada



(b) United States

Notes: The figure shows the reasons part-time by age group. Figure A represents Canada and Figure B represents the United States. **Source:** Canada Labour Force Survey and US Current Population Survey (1997 - 2019). No. Observations = 2,834,136 (Canada) and 2.723,076 (United States).

2.4.2 Gender and Part-Time Employment

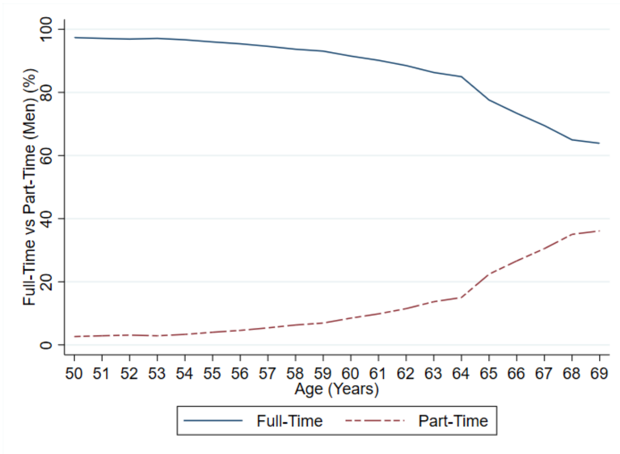
Gender impacts the preference for part-time employment amongst older workers. Table 2.3 shows that women are more likely to work part-time in both countries on average. The fraction of women employed part-time is 21.9% and 28.2% in Canada and the United States, respectively. Further, women in Canada are three times more likely to be in part-time employment than men, while they are two times more likely in the United States. An examination of the age profile shows that older women are more likely to work part-time than men of all ages (Figure 2.6). For men at age 50, though the share employed part-time in Canada starts more than 10 percentage points lower than men in the United States, the share increases more rapidly in Canada, with a dramatic increase seen after age 64. By age 69, the gap between full-time and part-time employment closes to around 30 and 20 percentage points in Canada and the United States, respectively.

For women, the proportion of part-time at age 50 is approximately 15 percentage points higher in the United States than in Canada. There is a slight increase in the proportion of working part-time for the age group 50 - 54 in both countries, and it starts increasing faster in Canada than in the United States between the ages of 55 and 64. At age 64, there is a considerable increase in the proportion of part-time workers in both countries; by age 66, approximately 50% of women are employed part-time in Canada and the United States. In Canada, the full-time and part-time proportion intersects at age 66, after which point part-time employment exceeds full-time, though the rate of increase decreased slightly at age 66. I see a similar pattern in the United States, but the full-time and part-time proportion intersects one year later at age 67.

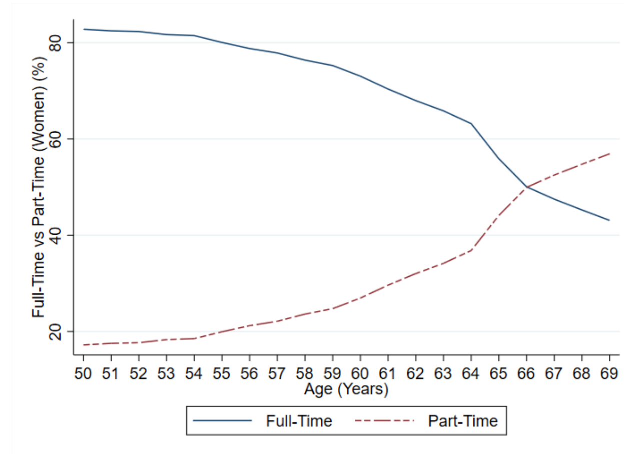
The higher fraction of women employed part-time is explained in the literature due to a longer life expectancy (Paz et al. 2017), caregiving and family responsibilities (Fagan 2003, O'Brien 2014). This finding is supported by Table 2.4, which shows that supply-side factors are the primary reason for being part-time in Canada and the United States. Personal preference tops the supply-side factors in Canada at 59.5%, followed by personal and/or family responsibilities (7.1%) and own-illness (6.2%). Somewhat surprisingly, caregiving only accounts for 3.1%. In the United States, personal and/or family responsibilities (35.3%), retired and/or social security limits (11.7%) and own illness (11.2%) make up the top three supply-side reasons given by women for part-time employment (Figure 2.7).

In summary, there is a greater prevalence of part-time employment in Canada and the United States. Furthermore, workers aged 65 - 69 are more likely to be part-time compared to workers 50 - 64 years. The main reason for part-time employment is personal preference in Canada and retirement and/or social security limits in the United States. This result indicates that supply-side considerations rather than demand-side factors account for the increase in part-time employment for older workers. Regarding gender, women are more likely than men to be employed part-time. Surprisingly, the main reason women provided for being part-time was not caregiving or family and personal responsibilities but rather a preference for part-time versus full-time arrangements.

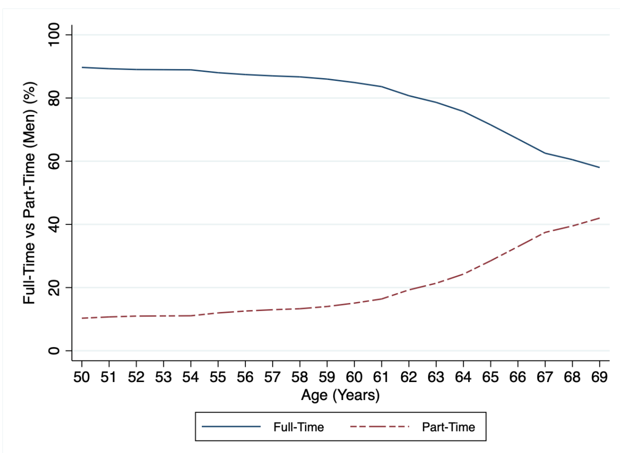
Figure 2.6: Full-Time versus Part-Time Employment by Age and Gender



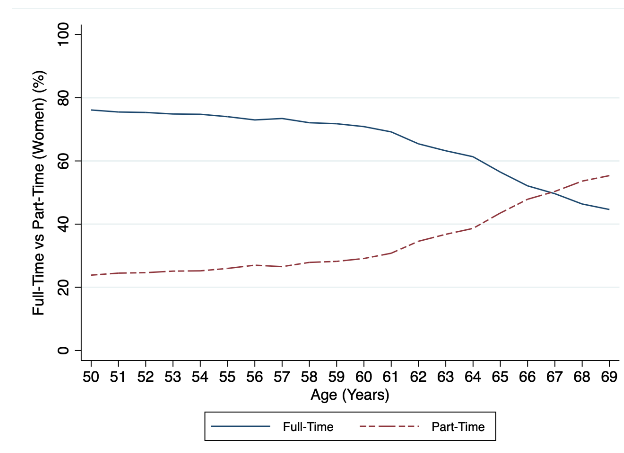
(a) Men - Canada



(b) Women - Canada



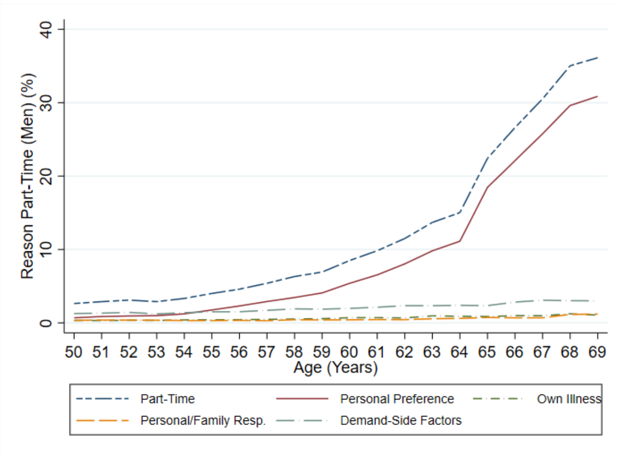
(c) Men - USA



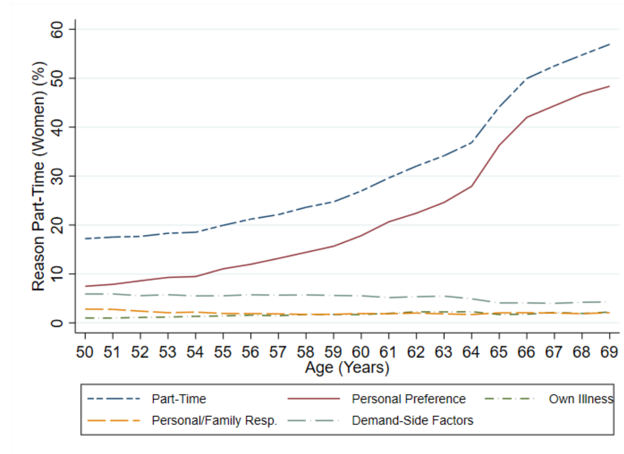
(d) Women - USA

Notes: The figure plots the proportion of workers employed full-time and part-time by age and gender in Canada and the United States. Figure A plots the proportions for Men - Canada, Figure B the proportions for Women - Canada, Figure C plots the proportions for Men - USA, Figure D the proportions for Women - USA. **Source:** Canada Labour Force Survey and US Current Population Survey (1997 - 2019). No. Observations = 2,834,136 (Canada) and 2.723,076 (United States).

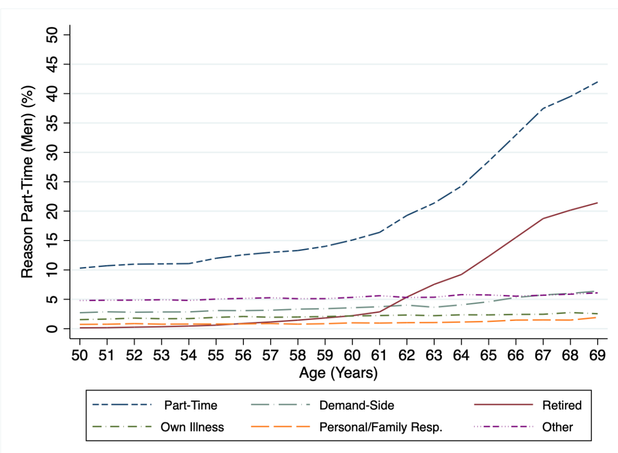
Figure 2.7: Reasons Employed Part-Time by Age and Gender



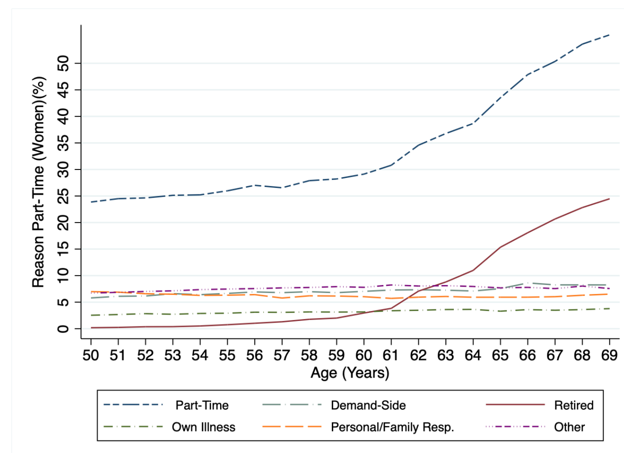
(a) Men - Canada



(b) Women - Canada



(c) Men - USA



(d) Women - USA

Notes: The figure plots the proportion of workers employed part-time by reason and gender in Canada and the United States. Figure A plots the proportions for Men - Canada, Figure B the proportions for Women - Canada, Figure C plots the proportions for Men - USA, Figure D the proportions for Women - USA. **Source:** Canada Labour Force Survey and US Current Population Survey (1997 - 2019). No. Observations = 2,834,136 (Canada) and 2.723,076 (United States).

2.5 Attachment of the Non-Employed

Turning my attention to non-employed older workers in Canada and the United States, I examine how the proportion of non-employed workers changes with age. Additionally, I review the reasons for non-employment in the literature in light of the responses to the survey questions in the LFS and CPS on the reasons for being unemployed and NILF. In this chapter, I focus on answering two questions:

1. Are older workers more likely to separate from the labour market as they age?

2. Is the non-employment of older workers driven by demand-side or supply-side factors?

According to Jones and Riddell (2019), workers classified as unemployed are seen as attached to the labour market, while those not in the labour market are non-attached. However, the proximity to eligibility age thresholds for public pension influences the decision whether to exit the labour market or remain attached for respondents classified as unemployed or NILF. To address question 1, I examine the proportion of workers unemployed and NILF by age. In the LFS and the CPS sample, all workers are employed in Month 1, and, on average, 1.6% and 1.3% of the sample are unemployed, and 2% and 3.2% are NILF in Canada and the United States, respectively (Table 2.2). Figure 2.8 shows that the proportion of unemployed workers in Canada and the United States declines with age while the proportion NILF increases. In Canada, between the ages of 50 to 54 years, the proportion of unemployed exceeds those of NILF, but the fraction of NILF increases significantly from age 54. From age 55 onwards, the fraction of respondents that are NILF exceeds those unemployed, with peaks seen at ages 55, 60 and 65. In contrast, in the United States the proportion NILF exceeds the proportion unemployed throughout the age profile, with peaks seen in the NILF at ages 55 and 62.

I see a similar trend for gender as seen for the total sample in Canada and the United States. In both countries, the proportion of unemployed men exceeds that of women (Figure 2.9). The predominance of men in primary and manufacturing industries with strong unions could account for this result. Additionally, women are more likely to be employed in service industries. The service industries in the US and Canada have grown in the past ten years, and there tend to be more jobs available.

The descriptive analysis shows that the proportion of non-employed workers increases with age. Interestingly, in Canada, the proportion of older men unemployed remains above those NILF until age 59, while in the United States, the change occurs at age 55. Figure 2.9 (panels b and d) also shows that the proportion of women who are NILF at age 50 is higher in the United States than in Canada. However, this proportion increases faster in Canada; by age 69, they are roughly the same. Additionally, the share of NILF remains above those unemployed throughout the entire age profile in the United States. For Canada, from age 54 onwards, the proportion of NILF exceeds the number of unemployed. Further, the peaks seen in the full sample are replicated for men and women in Canada and the United States.

Older workers decide whether to exit the labour market at the early and NRA in Canada and the United States. The earnings test in the United States and the work cessation test in Canada (before 2012) can explain the peaks seen at ages 60, 62 and 65. The US earnings test reduces pension benefits if the earnings of retirees are above the RET exemption threshold. This test could encourage older workers to leave the labour market to avoid the penalty on social security benefits.[7] In Canada, before 2012, workers had to show a reduced income level in the month they claimed CPA-QPP and the month after to be eligible for early retirement benefits. This work cessation test could account for the peaks in the proportion of NILF observed in the LFS at ages 60 and 65.

[7] After 2000, the Social Security earnings test applies only to workers who retire before their NRA.

However, what accounts for the peak seen at age 55? In the pension system in Canada and the United States, workers can retire earlier than the early age eligibility thresholds (60 and 62, respectively) in the following scenarios:

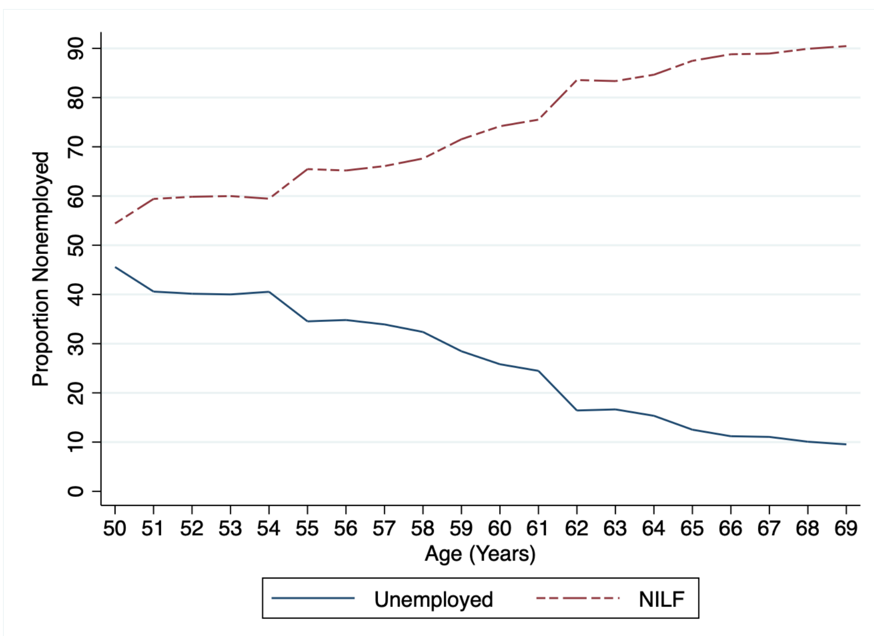
1. Retirement due to mental or physical disability that prevents working regularly.

2. Employed as a public sector worker.

Given that there is no specific age requirement for claiming disability pension benefits, workers leaving the labour force due to disability would not account for the spike observed in the NILF at age 55. Regarding public sector workers, In Canada, before 2012, public sector workers qualify for unreduced pension benefits at age 60 with two years of pensionable service or at age 55 with 30 years of pensionable service. In addition, they are eligible for reduced pension benefits at age 50 with at least two years of pensionable service. In the United States, the Minimum Retirement Age (MRA) for public sector workers is 55 if born before 1948 and increases by 2-month increments based on birth. Public sector workers can qualify for unreduced pension benefits with at least 30 years of service before age 60 and 20 years of service from age 60 onwards. Public sector pension benefits in the United States are subject to a 5% reduction if the retiree is under 62 and has less than 30 years of pensionable service. Therefore, public sector workers' decision to leave the labour market once becoming eligible for pension benefits explains the spike at age 55 in the NILF for Canada and the United States. According to Statistics Canada (2023), the average retirement age for public sector workers in 2019 was 62.6 years compared to 64.2 years for private sector workers.

Figure 2.8: Unemployed versus Not-In-the-Labour Force by Age

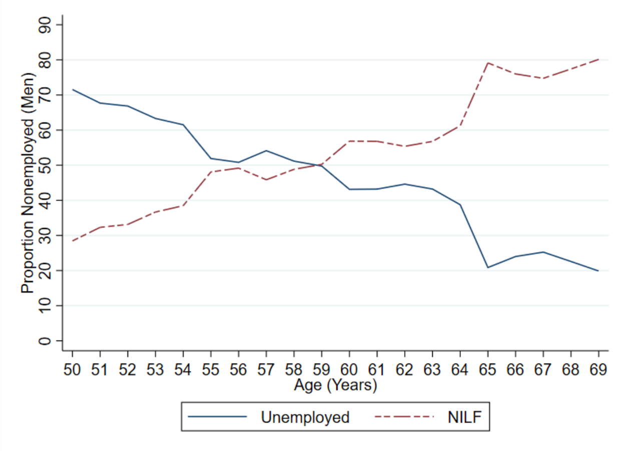
(a) Canada



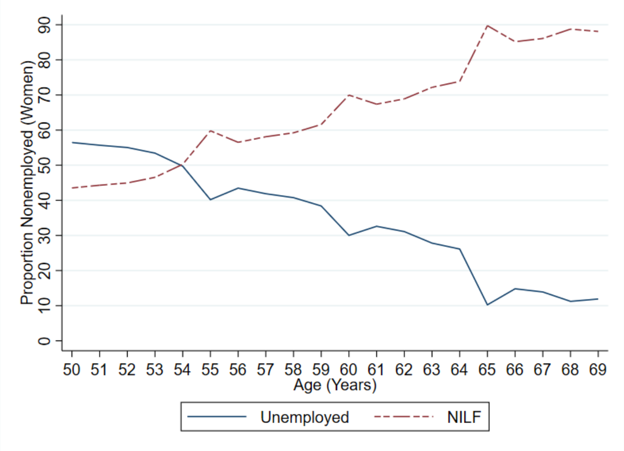
(b) United States

Notes: The figure plots the proportion of workers unemployed versus NILF by age. Figure A plots the proportions for Canada and Figure B the proportions for the United States. **Source:** Canada Labour Force Survey and US Current Population Survey (1997 - 2019). No. Observations = 2,834,136 (Canada) and 2.723,076 (United States).

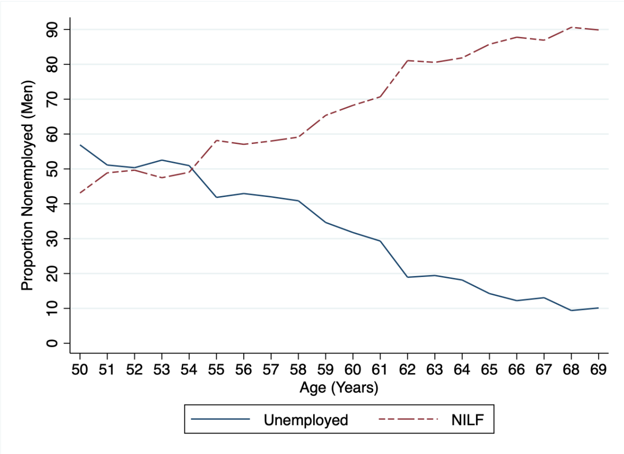
Figure 2.9: Unemployed versus Not-in-the-Labour Force by Age and Gender



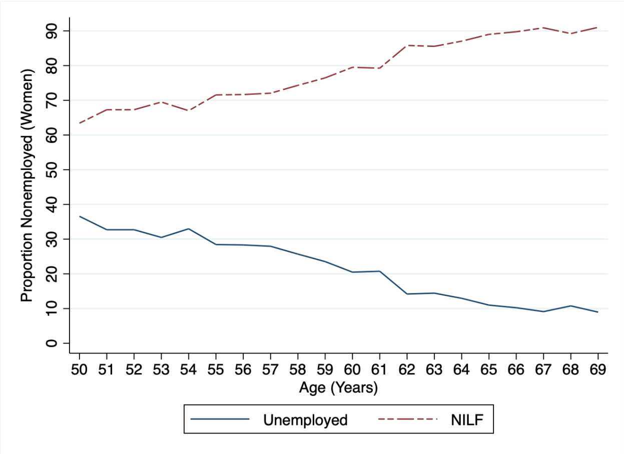
(a) Men - Canada



(b) Women - Canada



(c) Men - USA



(d) Women - USA

Notes: The figure plots the proportion of workers unemployed versus NILF by age and gender in Canada and the United States. Figure A plots the proportions for Men - Canada, Figure B the proportions for Women - Canada, Figure C plots the proportions for Men - USA, Figure D the proportions for Women - USA. **Source:** Canada Labour Force Survey and US Current Population Survey (1997 - 2019). No. Observations = 2,834,136 (Canada) and 2.723,076 (United States).

2.5.1 Reasons Non-Employed

In the existing literature, reasons given for older workers deciding not to remain in the labour market include lack of transferable skills, discrimination, lack of jobs, health issues, family responsibilities and financial incentives from public pension systems (Canada (2016, 2018); OECD (2019c). In this section, I use the responses to the questions on reasons non-employed to determine if the factors driving the non-employment of older workers in Canada and the United States align with the existing literature.

In Canada, the LFS provide responses to the question: *What is the main reason for being non-employed?* Table 2.5 presents the sample in each category for older workers unemployed and NILF in Canada. I group these reasons into (1) demand-side and (2) supply-side factors. In Table 2.5, demand-side factors account for 65.5% of total non-employed older workers in Canada. The main reasons for non-employment by these workers are business conditions and the end of short-term jobs (temporary, seasonal and casual).

In contrast, supply-side reasons account for 34.5%, with retirement as the primary reason. An examination of those unemployed provides a similar trend, with 89.3% due to demand-side factors. Business conditions account for 41.1%, and end of short-term jobs account for 39.9% (Table 2.5 (column 2)). This descriptive evidence suggests that unemployed older workers have a preference for short-term work jobs, which ties in with the findings from Section 2.4 that older workers have a greater desire for part-time work arrangements. For those classified as NILF, Table 2.5 (column 3) shows that supply-side factors (53.3%) are the drivers for those who did not look for a job in the past four weeks. On the demand side, the main reasons are due to the ending of short-term employment (37%).

In the United States, the CPS provide responses to two questions: (1) *What is the main reason for being unemployed?* and (2) *What is the main reason for not looking for a job in the past four weeks?* The CPS interviewers ask question 2 if respondents are NILF but indicate they want a job. Jones and Riddell (2019) define these workers that are NILF but want a job as marginally attached to the labour market. Table 2.6 gives the sample in each

Table 2.5: Reasons Non-Employed - Canada

Reasons	Unemployed	NILF	Total
End of Seasonal Job	20.9%	18.3%	19.5%
End of Casual Job	6%	10.7%	8.6 %
End of Temporary Job	13%	8%	10.2%
Demand-Side	49.4%	9.7%	27.2%
Quit	8.4%	13.4%	11.2%
Retired	2.3%	39.9%	23.3%
Share of sample	44%	56%	100%
No. Observations	2,834,136		
No. Individuals	472,356		
No. Waves (months)	6		

Notes: The table shows the average responses to the question: *What is the main reason you left your last job?* classified into six categories. All respondents were aged between 50 and 69 years.
Source: Canada Labour Force Survey (1997 - 2019)

category for workers unemployed and NILF (but want a job) in the United States. I group these reasons into (1) demand-side and (2) supply-side factors. For the unemployed workers in the United States, demand-side factors account for 85.3%, while for those NILF who want a job, demand-side factors account for 46.7%. The main reason for being unemployed is job separation due to layoff, the end of a temporary job or some other job loss.

Table 2.6: Reasons Non-Employed - United States

Reasons	Unemployed	Reasons	NILF
Layoff	40.5%	Demand-Side	23.3%
Othe Job Loser	33.8%	Lack of Skill	1 %
End of Temporary Job	11%	Discrimination	1.9%
Quit	7.3%	Own-illness	15.3%
Re-Entrant	7.2%	Personal	15.9%
New Entrant	0.2%	Other	42.6%
No. Observations	2,723,076		
No. Individuals	680,769		
No. Waves (months)	4		

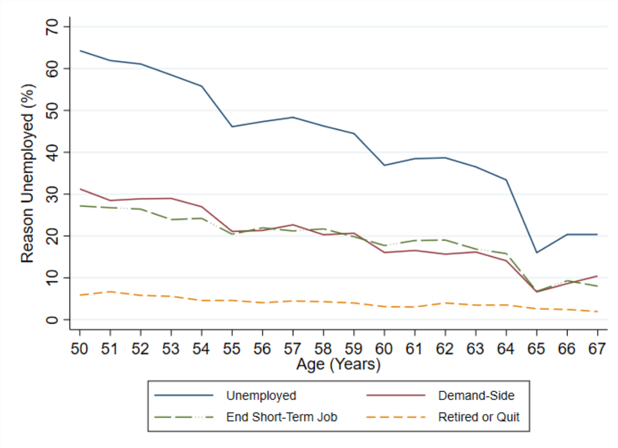
Notes: The table shows the average responses to the questions: (1) *What is the main reason you are unemployed?* classified into six categories and (2) *What is the main reason you were not looking for work during the previous four weeks?* classified into six categories. All respondents were aged between 50 and 69 years. All respondents were aged between 50 and 69 years. **Source:** US Current Population Survey (1997 - 2019).

Figure 2.10 depicts how the reasons given by the non-employed workers change with age in Canada and the United States. In Canada (Figure 2.10 (panel a), demand-side factors and end of short-term jobs move together over the age profile with slight deviations between ages 50 to 54 and 60 to 64 years. Between 50 and 54, demand-side factors exceed the end of short-term jobs, while between ages 60 and 64, the end of short-term employment is higher. This finding supports the evidence seen in Section 4 and Table 2.5 that unemployed older workers have a greater preference for flexible work arrangements.

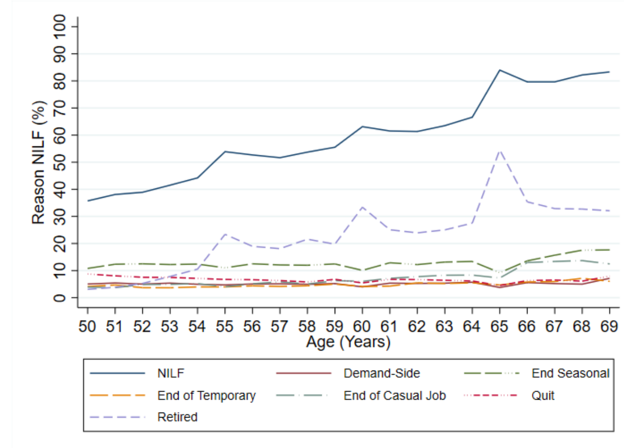
I see a similar pattern between layoffs and other job losers in the United States. Figure 2.10 (panel c) shows that the series move together from ages 50 to 56, but from age 56 onwards, layoffs exceed other job losers as the main reason given for being unemployed. This finding aligns with the literature in the United States that older workers are more likely to experience layoffs as they age (Chan and Stevens (2001, 2004, 1999)). In chapter 3 of my thesis, I analyse the impact of a layoff on the decision of older workers in Canada to remain attached or exit the labour market.

For workers, NILF in Canada aged 50 to 54 years, the end of seasonal jobs was the main reason. These workers might be waiting for recall to their previous employment once the season begins. Additionally, Figure 2.10 (panel b) illustrates that for Canadian workers NILF, retirement is the predominant reason given for workers ages 55 to 69. There are dramatic spikes in the retirement series at ages 55, 60 and 62. Age 60 corresponds to the early eligibility threshold for pension claiming, so the spike corresponds to the worker's decision to leave the labour market. Also, before 2012 workers deciding to claim early retirement had to show a reduced level of income in the month of pension claiming and the month after. The work cessation test could account for the spike in the analysis at age 62. The spike at age 55 could be due to eligible public sector workers deciding to retire at age 55.

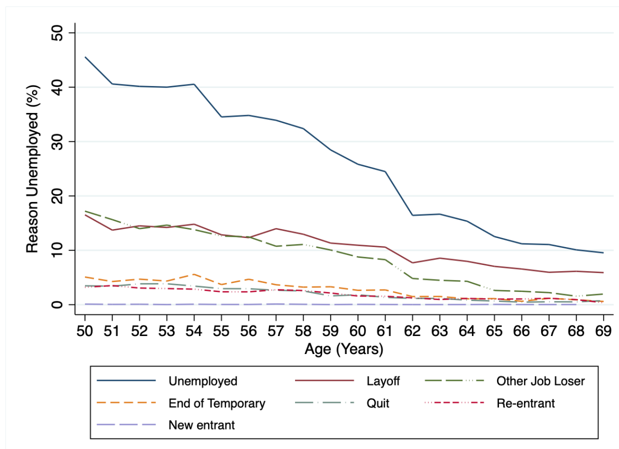
Figure 2.10: Reasons Non-Employed by Age



(a) Unemployment - Canada



(b) NILF - Canada



(c) Unemployed - USA

Notes: The figure plots the proportion of workers non-employed by age in Canada and the United States. Figure A plots the proportions Unemployed - Canada, Figure B the proportions NILF - Canada, Figure C plots the proportions Unemployed - USA. **Source:** Canada Labour Force Survey and US Current Population Survey (1997 - 2019). No. Observations = 2,834,136 (Canada) and 2.723,076 (United States).

Spike at Age 55

To examine the spike at age 55 for workers NILF, I focus on the responses given by respondents in the LFS for not looking for a job in the last four weeks. The question asked in the CPS only focuses on NILF workers who indicated they want a job, so the responses do not adequately represent all workers classified as NILF. Workers in the public sector tend to retire earlier and are less likely to return than private-sector workers. To determine if early retirement of public sector workers in Canada drives the spike seen at age 55, Figure 2.11 illustrates the reasons given for being NILF by each class of workers. For public sector workers, the spikes at the eligibility age thresholds of 55, 60 and 65 years are much higher than those for private sector workers at the same ages. A similar trend is observed across the age profile for older men and women in Canada. Figure 2.12 and Figure 2.13 illustrates these results. Overall public sector workers are more likely to retire due to NILF at the age eligibility thresholds of 55, 60 and 65 years than men and women in the private sector.

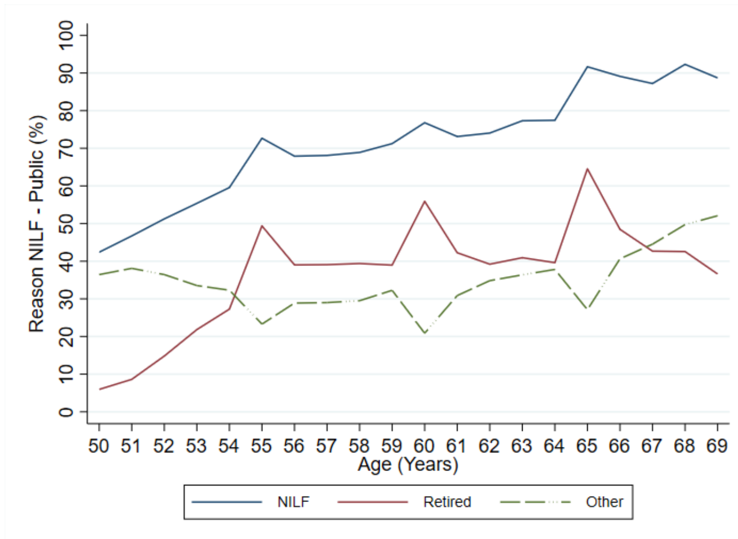
2.6 Conclusion

In this chapter, I examine the labour market attachment of older workers in Canada and the United States. In my LFS and CPS samples, all workers are employed in Month 1 of the panels. I focus on how the level of attachment changes with age for workers (1) employed, (2) unemployed and (3) NILF. I also explore whether demand-side or supply-side factors account for the changes in the labour market attachment in the samples. The descriptive analysis shows that older workers 50 - 69 years prefer part-time versus full-time work arrangements, which increases considerably between the ages 60 to 69.

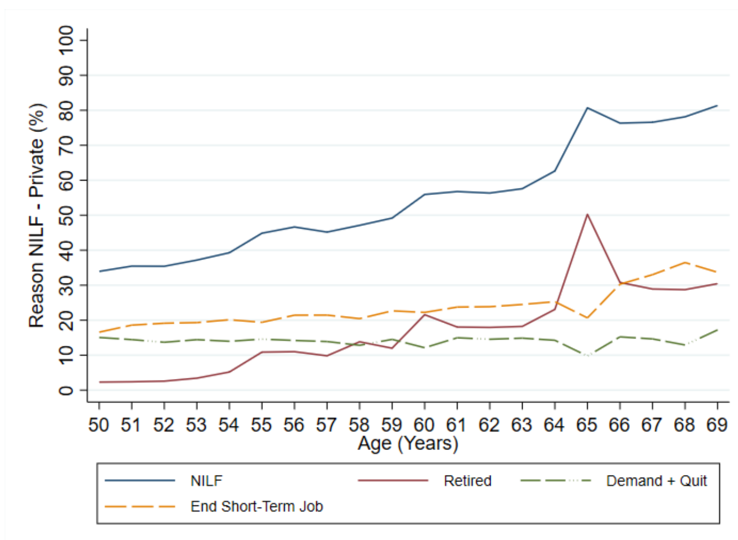
Further, women are more likely to be employed part-time than men. The reasons given for part-time employment are driven by supply-side rather than demand-side factors. In Canada, the primary reason is personal preference, while in the United States retirement and/or social security earning limits. Additionally, I find that for unemployed workers, even though demand-side factors are more dominant, workers are usually unemployed due to ending short-term jobs. I also find that proportion of workers NILF increases with age. Separated workers are more likely to be non-attached to the labour market.

Overall, the chapter supports the view that the labour market attachment of older workers weakens with age. Though the LFPR is increasing over time, a considerable share of employed and unemployed workers prefer short-term, flexible work arrangements. For the workers marginally attached or not attached to the labour market, the dominant factor seems to be retirement or the ending of short-term jobs rather than ageism or lack of skill.

Figure 2.11: Reasons NILF for Public and Private Sector Workers by Age



(a) Public Sector



(b) Private Sector

Notes: The figure plots the reasons given by public and private sector workers for NILF by age. Figure A plots the reasons for Public sector workers and Figure B the reasons for the Private Sector workers. **Source:** Canada Labour Force Survey (1997 - 2019). No. Observations = 2,834,136 (Canada). To meet the vetting requirement for the release of the graphs the other reasons given by public sector workers was combined to give the category "Other". Some of the categories were also combined for the private sector workers to give (1) demand and quit and (20 end of short-term jobs.

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2.7 Appendix

Table 2.7: Descriptive Statistics for Canada and the United States (Month 1)

	Canada	United States
Age	55.8 years	53.5 years
Age Group		
50 - 54	46%	68.5%
55 - 59	32.6	16.9
60 - 64	16.9	10.1
65 - 69	4.5	4.5
Gender (Male = 1)	50.1	50.8
Living with Partner	77	67.2
No. Individuals	472,356	680,769
Waves (Months)	6	4

Notes: The table provides the summary statistics for all employed workers aged 50 - 69 years in Month 1 of the sample. **Source:** Canada Labour Force Survey and US Current Population Survey (1997 - 2019)

Table 2.8: Institutional Framework in Canada and the United States

Institutional Framework	**Categories**	**Canada**	**United States**
Definitions	“Passive” search methods	Unemployed	NILF
	Job to start within the next four weeks	Employed	NILF
	Not available for work because of personal or family responsibilities	NILF	NILF
	Full-time students	NILF	Unemployed
	Working Age Population	15 years and older	16 years and older
Demographics	Median Age (2018)	40.8 years	38.2 years
	Share of population 65+ (2019)	18%	16%
	Life expectancy at birth (2019)	82 years	78.8 years
	Age dependency ratio (2019)	26%	24%
Health Care System		Government funded No co-payment	Self-funded and tied to employment Co-payment required Medicare access at age 65
Public Pensions	Funding	CPP/QPP - Defined benefit GIS - income requirement OAS - meet residency requirement	OASDI - Defined benefit
	Eligibility Age	Early - 60 years Normal - 65 years	Early - 62 years Normal - 65 years born 1937. Increases in 2 months increments after 1937
	Early Claiming Penalty	0.6% for each month before the NRA.	6 2/3% for each year before the NRA
	Pension Deferral	Maximum Age 70 CPP-QPP increases by 0.7% per month OAS increases by 0.6% per month	Maximum Age 70 Social security increase by 8% for each year

Institutional Framework in Canada and the United States (Cont'd)

Institutional Framework	**Categories**	**Canada**	**United States**
	Retirement Earnings Test (RET)	CPP/QPP - None Before 2012, reduction in earning required OAS - reduced by 15 cents for every dollar above the threshold OAS threshold (2019) - CAN$77,580	65 + - No penalty (since 2000) under 65 - 50% penalty over RET-exempt. RET-exempt 2019 = US$17,640
Private Pensions	Contribution (2023)	RRSP - before tax maximum CAN$31,560$ TFSA - after tax maximum CAN$7,500$	Traditional IRA - before tax contribution Roth IRA - after tax contribution IRA maximum US$6,500$ Traditional 401k - before tax maximum US$22,500 under 50$ maximum US$30,000 for 50+$ Roth 401k - after tax maximum US$7,500$
	Taxes	RRSP - tax paid at withdrawal TFSA - tax free withdrawal	Traditional IRA - tax paid at withdrawal Roth IRA - tax free withdrawal Traditional 401k - tax paid at withdrawal Roth 401k - tax free withdrawal
	Age Restrictions	RRSP - 71 years	IRA - 70.5 years (removed 2019)

Notes: The table provides a summary of the institutional framework in Canada and the United States.

Table 2.9: Part-Time Employment by Reason - United States

Reasons Part-Time	Responses
Demand-Side	Slack work, business condition; Labour dispute; Full-time work under 35 hours; Job started or ended during week; Could only find part-time; Seasonal work Weather affected job; Holiday
Own-illness	Own-illness; Health and/or medical limitation
Retired	Retired and/or Social Security limit on earnings
Personal/Family Responsibilities	Other family/personal responsibilities; School/training; Civic and/or Military duty; Vacation/Personal day
Care Giver	Child care problems
Other	Other

Notes: The table shows the responses to the question: *What is the main reason you are part-time?* classified into five categories. **Source:** US Current Population Survey (1997 - 2019).

Table 2.10: Part-Time Employment by Reason - Canada

Reasons Part-Time	Responses
Personal Preference	Personal preference
Own-illness	Own-illness or disability
Care Giver	Caregiving for own children; Caregiving for elder relative (60+)
Personal/Family Responsibilities	Other; Other family/personal responsibilities; Going to school
Business Conditions	Business conditions, looked for full-time, involuntary part-time; Business conditions, did not look;
Could not find	Could not find full-time, looked for full-time, involuntary part-time; Could not find full-time work, did not look

Notes: The table shows the responses to the question: *What is the main reason you are part-time?* classified into six categories. Source: Canada Labour Force Survey (1997 - 2019).

Table 2.11: Non-Employed by Reason - Canada

Reasons	Responses
End Seasonal	End seasonal job
End Casual job	End of casual job
End Temporary	End of temporary, term or contract (non-seasonal)
Demand-Side	company moved; company went out of business; business sold or closed down; business conditions; dismissal by employer
Quit	dissatisfied with job; caring for own children; caring for elder relative (60+); going to school; other personal or family responsibilities; other reasons; pregnancy; changed residence
Retired	retired

Notes: The table shows the responses to the question: *What is the main reason you left your last job?* classified into six categories. **Source:** Canada Labour Force Survey (1997 - 2019).

Table 2.12: Non-Employed by Reason - United States

Reasons	Responses
Layoff	Job loser - on layoff
Other Job Loser	Other job loser
End Temporary	Temporary job ended
Quit	Job leaver
Re-Entrant	Re-entrant
New Entrant	New entrant

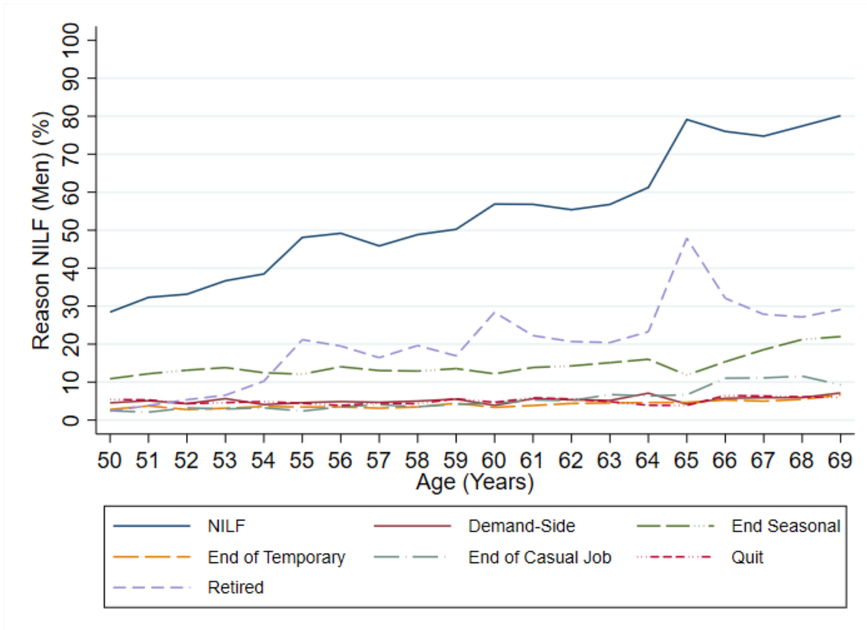
(a) Unemployed

Reasons	Responses
Demand-Side	Couldn't find any work ; Believes no work available in area of expertise ;
Lack of Skill	Lacks necessary schooling / training
Discrimination	Employers think too young or too old Other types of discrimination
Own-illness	Ill-health, physical disability
Personal/Family Responsibilities	Can't arrange childcare; Family responsibilities ; In school or other training
Other	Other - specify; Transportation problems

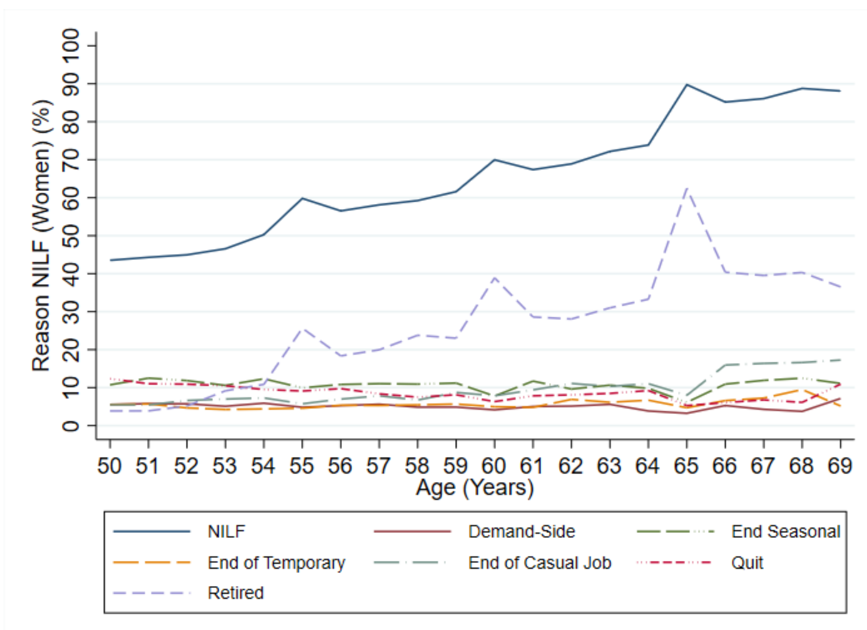
(b) Not-in-the-Labour-Force

Notes: The table shows the responses to the questions: (1) *What is the main reason you are unemployed?* classified into six categories and (2) *What is the main reason you were not looking for work during the previous four weeks?* classified into six categories. **Source:** US Current Population Survey (1997 - 2019).

Figure 2.12: Reasons NILF by Age and Gender



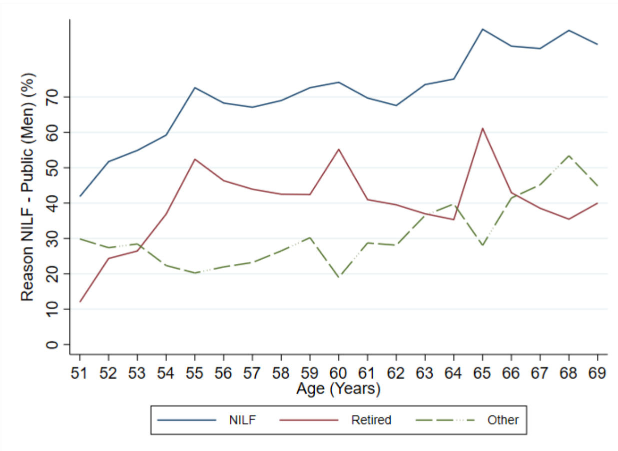
(a) Men - Canada



(b) Women - Canada

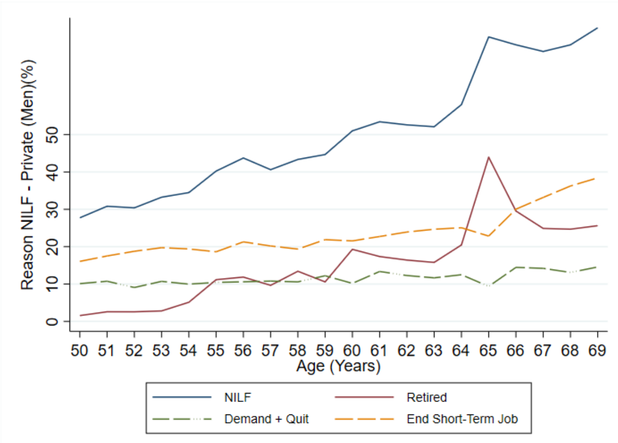
Notes: The figure plots the reasons given by workers NILF by age and gender in Canada. Figure A plots the reasons for Men, Figure B the reasons for Women- Canada. **Source:** Canada Labour Force Survey (1997 - 2019). No. Observations = 2,834,136 (Canada).

Figure 2.13: Reasons NILF for Public and Private Sector Workers by Age and Gender

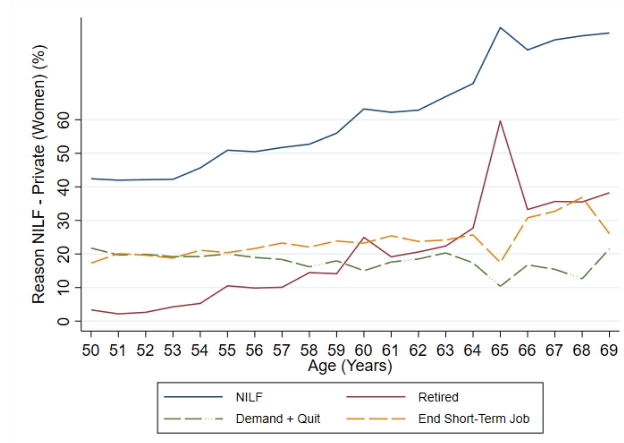


(a) Men - Public Sector

(b) Women - Public sector



(c) Men - Private Sector



(d) Women - Private sector

Notes: The figure plots the proportion of workers unemployed versus NILF by age and gender in Canada and the United States. Figure A plots the proportions for Men - Canada, Figure B the proportions for Women - Canada, Figure C plots the proportions for Men - USA, Figure D the proportions for Women - USA. **Source:** Canada Labour Force Survey and US Current Population Survey (1997 - 2019). No. Observations = 2,834,136 (Canada) and 2.723,076 (United States).

Connecting Text I

Chapters two and three explore how proximity to the pension eligibility thresholds influence the labour market decisions of older workers. In Chapter 2, I carry out a descriptive analysis of the labour market attachment of older workers in Canada and the United States. The chapter explores how older workers' preferences for part-time and non-employment changes with age and gender. Additionally, I compare the reasons provided for non-employment and working part-time to the findings from the literature.

The results from chapter 2, motivates the investigation of whether a job separation is a possible path through which older workers exit the labour market. In chapter 3, I examine how older workers react to an unexpected shock to their labour force status. This chapter studies the impact of an involuntary job separation on the probability of re-employment and labour market exit for workers aged 50 to 69 years in Canada.

Chapter 3

Older Workers and Involuntary Job Loss in Canada

3.1 Introduction

As the older labour force approaches the age for claiming retirement benefits, they face decisions about the timing of their labour market exit. In this paper, I examine the re-employment and labour market exit of older workers who experience an involuntary separation. Permanent and involuntary job separation, resulting from layoff, termination or job destruction, presents a complication for the older worker's decision-making process. First, the possibility exists that older workers will remain attached to the labour force longer to recoup the losses in earnings and retirement savings that might occur due to the job separation (Coile and Levine (2007)). This unexpected event results in an immediate loss of earnings, a period of unemployment and a decline in future earning potential for all affected workers. Meanwhile, older workers who experience a job separation face persistent lower wages (Chan and Stevens (1999)).

Second, proximity to retirement thresholds may make labour market exits more salient. The extended period of unemployment and the possibility of receiving social security benefits might increase the likelihood that older workers will exit the labour market earlier (or faster) than planned. Existing studies in the United States show that older unemployed workers leave the labour market at a significantly higher rate than their employed counterparts (Marmora and Ritter (2015)). Additionally, the "discouraged worker effect" is higher among older workers, as they experience a more extended period of unemployment than prime-aged workers. As a result, older workers are less likely to re-enter the labour market if they leave due to discouragement (Chan and Stevens (1999, 2001, 2004) and Farber (2005)). Chan and Stevens (1999, 2001, 2004) find that workers aged 55 who suffer an involuntary job separation have an employment rate 20% lower than similar workers who did not lose their jobs. Coile and Levine (2007) and Coile and Levine (2011) find that this percentage is higher for less educated workers. However, there is very little evidence from outside the United States.

This paper exploits the panel structure of the Canadian Labour Force Survey (LFS) from 1997 to 2019 to examine the short-term impact of a permanent, involuntary job separation on the re-employment and exit probabilities for workers aged $50-69$ years. I analyse the likelihood of re-employment and labour market exit by estimating Linear Probability Models (LPM) of returning to work and exiting the labour market. I use two different samples to examine the outcomes: re-employment and labour market exit.[1]

To estimate the probability of re-employment for older workers following a permanent and involuntary job separation I use a sample of all separated workers. The "separated" sample consists of all workers who experience at least one period of job separation. The analysis compares the probability of re-employment for older workers to workers who experience a separation for other reasons. The analysis is similar to the work done by Chan et al. (2011) who use the LFS to examine how the re-employment probability for individuals aged 15 to 64 evolves across recessions in Canada. Chan et al. (2011) did not distinguish between temporary and permanent job separations or between for cause, voluntary or involuntary separations. The definition of job loss that I use in my study includes only workers who

[1]Labour market exit is defined as transition from employed or unemployed to not-in-the-labour force.

experience permanent and involuntary separation. My definition provides a more accurate estimate of the likelihood of re-employment following an involuntary job loss. Workers experiencing a temporary job separation are more likely to return to employment, which would lead to an overestimation of the likelihood of re-employment. Additionally, individuals who left their jobs voluntarily might have been actively looking for a job before leaving so their inclusion would also increase the probability of re-employment. By focusing only on permanent and involuntary job separation, my analysis eliminates those individuals who left their job voluntarily or are waiting for recall to their previous employment.

Next, I use a sample of all workers to examine the effect of permanent and involuntary job separation on the labour market exit of older workers. Using the Canadian LFS data, I cannot classify individuals as being retired. Instead, I define a job market exit as individuals who move from employed or unemployed to NILF and remain in this state for the remainder of the sample.[2] I estimate the probability of leaving the labour force for workers who receive a permanent and involuntary job market separation. The analysis compares the probability of labour market exit for older workers to workers who experience no separation or a separation for other reasons. The procedure I use in this study is similar to Chan et al. (2011). They estimate the long-run effects of transitioning from work to nonwork for employed individuals who experienced a previous job loss versus those with no prior separation.[3] Finally, I examine whether the probability of re-employment and exit would differ due to individual characteristics such as gender, marital status and education.

My paper contributes to the literature on the effects of job separation on older workers. The existing literature primarily focuses on the impact of involuntary separation of workers younger than 55 years. In my paper, I focus on workers aged 50 - 69, who comprise a policy pertinent demographic. In Canada, older workers make up a significant share of the labour force. In December 2019, the Canadian labour force was 20.2 million. Prime-aged

[2] Ideally, I would like to examine the effect on the likelihood of retirement following a job separation, but the LFS does not provide information that allows me to determine whether the workers retire following a job separation.

[3] Individuals classified as nonwork are not working, but still in the labour force.

(25 to 54 years) made up 63.9% (12.9 million), and older (55 years and over) accounted for 21.8% (4.4 million) (StatisticsCanada (2022)). The older workers as a fraction of the labour force in 2019 represent a 44.4% increase in comparison to 2007, where older workers account for 15.1% of the labour force.[4] Given these numbers, understanding retirement behaviour for this particular demographic should be of primary policy importance. The decision by these workers to exit the labour market will significantly impact the composition of the labour market, government-funded pension plans, public health expenditures and individual well-being in retirement.

Additionally, as older workers age and transition into retirement, their labour market exits will exert downward pressure on the Canadian labour force participation rate. My paper examines involuntary separation as a possible path through which older workers exit the labour market. The main contribution of my paper is to estimate the probability of labour market exit for older workers following a permanent and involuntary job separation. To my knowledge, this is the first study to evaluate this probability in Canada.[5] My results show that a prior job separation increases the likelihood of labour market exit by approximately 4.7 percentage point and decreases the chance of re-employment by 5.3 percentage points.

In the remainder of the paper, Section 2 reviews the literature. Section 3 discusses the data and provides descriptive statistics for the main variables used in the study. Section 4 outlines the empirical strategy used to examine the effect of a permanent job separation on older workers in Canada. I examine the results of the empirical analysis in section 5, and section 6 concludes the paper.

[4] In December 2007 the Canadian labour force was 17.9 million and older workers accounted for 2.7 million (StatisticsCanada (2022)).

[5] Chan and Stevens (2001) examines the probability of older workers exiting employment after re-employment following a job separation.

3.2 Conceptual Framework

An involuntary job loss is defined theoretically as a separation due to external economic reasons. This type of separation is not influenced by individual behaviours or characteristics. The definition of involuntary job loss includes separations due to firm shutdowns and excludes separations due to the ending of contracts, resignations, separations caused by accidents or sickness, and just-cause dismissals. There are many definitions of involuntary job separation in the literature. The definitions differ based on the data availability and whether the job loss variable includes temporary and/or permanent separations.

An involuntary job loss results in persistent adverse effects on the economic outcomes of the workers. Workers experience a decline in earnings immediately after the separation and after re-employment. The earnings loss varies with age, gender, the seniority of the workers and the duration of the job separation (Chan and Stevens (1999, 2001); Morissette et al. (2007); Morissette and Bonikowska (2012); Schirle (2012)). For older workers in Canada, high-seniority displaced men experience long-term earnings losses between 18% and 35% compared to their female counterparts, whose earnings loss varies between 24% and 35% (Morissette et al. (2007)).[6] Though earnings recover slowly in the years following the separation, workers continue to suffer from persistent wage deficits (Chan and Stevens (1999, 2001); Morissette et al. (2007); Chen and Morissette (2010); Morissette and Bonikowska (2012)). These persistent deficits occur because workers lose valuable job-skill matches, lose wage premiums earned during employment and accept job offers and lower seniority and wage levels due to liquidity constraints (Coile and Levine (2007, 2011)).

Additionally, older workers who experience a separation have a longer duration of unemployment and may remain out of work for several years after the job loss (Chan and Stevens (1999, 2001, 2004); Farber (2005); Marmora and Ritter (2015); Merkurieva (2019)). This could be due to the difficulties of finding new employment due to the loss of firm-specific

[6] Morissette and Bonikowska (2012) classifies workers as having a high attachment if they have at least six consecutive years of positive wages and salaries before job loss.

skills, the employer's unwillingness to invest in workers near the end of their career or barriers such as age discrimination (Bélanger et al. (2016); OECD (2019b,a); StatisticsCanada (2018). Merkurieva (2019) finds that the employment rate of older workers falls by 57% in the year following the separation, and that an employment gap of 13% exists up to eight years after. Chen and Morissette (2010) find no significant increase in the average re-employment rates for men aged 50 to 54, but the re-employment rates for women increased over time.

Due to their shorter employment horizon, the reaction of older workers to an involuntary job loss differs in comparison to younger cohorts. The longer duration of unemployment may encourage workers to leave the labour force sooner than planned and access retirement benefits at an earlier age. Merkurieva (2019) finds that workers retire 14 months earlier on average than if they had not lost their jobs. Other workers may work longer to replenish retirement savings lost due to unemployment (Coile and Levine (2007)).

Given the empirical evidence on the longer duration of unemployment and the shorter employment horizon of these workers, this paper determines the probability of re-employment following a separation as well as whether an involuntary job separation is a possible path through which older workers exit the labour market in Canada. Previous work on job separation in Canada focuses mainly on the impact on wage losses after re-employment.

3.3 Data

3.3.1 Labour Force Survey

The primary data sets used to examine involuntary job separation in Canada are the Longitudinal Worker File (LWF) and the Labour Force Survey (LFS). Morissette et al. (2007), Chen and Morissette (2010), and Morissette and Bonikowska (2012) use the administrative data set from the Longitudinal Worker File (LWF) to estimate the earnings loss. The main drawback of the LWF is that the dataset does not provide information on the worker's labour force status, education, and occupation. It is also not stratified by province and industry. As a result, The LWF does not allow for analysis of the impact of these individual characteristics. Additionally, the LWF is only available up to 2016.

Similar to Chan et al. (2011), this study uses monthly microdata for twenty-three (23) years (1997 – 2019) from the master file of the Canadian Labour Force Survey (LFS). The LFS is a monthly household survey administered by Statistics Canada and is representative of the civilian, non-institutionalized population in Canada. It is the primary source of information for measuring employment and unemployment statistics in Canada. Statistics Canada conducts the LFS nationwide in provinces and territories. Each month, the survey collects labour force information from 56,000 households for all household members aged 15 and over. In addition, it also contains demographic information such as gender, age, and education. The LFS relies on a rotating panel design. The LFS interviews each household for six consecutive months and replaces one-sixth of the sample monthly with a new cohort of respondents (Statistics Canada (2017)).

I use the restricted access panel microdata of the LFS to measure the labour force transitions of older workers. I restrict the sample to individuals aged 50 – 69 from 1997 to 2019. I construct a unique identifier by concatenating dwelling, household, and individual-within-household codes to match individuals across periods. I create a series of 6-month panels using data for every month an individual is present in the LFS. I use the 6-month panels to create the two samples used in this paper.

To create the samples, I follow the sample selection procedure from Chan et al. (2011). The primary sample includes individuals who are:

1. present all six consecutive months in each panel.
2. age 50 to 69 in all six months.
3. employed as paid workers during the 1st month of the panel.[7]

The primary sample comprises 472,356 individuals between 50 and 69 years. In month 1, the

[7] This eliminates self-employed workers and includes individuals employed in the public and private sector.

average age is 56 years, with 78% of the sample younger than 60 years, 17% between ages 60 and 64 and 5% 65 years and older (Table 3.4). I use the primary sample to estimate the probability of labour market exit of older workers who experience an involuntary separation.

The "separated" sample includes individuals age 50 to 69 years, employed as a paid worker during the 1st month of the panel who experience a job separation at any point between the second and the sixth month.[8] The "separated" sample comprises 23,770 individuals (Table 3.5). I use the "separated" sample to estimate the probability of re-employment for older workers who experience an involuntary job loss. All descriptive and statistical analysis in this paper employs sampling weights from Statistics Canada.

3.3.2 Definition of Job Loss

In this paper I focus on the effect of permanent job separations. This includes involuntary, permanent breaks (do not return to the same employee) and not a termination due to "cause". Papers using the LWF define workers as displaced if they experience a permanent layoff and do not return to work with the same employer during the year of the dismissal or the following year (Morissette et al. (2007), Chen and Morissette (2010), Morissette and Bonikowska (2012)). Since the LWF contains a link between the Record of Employment (ROE) file and firm-level data, the data set allows the distinction between quits, layoffs and other types of separations. Morissette et al. (2007) focus on individuals displaced due to firm closures and mass layoffs, while Morissette and Bonikowska (2012) extend the definition to include all permanent layoffs that were not quits or dismissals due to cause.

Neill and Schirle (2009) and Schirle (2012), using the Survey of Labour Income Dynamics for 1993 – 2005, describe an individual as displaced if laid off due to a business slowdown not caused by seasonal conditions or his job ends because the company moves or goes out of business. Chan and Stevens (1999, 2001, 2004), using data from three waves (1992, 1994 and 1996) of the US Health and Retirement Survey, define an involuntary job separation as

[8] This includes individuals with multiple separations. Approximately 6% of individuals separated between month two and five had multiple separations.

individuals self-reported as “layoff” or “business closed” and treated as exogenous.

My primary independent variable differentiates between temporary and permanent job separations and captures individuals who experience a permanent job loss that is not voluntary or due to termination for cause. I define Job Loss = 1 using the individual's labour force status and responses to the question, *What was the main reason you stopped working at that job?*[9] I define an individual as experiencing a permanent job separation (or job loss) if employed in period *t - 1*, not expecting a recall from their previous employer, and responding that their reason for leaving was involuntary (business shut down or changed location) or due to economic conditions (Table 3.3). My definition differs from that of Chan et al. (2011). They determine an involuntary job separation based on responding “laid off” which includes temporary and permanent layoffs.

In this study, economic conditions are the primary reason for a job separation and account for 94% of individuals who experience at least one job separation between months 2 and 6. Figure 3.1 depicts the job loss rate by age and gender. I calculate the job loss rate as the number of individuals who experience an involuntary separation at each age divided by the number of persons employed. The figure shows that the job loss rate increases with the age of the individual up to age 64, after which it decreases.[10] The average job loss rate is approximately 1%. In contrast, Chan et al. (2011) assess that the average risk of a layoff between 2008 and 2011 was 2.0%. Furthermore, the job loss rate is substantially higher across the age range for males than females. A higher proportion of men employed in more labour-intensive occupations and industries predominantly affected by involuntary separations could account for this disparity.

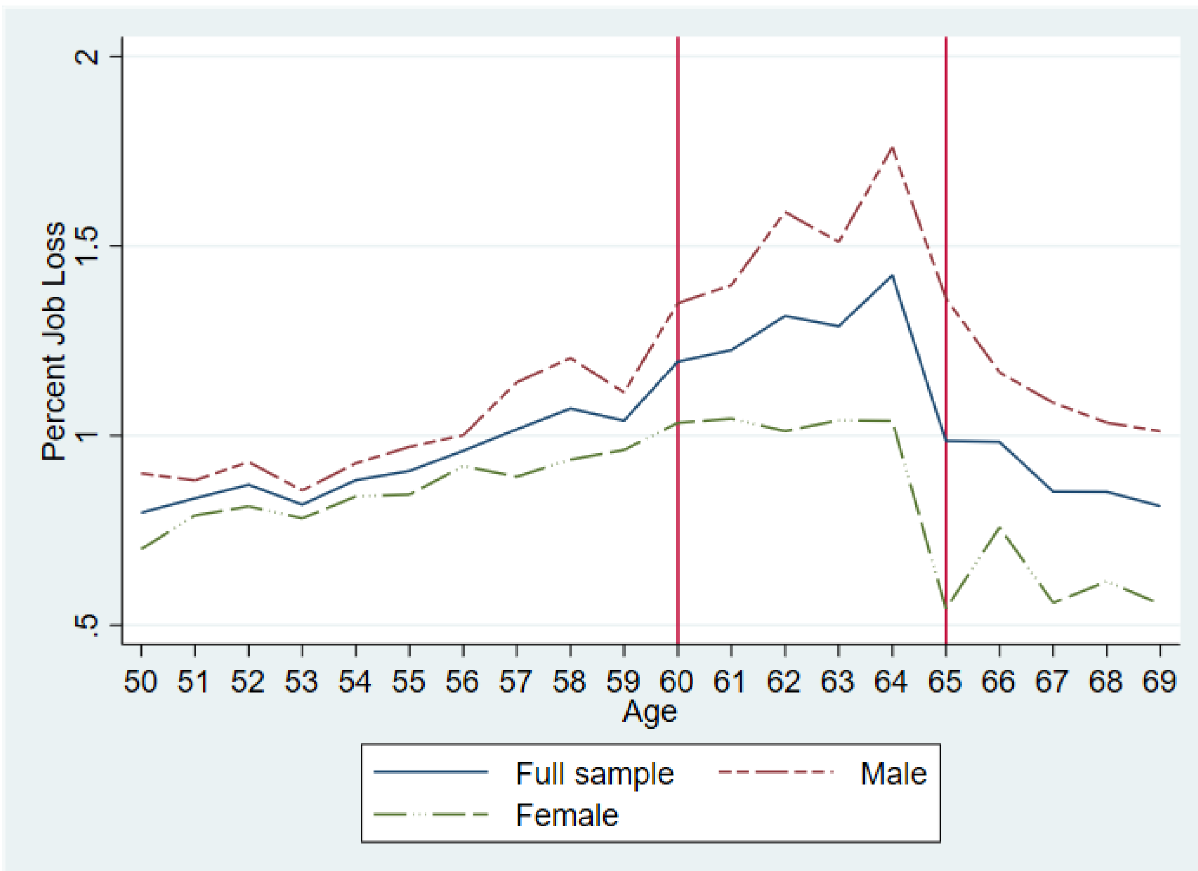
The prior job loss variable is the primary independent variable in the regression analysis.

[9]Based on the survey question, the LFS classified respondents into eighteen categories. These include: other; own illness or disability; caring for own children; caring for elder relative (60+); pregnancy; other personal or family responsibilities; going to school; end of seasonal job; end of temporary term or contract; causal job; company moved; company went out of business; business condition; dismissal by employee; other reasons; changed residence; dissatisfied with job and retired for a total of 118,850 observations.

[10]The reduction of the sample size for older workers in the age group 65 - 69 is a possible reason for the marked decrease in the job loss rate. Only 4.9% of the sample are in the age group 65 - 69, and this age group makes up 10% of workers who experience a job loss.

I define Prior Job Loss = 1 if the individual experiences an involuntary separation at any time within the past six months. Numerous studies in the United States find that older unemployed workers are more likely to retire or leave the labour market compared to their employed counterparts (Marmora and Ritter (2015); Farber (2005) and Chan and Stevens (1999, 2001, 2004)). Since an involuntary separation is unexpected or unplanned, I investigate if experiencing an involuntary separation at some point in the recent past increases the probability of older workers leaving the market.

Figure 3.1: Job Loss Rate by Age and Gender



Notes: The figure represents the proportion of individuals who experience an involuntary job separation conditional on being employed in month 1. Based on author's calculations ($JobLossRate = no.Jobloss/no.Employed$). The average Job Loss rate is 1%. Vertical lines at ages 60 and 65 represent the early and normal age for benefit claiming in Canada, respectively. **Source:** Labour Force Survey (1997 - 2019).

3.3.3 Dependent Variables

The two main dependent variables are re-employment and labour market exit. I define return = 1 if an unemployed worker is re-employed following a job separation. These are individuals whose job market status change from unemployed to employed. Table 3.6 (column 2) shows that the re-employed workers tend to be male, between 50 and 59 years and have a high school certificate or higher. I calculate the re-employment rate as the number of persons, in each age, re-employed in period t divided by the number of persons unemployed in period *t-1*. Figure 3.2 shows that the re-employment rate for unemployed workers who experience a job separation declines with age. There is a marked decrease in the re-employment rate at ages 53, 60, 65 and 67. The most dramatic drop occurs between ages 62 and 65 years. The trend in the re-employment rate was similar for males and females, but based on the number of individuals in the labour force, female workers have a higher re-employment rate than males for the entire age range.

In this paper, I determine if an involuntary job loss impacts the chance of a labour market exit. I define exit = 1 if the worker transitions from a labour force status (unemployed or employed) to NILF during months 2 to 5 and remain in this state for the remainder of the sample.[11] There are two main transitions: (1) unemployed-to-NILF and (2) employed-to-NILF. When jobloss = 0, the exit proportion is approximately 1.4%.[12]

Table 3.6 (column 1) shows that the workers who exit tend to be female between 55 and 69 years and have a high school certificate or higher. I calculate the exit rate as the number of persons, in each age, who exit divided by the number of persons in the labour force. Figure 3.3 illustrates a spike in the exit rate at ages 55, 60 and 65. The largest spike occurs at age 65 when the exit rate dramatically increases from 2.5% to 8%. These spikes support the view that reaching the age thresholds for benefit claiming impacts the decision

[11] The focus is on months 2 to 5 as in the sample all individuals are employed in month 1, and I cannot classify a move to NILF in month 6 as permanent as I do not have information on the individuals beyond month 6.

[12] This percentage will serve as the baseline line that I use for comparison in the empirical analysis.

of older workers to leave the labour market. The exit rate for men and women is comparable across the age range.

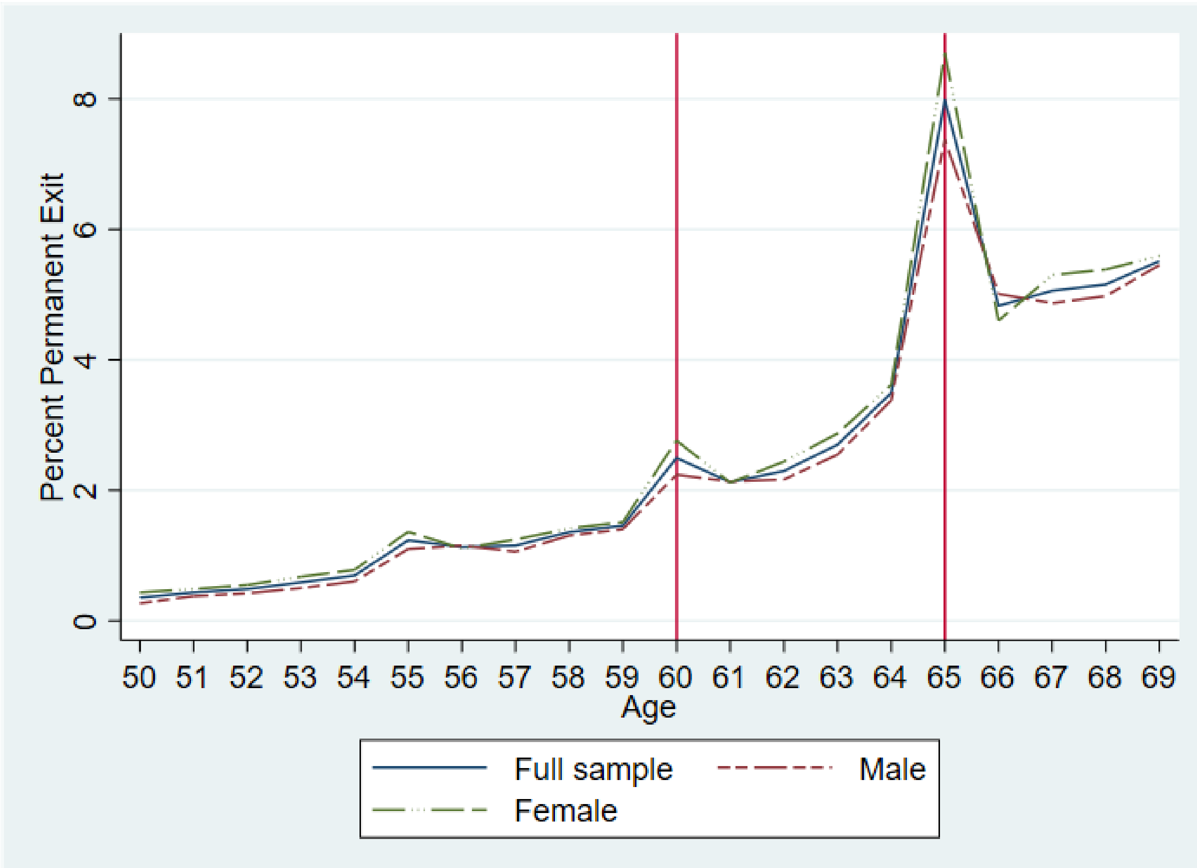
Figure 3.2: Re-employment Rate by Age and Gender

Notes: The figure shows the re-employment rate for workers who experience a job separation by age and gender. Based on author's calculations ($ReturnRate = No.return_t/No.Unemployed_{t-1}$). All respondents were aged between 50 and 69. Vertical lines at ages 60 and 65 represent the early and normal age for benefit claiming in Canada, respectively. **Source:** Labour Force Survey (1997 - 2019).

3.3.4 Control Variables

The probability of losing or finding a job is partly related to a worker's socioeconomic characteristics (Bernard and Galarneau (2010); Morissette et al. (2007)). Therefore, I include age, gender, level of schooling and whether living with a partner to investigate if the labour market transitions differ due to observed individual characteristics. I also include information on the individual's province, industry, and occupation. Education is an essential factor

Figure 3.3: Exit Rate by Age and Gender



Notes: This figure shows the labour market exit rate by age and gender for the entire sample of workers. Based on author's calculations ($ExitRate = No.Exit/No.LabourForce$). Vertical lines at ages 60 and 65 represent the early and normal age for benefit claiming in Canada, respectively. **Source:** Labour Force Survey (1997 - 2019).

associated with the type and quality of the individual's job. It is also associated with wealth and income, which plays a critical role in the decision of older workers about whether to exit the labour market. I capture the individual's level of schooling as (1) attaining less than a high school diploma, (2) attaining a high school diploma or trade certificate and (3) attaining post-secondary education.

I use two different age specifications in this study, a dummy variable for each age (50 to 69 years) and four age groups (50 - 54, 55 - 59, 60 - 64 and 65 - 69 years). Table 3.6 provides descriptive information on the average socioeconomic characteristics of the two dependent variables over 5-months. The summary statistics support the literature that (1) older workers are more likely to exit the labour market, while (2) younger workers are more likely to return to employment.

Additionally, the summary statistics show that the proportion of older workers that exit the labour market increases with age. In comparison, the proportion of older workers returning to employment declines with age. For the workers that exit, Table 3.6 (column 2) supports the literature that on average males remain connected to the labour market longer than females, and that reaching the early and full retirement age for claiming retirement benefits increases the likelihood of exit.

3.4 Empirical Strategy

This paper evaluates the impact of involuntary job loss on the labour market transitions of older workers in Canada (aged 50 – 69 years). The primary outcome variables are returning to employment and labour market exit (moving to NILF). I use two samples in this analysis: (1) the sample of all workers employed in a paid job in the first month of the panel, and (2) the sample of all individuals who experience at least one job separation over the 6-month panel.

Following Chan et al. (2011), I use Linear Probability Models to estimate the likelihood of re-employment. I also extend their analysis by assessing the probability of exiting the labour market following an involuntary job separation. I test two hypotheses: (1) The likelihood of re-employment decreases with prior job loss and the individual's age. (2) The likelihood of exiting the labour market increases with prior job loss and the individual's age. To test hypothesis one, I use the "separated" sample to compare the likelihood of re-employment for individuals who experience an involuntary separation against those who were separated for other reasons. To test hypothesis two, I use the primary sample to compare the likelihood of labour market exit for older workers against those who experience no separation or are separated for other reasons. The panel format of the LFS dataset only allows the examination of each worker over six months, so the results give the short-run effects of an involuntary job separation.

3.4.1 Identification

The threats to identification in this analysis are the possibility of reverse causality between the dependent variables and leading independent variable (prior job loss), self-selection into retirement, omitted variables and multicollinearity between the model covariates. Multicollinearity occurs when the correlation is high between two or more independent variables. The presence of multicollinearity could result in large standard errors and make it difficult to determine the effect of the independent variables on the dependent variables. Additionally, omitted variable bias might occur due to differences in unobserved individual characteristics that influence the possibility of re-employment and labour market exit and job separation. I estimate an earnings equation using a linear combination of the control variables (age, gender, living with a partner and level of education). The earnings equation estimates the impact of the individual characteristics of workers on their weekly wage. I then use the predicted value of the weekly log earnings as a covariate to estimate the likelihood of re-employment and labour market exit. This estimation examines if weekly earnings is a predictor of the likelihood of re-employment and exit.

Reverse causality occurs if the dependent variables: the likelihood of re-employment and exit, impacts whether a worker experiences an involuntary separation. An increase in the probability of re-employment could influence workers on casual or temporary contracts to opt not to renew their contracts with the same employers. Additionally, more productive workers could leave a poor-performing firm and start looking for another job before the firm shuts down. The definition of job loss that I use in this study focuses on individuals who are separated due to economic conditions. Though the literature commonly views involuntary job separation as exogenous, I carry out sensitivity analyses to account for this. I re-estimate the LPM, using only workers who experience a prior job loss due to a business shutting down or a location change.

3.4.2 Model

To determine the re-employment probability, I focus on the sample of workers who experience a job separation in a given month and estimate a linear probability model for "return" conditional on age, prior job loss status and other control variables.

$$\begin{aligned} return_{it} = \beta_0 + \beta_1 age_{it} + \beta_2 prior_{it} + \beta_3 prior * age_{it} \\ + \beta_4 X_{it} + \beta_5 year_{it} + \mu_{it} \end{aligned} \tag{3.1}$$

where X_{it} represents individual characteristics. *Prior* indicates whether the respondent experiences an involuntary job loss anytime in the previous months. *Year* represents time dummies for each year (1997 - 2019) in the sample. The model also includes interaction terms with age and prior job loss.

This equation gives the likelihood that an individual i returns to employment in month t, conditional on age and whether the individual lost a job in the past. Like Chan and Stevens (2001), this captures individuals unemployed for reasons other than an involuntary job loss. The coefficient on the prior job loss variable indicates the probability of returning to work for workers experiencing a separation due to involuntary job loss relative to those who lost jobs for other reasons (Unemployed to Employed transition).

I determine the probability of permanently exiting the labour market conditional on not working (Unemployed to NILF transition) and compare it to the likelihood of workers who did not experience a separation moving to NILF (Employed to NILF transition). In Equation 3.2, I estimate a second linear probability model.

$$\begin{aligned} exit_{it} = \alpha_0 + \alpha_1 age_{it} + \alpha_2 prior_{it} + \alpha_3 prior * age_{it} \\ + \alpha_4 X_{it} + \alpha_5 year_{it} + \xi_{it} \end{aligned} \tag{3.2}$$

The "exit" equation estimates the probability of labour market exit conditional on age, prior job loss status and other control variables. To determine the exit probability, I use the entire sample of workers aged 50 - 69 years. This sample allows me to determine if the likelihood of exit is different for those who experience a prior job loss versus those who experience no separation or lost a job for other reasons. The coefficient on the prior job loss variable indicates the probability of exit for workers experiencing a separation due to involuntary job loss. I expect unemployed workers or those NILF in an earlier period to have a higher chance of exiting the labour market than those employed.

To determine if weekly earnings is a predictor of the likelihood of re-employment and exit, I use Ordinary Least Squares (OLS) to estimate the linear relationship between the covariates and the weekly log earnings of the workers. I then use the predicted value of log earnings as a covariate in the LPM. Equation 3.3 and Equation 3.4 provide the model specification.

$$\begin{aligned} lnearnings_{it} &= \gamma_0 + \gamma_1 gender_{it} + \gamma_2 lvlschool_{it} + \gamma_3 marital_{it} \\ &+ \gamma_4 occupation_{it} + \gamma_5 industry_{it} + \gamma_6 regionit + \gamma_7 year_{it} + \epsilon_{it} \end{aligned} \tag{3.3}$$

$$\begin{aligned} Y_{it} &= \psi_0 + \psi_1 age_{it} + \psi_2 prior_{it} + \psi_3 prior * age_{it} \\ &+ \psi_4 \widehat{lnearnings}_{it} + \psi_5 year_{it} + \zeta_{it} \end{aligned} \tag{3.4}$$

where Y_{it} represents the outcome variables: re-employment and labour market exit.

$\widehat{lnearnings}_{it}$ represents the predicted values of log weekly earnings from Equation 3.3.

I use two different age specifications in the estimation models for return and exit. A dummy variable for each age from 50 to 69 years and four age groups: 50 - 54, 55 - 59, 60 - 64 and 65 - 69 years.[13] Both age specifications capture the respondents' return and exit age effect and the incentive effects of reaching the early and normal age for benefit claiming in Canada (60 and 64 years, respectively). The individual characteristics I use in the regressions are gender, marital status, level of education, occupation, industry, and region of residence.

[13]For convenience, the results will be reported using the age groups, but I obtain similar results from the estimation using age dummies.

3.5 Empirical Results

3.5.1 Return to Employment

The return to employment estimation uses the sample of all workers who experience a job separation. Table 3.1, (column 1) reports the likelihood of a return to employment for older workers who experience a job separation in previous periods. The coefficient for prior job loss indicates that workers who experience an involuntary job loss are 5.3 percentage points less likely to return to employment than workers who did not experience an involuntary job loss (Table 3.1 (column 1)). This finding is in contrast to Chan and Stevens (2001), who find that experiencing a prior job loss increases the hazard of returning to employment by 0.24 for men and 0.31 for women in comparison to those who did not experience an involuntary job loss. Chan and Stevens (2001) explain the increase in the hazard of returning to employment as attributable to non-displaced workers in the Health and Retirement Survey retiring voluntarily, so reduces the likelihood of non-displaced workers returning to work.

A decrease in the re-employment probabilities following an involuntary job loss can have a major impact on older workers. The longer duration of unemployment paired with the financial incentive of retirement benefit claiming can lead to discouragement and labour market exit. This impacts the labour force participation rate for the Canadian economy, so it is of vital interest to policymakers. The effect of a prior job loss on the likelihood of re-employment varies with age, as seen in the interaction between experiencing a prior job loss and the age groups. The interaction terms are negative and statistically significant only for the age group 65 - 69. The coefficient on the interaction term indicates that the age group 65 - 69 significantly alters the effect of prior job loss on the likelihood of returning to employment in the short term.

Table 3.1 (columns 2 and 3) show that the probability of re-employment is higher for workers who return to part-time versus full-time employment. The coefficients on the age groups in the return full-time regression Table 3.1 (column 2) are comparable to those from (column 1) and indicate that older workers in age groups 60 - 64 and 65 - 69 are less likely to return to full-time employment. Separated workers are more likely to return to part-time employment in age groups 55 - 59, 60 - 64 and 65 - 69 years relative to the younger 50 - 54 cohorts. This finding could be due to older workers having to work longer to replace the retirement savings they lost while unemployed.

Surprisingly, older male workers are 1.1 percentage points less likely to return to employment after a job separation than female workers. However, male workers that return to employment are more likely to return to full-time work. This finding contrasts with the existing literature. Chan and Stevens (2001) find that men aged 61 and 62 were more likely to return to work quicker than women. Women who experience an involuntary job separation would leave the labour market to become full-time caregivers for older parents or grandchildren.

The finding that men are less likely to return to employment could be due to men being employed in manufacturing and primary and construction industries which are very unionized in Canada, so it would be more difficult for them to be re-employed in the same industry following a separation. In comparison, women are employed in the services industries, which have a higher probability of re-employment. Also, in Canada, the labour force participation rate of men is declining overtime in comparison to women. This decline in the labour force participation of men could also account for men being less likely to return to employment following an involuntary separation.

The results for the age groups and age dummy specifications align with the literature and my expectations for the study. Older workers are less likely to be re-employed following an involuntary separation. The coefficients for the age groups indicate that older workers are less likely to return to employment following a separation than their younger counterparts. The effect of age on the re-employment probabilities in this study are similar to those of Chan et al. (2011) and Chan and Stevens (2001). Chan and Stevens (2001) find that the probability of returning to work declines after age 55 and decreases significantly after age

Table 3.1: Re-employment Regressions

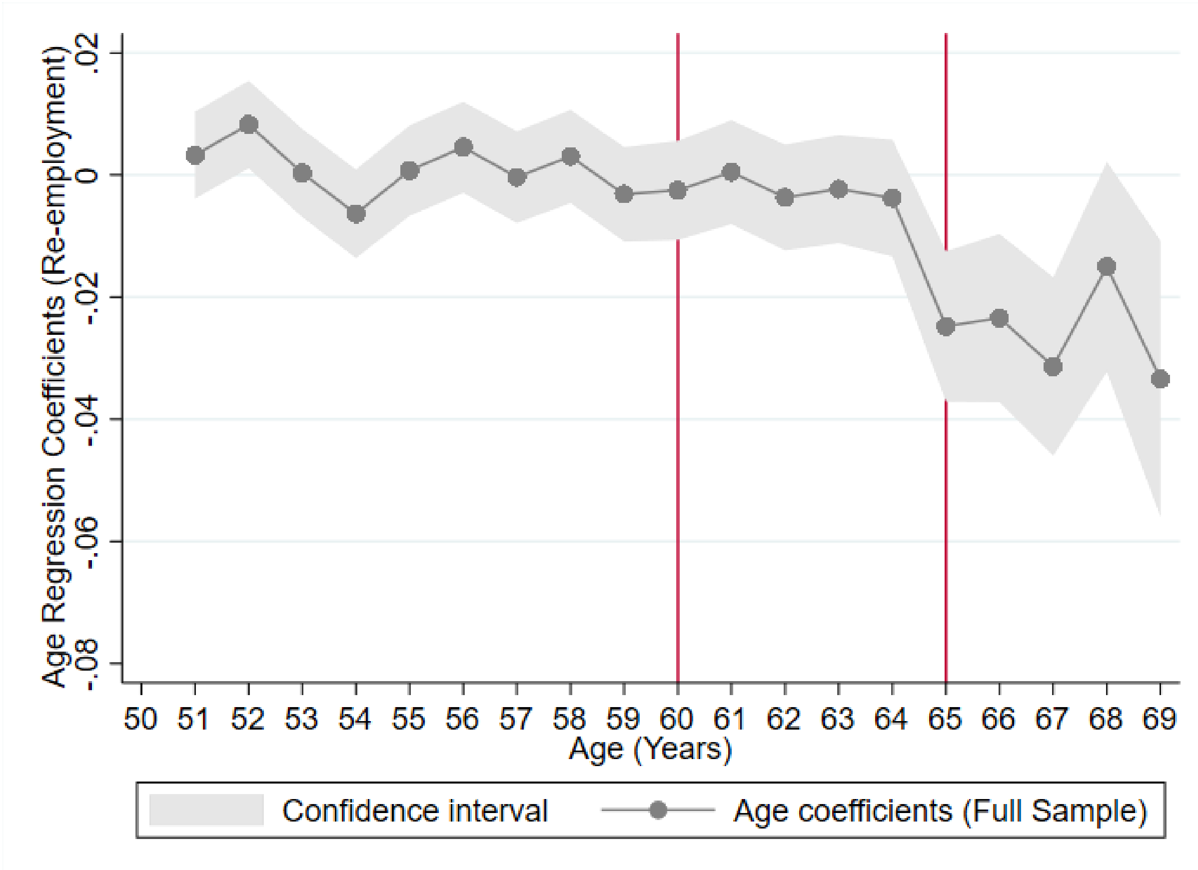
Variables	Return (1)	Return Full-time (2)	Return Part-time (3)	Return (Log Earnings) (4)
Prior Job Loss	-0.0527***	-0.0390***	-0.0137***	-0.0519***
	(0.0052)	(0.0045)	(0.0029)	(0.0053)
Age Group (reference: 50 - 54)				
55 - 59	0.0001	-0.0025	0.0026**	-0.0001
	(0.0017)	(0.0016)	(0.0010)	(0.0017)
60 - 64	-0.0026	-0.0135***	0.0108***	-0.0041**
	(0.0020)	(0.0018)	(0.0013)	(0.0020)
65 - 69	-0.0248***	-0.0292***	0.0044**	-0.0299***
	(0.0035)	(0.0029)	(0.0022)	(0.0034)
Prior*55-59	-0.0070	-0.0076	0.0006	-0.0083
	(0.0074)	(0.0064)	(0.0041)	(0.0075)
Prior*60-64	-0.0132	-0.0021	-0.0011**	-0.0152*
	(0.0083)	(0.0071)	(0.0046)	(0.0084)
Prior*65-69	-0.0380***	-0.0256**	-0.0124	-0.0410***
	(0.0141)	(0.0115)	(0.0076)	(0.0142)
Gender (Male = 1)	-0.0112***	0.004**	-0.0153***	
	(0.0019)	(0.0016)	(0.0011)	
$\widehat{lnearnings}$				-0.0092***
				(0.0021)
Constant	-0.0504***	-0.0257***	-0.0248***	0.0567***
	(0.0069)	(0.0060)	(0.0038)	(0.0136)
Controls	YES	YES	YES	NO
Year Dummies	YES	YES	YES	YES
Observations	118,406	118,406	118,406	118,406
R-squared	0.336	0.236	0.0998	0.355

Source: Labour Force Survey (1997 - 2019). All respondents aged between 50 and 69 years.
Significance: ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$. Robust standard errors are in parentheses.
Controls include living with a partner, level of schooling, occupation, industry, and region of residence.

60. Chan et al. (2011) find that the re-employment probabilities are low at ages 62 and 65, which correspond to the early and normal age for benefit claiming in the United States.

An analysis of the age dummies specification for the model in Figure 3.4 shows that overall, the likelihood of re-employment trends downwards over the age range 50 to 69.

Figure 3.4: Re-employment LPM Regression Age Dummies



Notes: Data from the Labour Force Survey (1997 - 2019). Vertical lines at ages 60 and 65 represent the early and normal age for benefit claiming in Canada, respectively. See Table 3.8 (column 1) for regression table.

The probability of re-employment decreases in the year before reaching ages 60 and 65, the age of early and normal pension benefit claiming in Canada. The most considerable reduction in the probability of re-employment occurs between ages 64 and 65. This suggests that reaching the age for normal retirement benefit claiming has a negative effect on the likelihood of returning to work. The low rates of re-employment at the benefit eligibility threshold suggest that workers decide between remaining attached to the labour market and retirement. Additionally, Figure 3.5 shows that men have a higher likelihood of re-employment over the age range of 52 - 69.

The covariates from the re-employment estimation living with a partner, level of schooling, occupation and industry all positively impact on the likelihood of re-employment (Table 3.7). For the region of residence, the likelihood of re-employment increases for British Columbia, Alberta, and the Prairie Provinces but decreases for Quebec and the Atlantic Provinces relative to the reference group (Ontario). The differences in the likelihood of re-employment could be due to the population size and the labour market density in these regions. The return full-time and part-time estimations provide comparable results for the covariates.

To determine if weekly earnings is a predictor of the likelihood of re-employment, I also estimate a linear probability model for re-employment using a two-step procedure. I first assess the effect of the covariates on the log earnings of the workers, and then I use the predicted log earnings as a control variable in the re-employment regression.[14] Table 3.1 (column 4) shows that the probability of re-employment is comparable to the estimation using the individual covariates. In fact, there is a slight increase in the probability of re-employment. Overall, Table 3.1 supports the hypothesis that experiencing an involuntary job loss reduces the possibility of returning to employment. These results vary across gender, marital status, level of schooling, industry, occupation, and region.

3.5.2 Labour Market Exit

The labour market exit estimation uses the sample of all workers employed and unemployed, Table 3.2, (column 1) shows results from the estimation of the probability of leaving the labour market (exit) for older workers. The coefficient for prior job loss shows that workers who experience a permanent, involuntary separation are approximately 4.7 percentage points more likely to exit the labour market than workers who experience no involuntary job loss. This result is in line with the finding by Merkurieva (2019) that workers who experience an involuntary job loss retire on average 14 months earlier.

An increase in the likelihood of labour market exit negatively impacts the labour market

[14] See Table 3.9 (column 1) for the Earnings regression table.

participation rate for older workers. Workers in all age groups are more likely to exit the labour market than those 50 - 54 years. The likelihood of exit was highest for age groups 60 - 64 and 65 - 69, with a percentage point increase of 2.3 and 7.2, respectively. This finding is in line with my a priori expectations. As workers get closer to or exceed the thresholds for retirement benefit claiming, considerations about retirement timing positively impact the probability that older workers will exit the labour market.

An examination of the age dummies in Figure 3.6 shows that the likelihood of labour market exit increases with age. There is a large increase in the likelihood of exit before the early and normal retirement age for benefit claiming. Reaching the early and normal age for retirement benefit claiming increases the probability of exit by 2.6 and 8.5 percentage points, respectively. The age dummies also show a reduction in the likelihood of exit following the benefit claiming ages. This reduction suggests that if workers did not decide to retire at the benefit claiming thresholds, they would be less likely to exit the labour market in the preceding year.

The interaction term between experiencing a prior job loss and age group is positive and statistically significant. The coefficient on the interaction equation indicates that the age group does alter the effect of prior job loss on the likelihood of exiting the labour market in the short term. The controls for the worker's demographic characteristics have the expected results. Gender, level of schooling and industry have a negative effect on the likelihood of labour market exit, while living with a partner, occupation, and region have a positive impact. An examination of the age dummies by gender (Figure 3.7) shows that the age effect on the exit probabilities differs based on gender. Men are less likely to exit the labour market at all ages than women. There is a substantial difference in the probability of exit for females between the ages of 64 and 65. This difference could be that women are more likely to be full-time caregivers for grandchildren or aging parents, so they have a higher probability of labour market exit. Additionally, women often have partners a few years older, so they tend to retire at the same time.

Estimating the probability of labour market exit using predicted log earnings (Table 3.2 (column 2)) gives comparable results. Table 3.2 (column 2) provides the estimates for this analysis.[15] The estimation with predicted log earnings slightly increases the effect of a prior job loss on the probability of exit. This change also decreases the impact of age groups on labour market exit. Overall, the findings in Table 3.2 support the possibility that experiencing an involuntary job separation is a potential avenue for labour market exit for older workers.

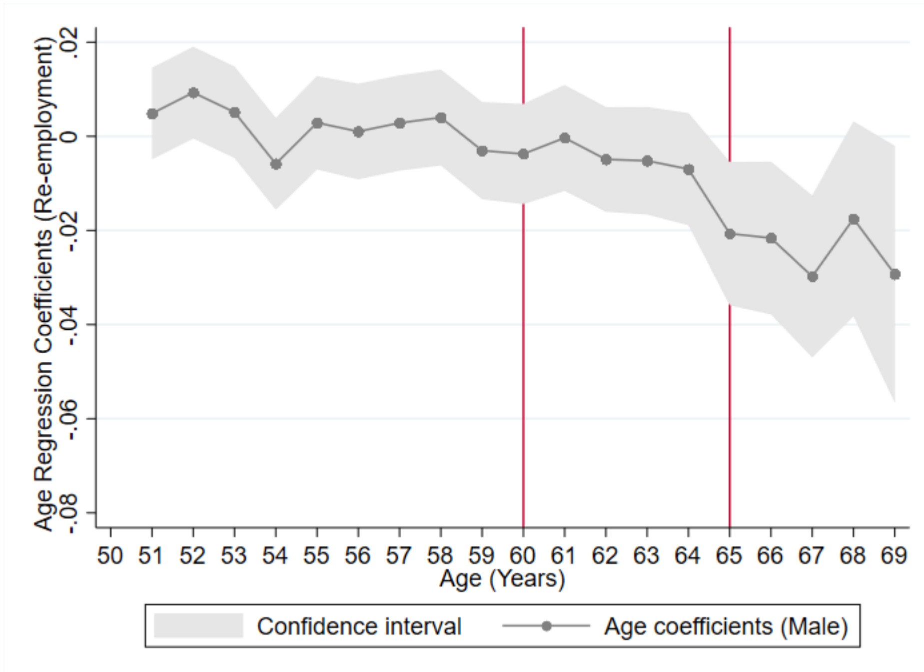
[15] See Table 3.9 (column 2) for the Earnings regression table.

Table 3.2: Exit Regressions

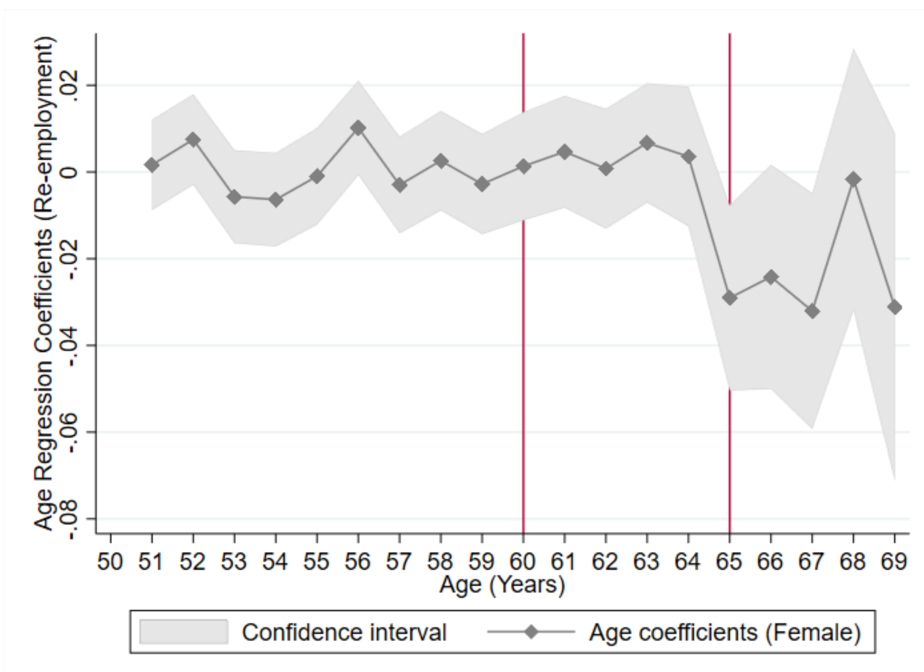
Variables	Exit (1)	Exit (Log Earnings) (2)
Prior Job Loss	0.0472***	0.0475***
	(0.0023)	(0.0023)
Age Group (reference: 50 - 54)		
55 - 59	0.0081***	0.0081***
	(0.0003)	(0.0003)
60 - 64	0.0227***	0.0225***
	(0.0006)	(0.0006)
65 - 69	0.0717***	0.0710***
	(0.0017)	(0.0017)
Prior*55-59	0.0132***	0.0132***
	(0.0034)	(0.0034)
Prior*60-64	0.0361***	0.0364***
	(0.0040)	(0.0040)
Prior*65-69	0.0654***	0.0653***
	(0.0063)	(0.0063)
Gender (Male = 1)	-0.0043***	
	(0.0004)	
$\widehat{lnearnings}$		-0.0011***
		(0.0004)
Constant	0.0080***	0.0087***
	(0.0014)	(0.0028)
Controls	YES	NO
Year Dummies	YES	YES
Observations	2,361,780	2,361,780
R-squared	0.0271	0.0250

Source: Labour Force Survey (1997 - 2019). All respondents aged between 50 and 69 years.
Significance: $***p < 0.01$, $**p < 0.05$, $*p < 0.1$. Robust standard errors are in parentheses.
Controls include living with a partner, level of schooling, occupation, industry, and region of residence.

Figure 3.5: Re-employment LPM Regression Age Dummies by Gender



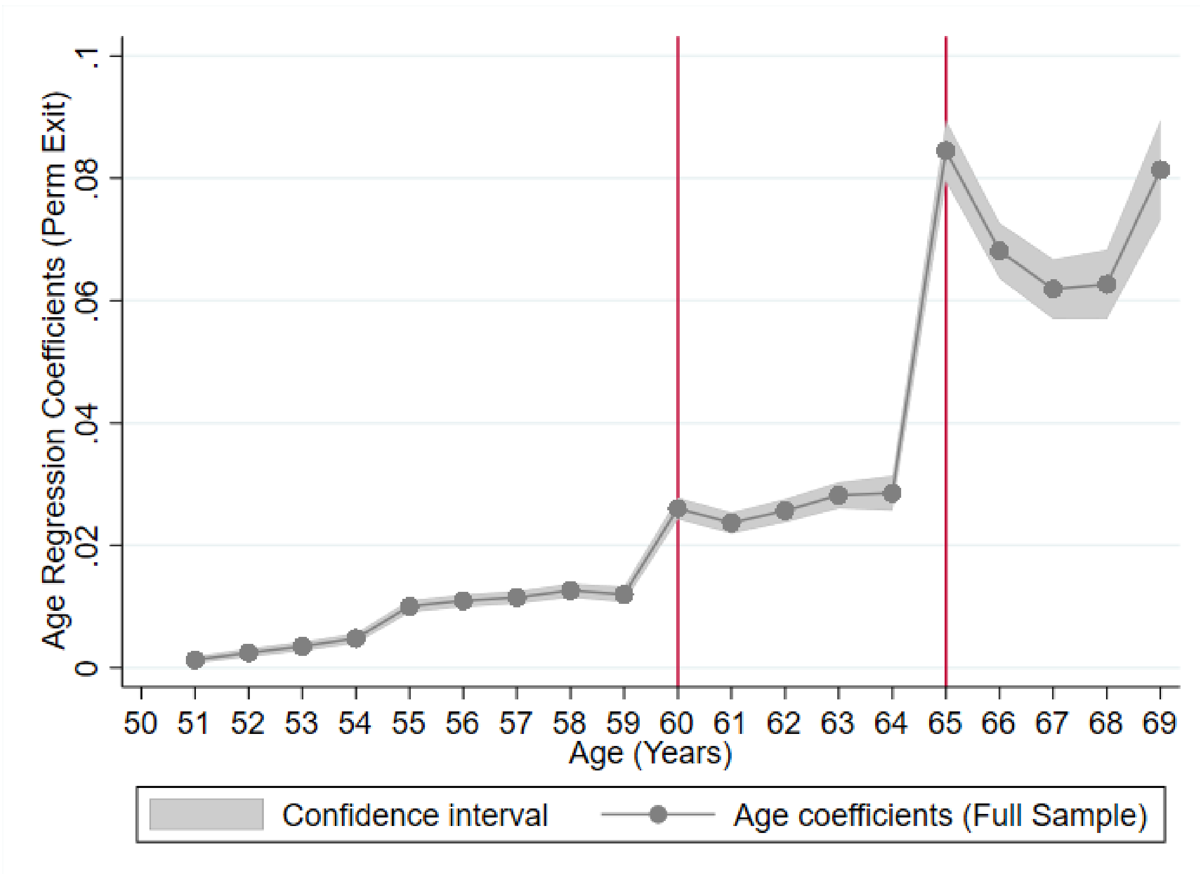
(a) Male



(b) Female

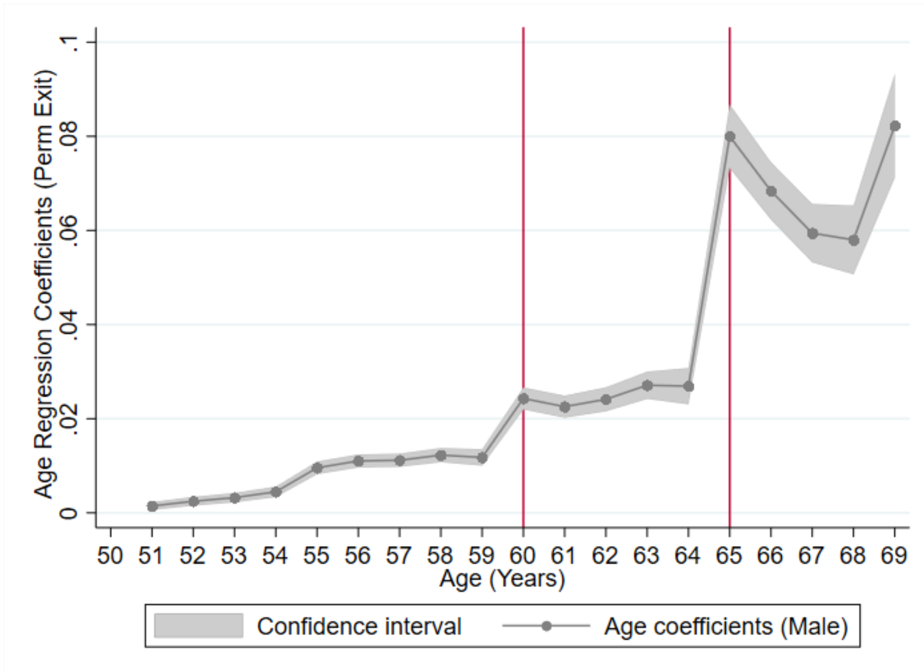
Notes: Data from the Labour Force Survey (1997 - 2019). Vertical lines at ages 60 and 65 represent the early and normal age for benefit claiming in Canada, respectively.

Figure 3.6: Exit LPM Regression Age Dummies



Notes: Data from the Labour Force Survey (1997 - 2019). Vertical lines at ages 60 and 65 represent the early and normal age for benefit claiming in Canada, respectively. See Table 3.8 (column 2) for regression table.

Figure 3.7: Exit LPM Regression Age Dummies by Gender



(a) Male

(b) Female

Notes: Data from the Labour Force Survey (1997 - 2019). Vertical lines at ages 60 and 65 represent the early and normal age for benefit claiming in Canada, respectively.

3.6 Conclusion

This paper explores the association between experiencing an involuntary job separation and the likelihood of re-employment and labour market exit. The results show a significant association between the respondent's age, whether they experienced a prior job loss and the possibility of a return to employment and labour market exit. Experiencing an involuntary job separation in the last six months decreases the probability of re-employment by 5.3 percentage points. It also increases the likelihood of labour market exit by 4.7 percentage points compared to workers who do not experience an involuntary job loss.[16] These findings suggest that older workers, being close to retirement thresholds, have a shorter expected remaining working life and respond to an involuntary separation by increasing their retirement rate and decreasing their labour force participation.

Given that older workers make up a significant portion of the Canadian labour force, policies geared at prolonging their attachment to the labour force are critical. Possible policy measures can involve retraining older workers to learn new skills and adapting firm-specific skills gained while employed to new industries or occupations. Over the past decades, the Canadian government has implemented several employment schemes to retain and reintegrating older workers into the labour force. One such program, the Employability Improvement Project (EIP) instituted in 1991, was geared at providing employment development services. This program increased weeks worked and annual earnings among older workers. Encouraging increased investment in skills development and improving the attractiveness of training are some key recommendations in the OECD policy report on promoting employability throughout working life.

Another hindrance to the re-employment of older workers is access to information on jobs that match their skill sets or that offer appropriate workplace accommodations. Given the finding that older female workers that return to work are more likely to be employed part-time, policymakers could consider implementing policies to encourage part-time versus

[16] Table 3.1 (column 3) and Table 3.2 (column 2), respectively.

full-time employment for these workers. A focus on increasing awareness of the demand side incentives for hiring older workers can also increase and extend the labour market participation of workers over 50 years.

The need for more empirical evidence on the pathways to labour market exits and the retirement decision-making of workers over 50 makes it difficult for policymakers to implement programs targeted to these individuals. It is also difficult to predict the effectiveness of policy initiatives to retain and reintegrate older workers into the labour market. The findings of my study provide additional information to aid policymakers in this endeavour.

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3.7 Appendix

Table 3.3: Reasons Separated (Why Left) by Involuntary and Economic Conditions

Reasons Separated (Why Left)	Involuntary	Economic Conditions
Other		
Own Illness		
Caring for own children		
Caring for elder relative (60+)		
Pregnancy		
Personal or Family Responsibilities		
Going to school		
End Seasonal Job		
End Temporary Job		X
End Casual Job		X
Company moved	X	
Company went to of business	X	
Business conditions		
Dismissal		
Other reasons		
Business sold or closed down	X	
Change residence		
Dissatisfied with job		
Retired		

Note: This table categorizes the reasons why separated into involuntary and economic conditions for respondents who are non-employed but were employed in the past 12 months.
Source: Labour Force Survey (1997 - 2019)

Table 3.4: Descriptive Statistics for Employed Sample in Month 1

	Full Sample - Month 1
Age	55.81 years
Age Group	
50 - 54	46%
55 - 59	32.6
60 - 64	16.9
65 - 69	4.5
Gender (Male = 1)	50.1
Living with Partner	77
Level of Schooling	
Less than High School	13.5
High School or Trade	39.6
Post-Secondary	46.9
Occupation	
Management	8.67
Professionals	17.6
Semi-Professionals and Technicians	7.2
Clerical, Sales, and Service	43
Manual workers & Trade	23.5
Industry	
Primary and Construction	8.4
Manufacturing	14.9
Retail, Accommodation and Food	13.6
High-Skill	15.3
Public Services	32.7
Other Services	15.1
Region	
Ontario	39.1
Atlantic Provinces (NB, NS, PEI, NL)	7.3
Québec	24.1
Prairie Provinces (SASK, MAN)	6.6
Alberta	10.3
British Columbia	12.6
No. Individuals	472,356

Note: Data from the Labour Force Survey (1997 - 2019). All respondents were aged between 50 and 69.

Table 3.5: Descriptive Statistics for Separated Sample

	Separated Sample	Job Loss = 1	Job Loss = 0
Age	57.9 years	57.4 years	58.2 years
Age Group			
50 - 54	29.9%	34.1%	27.2%
55 - 59	32.3	32.3	31.8
60 - 64	25.8	23.6	26.8
65 - 69	12.1	10	14.1
Gender (Male = 1)	52.4	47.3	52.1
Living with Partner	76.8	77.8	77.3
Level of Schooling			
Less than High School	21.5	21.8	21
High School or Trade	40.6	38.9	40.4
Post-Secondary	37.8	39.3	38.6
Occupation			
Management	4.7	2.2	5.2
Professionals	14.5	16.9	15.3
Semi-Professionals and Technicians	5.9	6.7	5.8
Clerical, Sales, and Service	38.6	37.7	39.5
Manual workers & Trade	36.2	36.5	34.2
Industry			
Primary and Construction	17.5	18.8	16
Manufacturing	13.6	8.4	14.4
Retail, Accommodation and Food	12.6	10.2	12.7
High-Skill	12.9	10.5	13.2
Public Services	28.5	35.9	29.1
Other Services	15	16.4	14.6
Region			
Ontario	33	30.3	34.2
Atlantic Provinces (NB, NS, PEI, NL)	13	15.1	11.7
Québec	27.9	27.7	27.5
Prairie Provinces (SASK, MAN)	6.3	6.9	6.4
Alberta	8.7	8.7	9.1
British Columbia	11	11.3	11.1
No. Observations	118,850	26,828	92,022
No. Individuals	23,770		
No. Waves (months)	5		

The table shows the average statistics for the separated sample between months 2 and 6. All workers are employed in Month 1. All respondents were aged between 50 and 69. **Source:** Labour Force Survey (1997 - 2019). "Job Loss" refers to individuals who have suffered at least one permanent involuntary job separation.

Table 3.6: Descriptive Statistics for Outcome Variables (Return and Exit)

	Exit = 1	Return = 1
Age	59.76 years	56.15 years
Age Group		
50 - 54	15.8%	41.7%
55 - 59	30.5	34.5
60 - 64	32.5	20.7
65 - 69	21.2	3.12
Gender (Male = 1)	48.4	53
Living with Partner	79.3	75.7
Level of Schooling		
Less than High School	19.2	21.2
High School or Trade	39.4	42.2
Post-Secondary	41.4	36.6
Occupation		
Management	6.68	2.24
Professionals	18.8	12.5
Semi-Professionals and Technicians	6.69	6.13
Clerical, Sales, and Service	42	34.5
Manual workers & Trade	25.9	44.6
Industry		
Primary and Construction	11.9	19.1
Manufacturing	11.8	15.2
Retail, Accommodation and Food	12.4	10.5
High-Skill	14.5	10.5
Public Services	35.7	29.2
Other Services	13.8	15.5
Region		
Ontario	34.6	32.5
Atlantic Provinces (NB, NS, PEI, NL)	10.7	13.1
Québec	28.1	27.9
Prairie Provinces (SASK, MAN)	7.25	6.03
Alberta	8.42	8.86
British Columbia	10.9	11.6
No. Observations	37,414	13,343
No. Waves (months)	5	

The table shows the average statistics for Return = 1 and Exit = 1 during months 2 to 6. All workers are employed in Month 1. All respondents were aged between 50 and 69.
Source: Labour Force Survey (1997 - 2019).

Table 3.7: LPM Regression with Covariate Coefficients

Variables	Return (1)	Exit (2)
Living with partner	0.0076***	0.0032***
	(0.0018)	(0.0004)
Level of School		
(reference: less than high school)		
High School and Trade	0.0069***	-0.0046***
	(0.0019)	(0.0005)
Post-Secondary	0.0064***	-0.0054***
	(0.0023)	(0.0006)
Industry		
(reference: Primary and Construction)		
Manufacturing	0.0097***	-0.0061***
	(0.0026)	(0.0007)
Retail Trade, Accommodation, and Food	0.0055*	-0.0076***
	(0.0033)	(0.0007)
High-Skill Services	0.0043	-0.0053***
	(0.0034)	(0.0007)
Public services	0.0544***	-0.0041***
	(0.0029)	(0.0007)
Other services	0.0110***	-0.0064***
	(0.0026)	(0.0007)
Occupation		
(reference: Management)		
Professionals	0.0375***	0.0017***
	(0.0048)	(0.0006)
Semi-Professionals and Technicians	0.0317***	0.0009
	(0.0052)	(0.0007)
Clerical, Sales, and Services	0.0295***	-0.0011**
	(0.0043)	(0.0005)
Manual Workers and Trade	0.0480***	0.0019***
	(0.0043)	(0.0006)
Region		
(reference: Ontario)		
Atlantic Provinces	-0.0179***	0.0063***
	(0.0021)	(0.0005)
Quebec	-0.0037	0.0050***
	(0.0023)	(0.0005)
Prairie Provinces	0.0021	0.0002
	(0.0027)	(0.0005)
Alberta	0.0052	-0.0029***
	(0.0032)	(0.0005)

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Variables	Return (1)	Exit (2)
British Columbia	0.0034 (0.0030)	-0.0014*** (0.0005)
Year dummies	YES	YES
Observations	118,406	2,361,780
R-squared	0.336	0.0271

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors are in parentheses. Controls include living with a partner, level of schooling, occupation, industry, and region of residence.

Table 3.8: LPM Regression with Age Dummies

Variables	Return (1)	Exit (2)
Prior Job Loss	-0.0511***	0.0324***
	(0.0127)	(0.0049)
Gender (Male = 1)	-0.0111***	-0.0044***
	(0.0019)	(0.0004)
Age Dummies (reference:Age 50)		
51	0.0028	0.0013***
	(0.0036)	(0.0003)
52	0.0085**	0.0025***
	(0.0036)	(0.0004)
53	0.0004	0.0035***
	(0.0036)	(0.0004)
54	-0.0064*	0.0048***
	(0.0037)	(0.0004)
55	0.0009	0.0101***
	(0.0037)	(0.0005)
56	0.0046	0.0110***
	(0.0038)	(0.0005)
57	-0.0004	0.0116***
	(0.0038)	(0.0005)
58	0.0036	0.0127***
	(0.0039)	(0.0006)
59	-0.0026	0.0120***
	(0.0039)	(0.0007)
60	-0.0017	0.0261***
	(0.0041)	(0.0009)
61	0.0011	0.0237***
	(0.0043)	(0.0009)
62	-0.0032	0.0256***
	(0.0044)	(0.0010)
63	-0.0004	0.0286***
	(0.0045)	(0.0011)
64	-0.0031	0.0287***
	(0.0048)	(0.0014)
65	-0.0227***	0.0846***
	(0.0063)	(0.0026)
66	-0.0219***	0.0684***
	(0.0070)	(0.0023)
67	-0.0304***	0.0619***
	(0.0074)	(0.0025)

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Variables	Return (1)	Exit (2)
68	-0.0133	0.0626***
	(0.0087)	(0.0029)
69	-0.0316***	0.0811***
	(0.0116)	(0.0042)
Constant	6.4656***	6.5751***
	(0.0388)	(0.0073)
Controls	YES	YES
Year dummies	YES	YES
Interaction	YES	YES
Observations	119,734	2,380,875
R-squared	0.258	0.0234

Note: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$. Robust standard errors are in parentheses. Controls include living with a partner, level of schooling, occupation, industry, and region of residence.

Table 3.9: Earnings Regression

Variables	Earnings_return (1)	Earnings_exit (2)
Gender (Male = 1)	0.4039***	0.3556***
	(0.0090)	(0.0020)
Living with partner	0.0533***	0.0209***
	(0.0081)	(0.0018)
Level of Schooling (reference: Less than High School)		
High School and Trade	0.0973***	0.1523***
	(0.0078)	(0.0024)
Post-Secondary	0.1223***	0.2337***
	(0.0108)	(0.0028)
Industry (reference:Primary and Construction)		
Manufacturing	-0.1318***	-0.0809***
	(0.0108)	(0.0041)
Retail Trade, Accommodation, and Food	-0.5112***	-0.4601***
	(0.0179)	(0.0052)
High-Skill Services	-0.3313***	-0.2050***
	(0.0180)	(0.0051)
Public Services	-0.2711***	-0.1249***
	(0.0166)	(0.0045)
Other services	-0.3416***	-0.2334***
	(0.0135)	(0.0046)
Occupation (reference: Management)		
Professionals	-0.1731***	-0.1196***
	(0.0275)	(0.0044)
Semi-Professionals and Technicians	-0.4108***	-0.3491***
	(0.0295)	(0.0052)
Clerical, Sales, and Services	-0.5828***	-0.5652***
	(0.0236)	(0.0041)
Manual Workers and Trade	-0.4615***	-0.4669***
	(0.0238)	(0.0044)
Region (reference: Ontario)		
Atlantic Provinces	-0.1064***	-0.1433***
	(0.0095)	(0.0022)
Quebec	-0.0809***	-0.0956***
	(0.0103)	(0.0022)
Praire Provinces	-0.0643***	-0.0653***
	(0.0133)	(0.0024)
Alberta	0.1451***	0.0901***
	(0.0147)	(0.0028)

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Variables	Earnings_return	Earnings_exit
	(1)	(2)
British Columbia	0.0461***	0.0234***
	(0.0145)	(0.0027)
Constant	6.4656***	6.5751***
	(0.0388)	(0.0073)
Year dummies	YES	YES
Observations	97,615	2,741,104
R-squared overall	0.316	0.372

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Robust standard errors are in parentheses. Controls include living with a partner, level of schooling, occupation, industry, and region of residence.

Connecting Text II

The last two chapters focus on the labour market decisions of older workers. The second chapter highlights that older workers in Canada and the United States have an increased preference for part-time employment, which increases with age and gender. Chapter 2 indicates that that the decision for non-employment and working part-time in Canada and the United States is due to the choices of the older workers rather than ageism, lack of jobs or lack of skills.

The third chapter extends the analysis from chapter 2 and estimates the likelihood that older workers in Canada will exit or remain attached to the labour market following an involuntary job separation. Chapter 3 shows that the probability of labour market exit increases with age and the highest probability is observed at the NRA (Table 3.8). The chapter concludes that involuntary separation is a possible pathway through which older workers exit the labour market.

For the next chapter, I focus on the health outcomes of retired older workers aged 50 to 75 in the United States. In particular, I study whether engaging in physical activity is protective of the cognitive functioning of older workers. The importance of age in the decision making of older workers motivates the use of the retirement eligibility thresholds as instrumental variables in the empirical analysis for chapter 4.

Chapter 4

Cognitive Functioning in Retirement: Does physical activity improve the impact?

4.1 Introduction

In 2016, the average life expectancy at birth for the global population was 72.0 years, in comparison to 66.5 in 2010. Longer life expectancy and an average retirement age of 63 years suggests that individuals are spending approximately six years longer in retirement. As more of the "Baby Boomers" (born 1946 - 1965) start retiring, there will be a significant impact on government-funded pension plans, health services and individual well-being in retirement. Some countries have implemented policies to increase the retirement age for claiming pension benefits to reduce the pressure on government-funded pension plans. These policies have been somewhat successful in delaying worker retirement and reducing the expenditure of government-funded pensions (Staubli and Zweimüller (2013); Mastrobuoni (2009)).

Increases in the retirement age for pension claiming have brought into question the influence of delayed retirement on individual health outcomes. Studies examining the relationship between retirement and health (defined as physical or mental health) yield heterogeneous effects. Some studies conclude that retirement has a positive effect on health and health outcomes (Neuman (2008); Johnston and Lee (2009); Coe and Zamarro (2011); Insler (2014); Eibich (2015); Kämpfen and Maurer (2016); Shai (2018)) while others found that retirement has a negative or no effect (Rohwedder and Willis (2010); Mazzonna and Peracchi (2012); Bonsang et al. (2012); Bingley and Martinello (2013); Celidoni et al. (2017); Heller-Sahlgren (2017)). According to Nishimura et al. (2018), the impact of retirement on health differs based on (1) the method of analysis used in the study (RDD, DID, IV-FE), (2) the measures of health used (cognitive, self-reported, ADL, BMI, Depression), (3) institutional heterogeneity across countries studied (pensions, taxes, and disability policy) and (4) heterogeneity across individuals.

This paper investigates whether engaging in healthy behaviours during retirement will improve the impact of retirement on health outcomes. A major channel through which retirement impacts health is through the health behaviours that individuals invest in during retirement. Following Grossman (1972), individuals demand health for consumption and investment purposes. Additionally, Dave et al. (2006) indicate that the increase in leisure time that comes with retirement might positively impact the consumption value of health so that individuals would demand a higher level of health. They can achieve this by investing more in healthy behaviours and/or consuming more health care services. This choice depends on the cost of the health services and the shadow price of their time. Alternatively, retirement could remove the incentive for health investments since retirement income (pension) does not depend on the individual's health. Empirical studies reveal that retirement positively impacts individual investment in health behaviour such as smoking, alcohol consumption, physical activity, sleep time on weekdays, and doctor's visits (Insler (2014); Eibich (2015); Kämpfen and Maurer (2016); Zhao et al. (2017); Müller and Shaikh (2018); Kesavayuth et al. (2018)).

In this study, the health outcome of interest is cognitive functioning. Coe et al. (2012);

Bonsang et al. (2012); Celidoni and Rebba (2017) and Atalay et al. (2019) studied the impact of retirement on cognitive function. Cognitive functioning is crucial because it influences an individual's ability to process information and make decisions. Moreover, cognitive function declines with age and is one of the first health outcomes to be impacted by the retirement decision (Cadar (2018)). The health behaviour I use in this study is physical activity, as there is an established causal link between retirement and physical activity in the existing literature (Insler (2014); Eibich (2015); Kämpfen and Maurer (2016); Zhao et al. (2017); Müller and Shaikh (2018); Kesavayuth et al. (2018)). Also, there is a consensus in the medical literature on the positive association between physical activity and cognitive performance (Weuve et al. (2004); Frith and Loprinzi (2017)).

I use data from the U.S. Health and Retirement Study (HRS), 1998 - 2014, to determine the relationship between retirement, physical activity and cognitive functioning for individuals aged 51 to 75. I examine three (3) outcomes in this paper, the impact of (1) retirement on cognitive function, (2) retirement on physical activity, and finally (3) the combined impact of retirement and physical activity on cognitive function. I use Two-Stage Least Squares (2SLS) to determine the causal link between retirement and cognitive function and between retirement and physical activity. This analysis is similar to the work done by Bonsang et al. (2012); Celidoni et al. (2017); Heller-Sahlgren (2017) and Kämpfen and Maurer (2016).

Since individuals choose when to retire, retirement may not be exogenous to physical activity and cognitive scores. Poor cognitive functioning or the level of engagement in physical activity could influence the retirement decision and result in an endogeneity bias. I address the endogeneity of retirement using an instrumental variable approach. This strategy relies on the eligibility ages for pension claiming to provide a source of exogenous variation in the retirement decision. Similar to Bonsang et al. (2012); Celidoni et al. (2017); Heller-Sahlgren (2017) and Kämpfen and Maurer (2016), I use the eligibility ages for early and full retirement as instrumental variables for the retirement status. Next, I estimate the joint impact of retirement and physical activity on cognitive function, correcting for the possible endogeneity of retirement.

This paper contributes to the literature on the causal effect of retirement on the cognitive function of individuals aged 51 to 75. The main contribution of my paper is to determine whether engaging in physical activity is a mechanism through which retirement impacts cognitive function. In the literature, Bonsang et al. (2012); Rohwedder and Willis (2010); Mazzonna and Peracchi (2012) find that retirement negatively impacts cognitive function but positively impacts the individual's health behaviours and lifestyle choices. Given the positive association between physical activity and cognitive function, including physical activity as a mechanism could positively affect the short-run and long-run relationship between retirement and cognitive function. This positive impact would support the view that engaging in positive health behaviours in retirement is protective of cognitive function and helps to slow its deterioration. There is, however, little empirical evidence on the mechanisms through which retirement affects cognitive functioning. Also, I examine whether the collective impact of physical activity and retirement differs based on three individual characteristics - gender, marital status and education.

My results show that the retirement duration of at least one year decreases the cognitive scores of individuals aged 51 to 75 by 5.6% compared to the sample average of 10.6 points. Retirement also increases the likelihood of meeting the US 2018 Physical Activity Guidelines by 12.6 percentage points. The combined effect of retiring and meeting the physical activity guidelines increases cognitive scores by approximately 0.5 points. It corresponds to a 4.5% increase in the cognitive score compared to the HRS sample average.

The remainder of the paper is organized as follows. Section 2 provides the institutional setting. Section 3 outlines the empirical methodology and Section 4 discusses the data and provides descriptive statistics for the measures of retirement, cognitive ability and physical activity. Section 5 outlines the estimation results, and Section 6 concludes the paper.

4.2 Institutional Framework

In the US, the Social Security Act of 1935 provides information on retirement benefits to individuals 65 years or older who work in jobs covered by the Social Security system. The earliest age at which individuals may start receiving retirement benefits is 62 years for both men and women.[1] The early retirement benefit, at age 62, is at a reduced level to account for the more extended period that individuals would collect retirement benefits. Faced with untenable financing of the Social Security programme, the US amended the Social Security Act in 1983. The changes took effect in 2000 and resulted in delayed retirement for both men and women (Mastrobuoni (2009)). The amendment gradually increases the normal retirement age (NRA) for benefit claiming from 65 to 67 years. The NRA increased from 65 to 66 years in 2009, with a further increase from 66 to 67 years by 2027. The NRA for benefit claiming depends on the individual's birth year. For individuals born before 1937, the full retirement age is 65 years, while the full retirement age increases in 2-month increments for those born after 1937 (Mastrobuoni (2009)). I use the eligibility thresholds as instruments to establish the causal relationship between retirement and cognitive function and retirement and physical activity. The age eligibility thresholds for claiming retirement benefits impact the decision to retire, but have little impact on the respondents' cognitive scores or their decision on whether to engage in physical activity.

4.3 Econometric Analysis

This paper examines the impact of physical activity on the causal link between cognitive function and retirement. Before carrying out this analysis, I first establish the theoretical and empirical relationships that exist between (1) retirement and cognitive function, (2) retirement and physical activity and (3) physical activity and cognitive function. Retirement negatively impacts cognitive function. When individuals separate from the labour market,

[1] A 1961 amendment allowed men to retire as early as age 62.

they often leave mentally stimulating jobs that protect against cognitive decline (Bonsang et al. (2012); Rohwedder and Willis (2010); Mazzonna and Peracchi (2012)). The existing studies analysing the association between retirement and cognitive scores focus on data from the US and Europe. The three primary surveys, the Survey of Health, Ageing and Retirement in Europe (SHARE), The English Longitudinal Study of Ageing (ELSA) and the Health and Retirement Survey (HRS), have comparable measures of cognitive functioning based on an episodic memory word recall test. In the literature, Rohwedder and Willis (2010); Mazzonna and Peracchi (2012); Bonsang et al. (2012); Celidoni et al. (2017), and Atalay et al. (2019) examine the relationship between retirement and cognitive functioning and find that on average, retirement negatively affects cognition. Additionally, Atalay et al. (2019) find that the rate of cognitive decline with age is more significant for men than women. Mazzonna and Peracchi (2012) also find that the decline in cognitive scores after retirement is more substantial for less educated women than highly educated women.

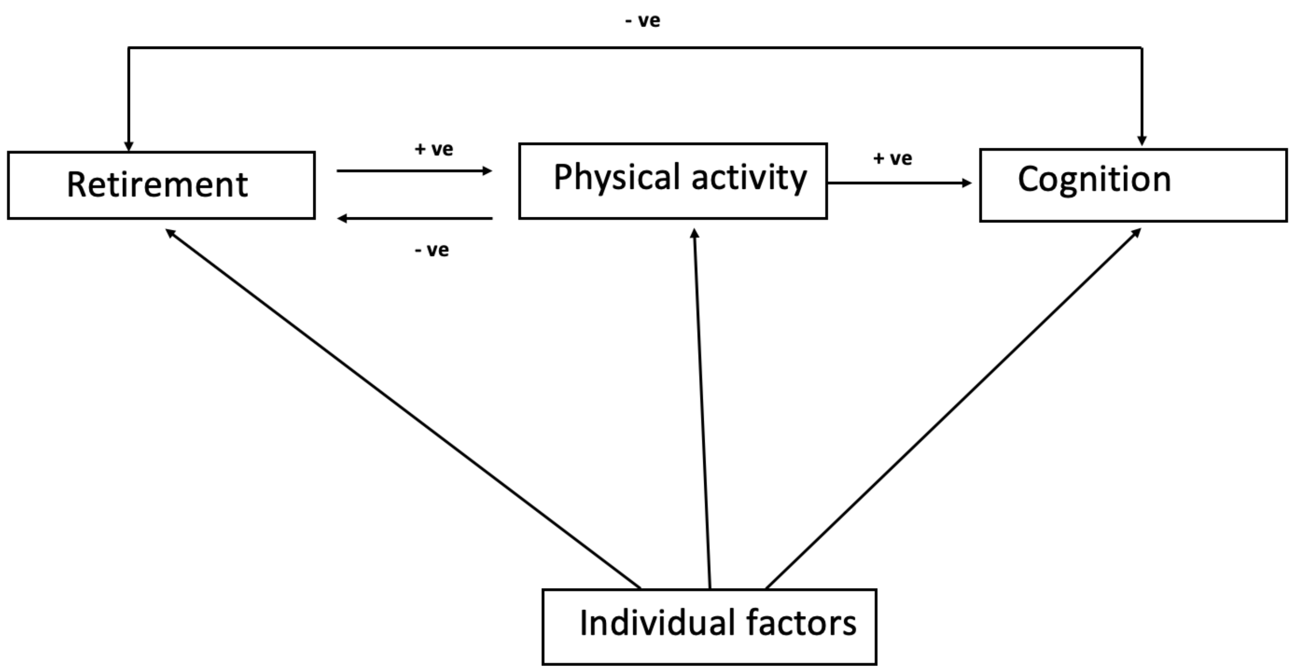
Scientists do not view the impact of retirement on cognitive function to be instantaneous (Gall et al. (1997); Ekerdt et al. (1985); Rohwedder and Willis (2010); Mazzonna and Peracchi (2012)). They do not expect individuals to experience a significant decline in their cognitive functioning the instant they retire from the labour force. Instead, individuals take time to respond to the changes in lifestyle and activities that arise as they transition from work to retirement (Bonsang et al. (2012)). This "honeymoon phase" is usually expected to last between six months to two years (Gall et al. (1997); Ekerdt et al. (1985)). The effect of retirement would also depend on how long the individual has been retired (Celidoni et al. (2017); Heller-Sahlgren (2017)). Recognizing that the time since retirement impacts the level of cognitive functioning, the primary retirement variable in this analysis is a dummy capturing retirement duration. The variable is equal to 1 if the individual is retired for at least one year, and zero otherwise.

In examining the association between health behaviours and cognitive functioning, evidence from epidemiology and neurological studies suggest that smoking, alcohol consumption and poor diet results in a faster decline in cognitive functioning, while engagement in physical and social leisure activities is protective. I use physical activity in this study, as there is a consensus in the medical literature on the positive association between physical activity and cognitive performance (Weuve et al. (2004); Frith and Loprinzi (2017)). The literature on the impact of other health behaviours (smoking and alcohol consumption) is inconclusive, with some studies finding a negative, positive or no association. Physical activity improves cognitive function by increasing cerebral blood flow, promoting nerve cell growth and enhanced cerebrovascular function (Cadar (2018)). Physical activity also impacts depression, sleep, appetite, and energy levels, preventing age-related diseases (diabetes, hypertension) that affect cognition.

Additionally, the literature has an established relationship between retirement and physical activity. Insler (2014); Eibich (2015); Kämpfen and Maurer (2016); Zhao et al. (2017); Müller and Shaikh (2018); Kesavayuth et al. (2018) find that retirement has a positive effect on physical activity. With the increase in leisure time during retirement, individuals will have more time to engage in physical activity.

Figure 4.1 illustrates the associations between retirement, physical activity and cognitive function. Retirement positively impacts the level of physical activity, but negatively impacts cognitive functioning. In addition, physical activity has a positive impact on cognitive function. Heterogeneous individual characteristics such as the level of education, gender and marital status etc., also impact retirement, physical activity and cognitive function. This paper hypothesises that physical activity is a channel through which retirement affects cognitive function. Hence, engaging in physical activity should reduce the negative impact of retirement on cognition. If this is the case, the estimated effect of retirement on cognitive function should decrease once I account for physical activity.

Figure 4.1: Association between retirement, physical activity and cognition



4.3.1 Identification

The main empirical challenge is that retirement and physical activity may not be exogenous to cognitive scores. Three possible causes of endogeneity that might bias the results are omitted variable bias, measurement error and reverse causality. Measurement error might occur due to how the HRS panel data measures retirement, cognitive function and physical activity variables. If classical, the presence of measurement error would produce downward bias and underestimate the effect of retirement and physical activity on the cognitive measure. Omitted variable bias might occur due to differences in unobserved individual characteristics that influence cognitive function, physical activity and the retirement decision. I include individual and time-fixed effects to eliminate any potential bias due to unobserved time-invariant individual heterogeneity (e.g. gender and years of schooling) and to take account of the biennial nature of the panel data set.

I use instrumental variable analysis to address the possibility of time-varying omitted variables and reverse causality in retirement. Reverse causality might occur because individuals decide when to retire and how much physical activity to engage in. An increase in the time spent in retirement may reduce cognitive scores, or having low cognitive scores may increase the probability that the individual retires. The presence of reverse causality would overestimate the impact of retirement on cognition, as retirement would be associated with lower cognitive performance even without any effect of retirement on cognition. For physical activity, retirement increases the time available to spend on physical activity. However, engaging in physical activity can improve the individual's health and may decrease the probability that the individual will retire.

Additionally, reverse causality and omitted variable bias can also exist between physical activity and cognitive functioning. Engaging in physical activity positively affects cognitive scores, but having low cognitive scores could reduce the individual's ability to engage in physical activity. I have not identified any literature that examines reverse causality between cognitive function and physical activity. In this paper, I use the HRS memory recall score to measure the individual's cognitive function. There is no empirical evidence that lower memory recall scores in the short term influence or impact an individual's ability to engage in physical activity. As a result, I do not correct the possible endogeneity of physical activity. Failure to correct for the endogeniety of physical activity can lead to biased estimates.

4.3.2 Instruments

To construct instruments for the retirement variable, I follow Bonsang et al. (2012); Kämpfen and Maurer (2016); Celidoni et al. (2017) and use the eligibility ages for social security benefits to construct two instrumental variables. (1) Early represents exceeding the early age for benefit claiming. Since the earliest age for claiming benefits in the United States is 62 years, individuals exceed the early retirement age if they are older than 62 but younger than the full retirement age. The instrument (early) is valid for ages 62 to 65. (2) Normal, represents exceeding the full age for benefit claiming. The full retirement age is 65 for individuals born before 1937, and increases in 2-month increments for those born after 1937.

In order for my instrumental variable analysis to be valid the instruments, Early and Normal, must be relevant and satisfy the exclusion restriction. The findings from chapters 2 and 3 support the relevance of the benefit claiming thresholds. My findings indicate that proximity to the benefit claiming thresholds influence the decision of older workers to remain attached or to exit the labour market. Additionally, Bonsang et al. (2012); Kämpfen and Maurer (2016); Celidoni et al. (2017); Heller-Sahlgren (2017) find that the instruments (Early and Normal) are strong predictors of the retirement behaviour of the respondents and cause significant variation in the proportion of individuals that retire conditional on other controls. In this chapter, I test the relevance of the instruments, Early and Normal, in the first stage regression (Equation 4.5). The instruments are relevant if the F-statistic of the first-stage regression estimator is greater than 10 (Staiger and Stock (1997)).

In addition, the use of early and full retirement satisfies the exclusion requirement. The exclusion restriction requires that the instrument be uncorrelated with the error term of the regression. Hence, the instruments should affect cognitive abilities indirectly through the effects on the retirement benefit-claiming age. Ages 62 and 65 are thresholds in the US Social Security Program and do not impact or respond to changes in individual characteristics such as their cognitive score or health behaviours. Therefore, any change in the cognitive score for an individual would represent the causal effect of retirement.

4.3.3 Model

The central empirical equation in this paper takes the form:

$$C_{it} = \alpha_0 + \alpha_1 R_{it} + \alpha_2 P_{it} + \alpha_3 R_{it} * P_{it} + \alpha_4 X_{it} + \iota_i + \varepsilon_{it} \tag{4.1}$$

where C_{it} is a measure of the individual's i cognitive ability at time t, R_{it} is a dummy indicating retirement duration of at least one year; P_{it} is a dummy indicating meeting the 2018 U.S. Physical Activity Guidelines and X_{it} contains age and age squared. The variable

ι_i denotes time-invariant individual-specific effects correlated with the explanatory variable, and ε_{it} is the error term. The coefficient on the retirement duration, α_1, captures the impact of retirement on cognitive function and the coefficient on physical activity, α_2, captures the effect of physical activity on cognition. To determine if physical activity changes the impact of retirement on cognitive function, I focus on α_3, which captures the combined effect of retirement and physical activity on cognitive function. I estimate this equation using Two-stage Least Squares (2SLS) regressions using the early and full retirement eligibility ages as instruments for retirement duration.

I implement the empirical analysis in four steps. In step 1, I estimate three simple OLS regressions for (1) the impact of retirement on cognitive function, (2) the impact of retirement on physical activity and (3) the impact of physical activity on cognitive function. The OLS regressions provide descriptive evidence of the relationship between physical activity, cognitive function and retirement. These regressions include age controls (X_{it}) and individual-fixed effects but do not correct for the endogeneity of the retirement decision, which might bias the results.

$$C_{it} = \lambda_0 + \lambda_1 R_{it} + \lambda_2 X_{it} + \zeta_i + \xi_{it} \tag{4.2}$$

Equation 4.2 estimates the impact of being retired for at least one year on the individual's cognitive ability. ζ_i is the time-invariant individual effects, and ξ_{it} is the error term.

$$P_{it} = \beta_0 + \beta_1 R_{it} + \beta_2 X_{it} + \theta_i + \mu_{it} \tag{4.3}$$

Equation 4.3 estimates the impact of being retired for at least one year on whether the individual meets the physical activity threshold. θ_i is the time-invariant individual effects, and μ_{it} is the error term.

$$C_{it} = \gamma_0 + \gamma_1 P_{it} + \gamma_2 X_{it} + \varsigma_i + \phi_{it} \tag{4.4}$$

Equation 4.4 estimates the impact of meeting the 2018 U.S. Physical Activity guidelines on the individual's cognitive ability. ς_i is the time-invariant individual effects, and ϕ_{it} is the error term.

In step 2, I estimate the first stage of the 2SLS regression. Using Equation 4.5, I predict being retired using the instruments (early and normal) in addition to the age controls, X_{it}, and the error terms. Z_{it} represents the instrumental variables - *early* and *normal* for retirement duration, κ_i captures the time-invariant individual effects and ν_{it} is the error term.

$$R_{it} = \rho_0 + \rho_1 Z_{it} + \rho_2 X_{it} + \kappa_i + \nu_{it} \tag{4.5}$$

In step 3, I estimate the causal link between retirement and cognitive function (Equation 4.6), and retirement and engaging in physical activity (Equation 4.7). The second stage analysis uses the predicted value of the retirement variable, $\widehat{R}_{it}$, from the first stage regression in addition to the age controls, X_{it}, and error terms. τ_i and ϑ_i captures the time-invariant individual effects from the cognitive function and physical activity second stage regressions, respectively. While υ_{it}, and ω_{it} capture the error terms. This estimation addresses any reverse causality between retirement and cognitive function and retirement and physical activity once the instrumental variables are relevant and satisfy the exclusion restriction.

$$C_{it} = \varrho_0 + \varrho_1 \widehat{R}_{it} + \varrho_2 X_{it} + \vartheta_i + \omega_{it} \tag{4.6}$$

$$P_{it} = \pi_0 + \pi_1 \widehat{R}_{it} + \pi_2 X_{it} + \tau_i + \upsilon_{it} \tag{4.7}$$

In step 4, I establish the paper's main result using Equation 4.1 to estimate the second stage of the 2SLS regression estimation. I regress the cognitive score on the predicted value of the retirement and physical activity variables and the age controls. This estimation captures the heterogeneous treatment effects by interacting retirement and physical activity.

4.4 Data

4.4.1 Health and Retirement Study

This study uses data from nine waves (1998 - 2014) of the Health and Retirement Study (HRS). The HRS is a biennial national longitudinal household survey conducted by the Institute of Social Research at the University of Michigan in the United States. The HRS provides information on the economic, health and family status, as well as public and private support systems of Americans and their partners. The HRS sample comprises six entry cohorts: the original 1992 HRS cohort born 1931 to 1941; the 1993 Asset and Health Dynamics Among the Oldest Old (AHEAD) cohort born 1923 or earlier; the War Baby cohort (born 1942 to 1947) and the Children of the Depression (born 1924 to 1930) entering in 1998; the Early Baby Boomers born 1948 to 1953 entering in 2004 and the Mid-Baby Boomers born 1954 to 1959 entering in 2010.

I use the RAND HRS Longitudinal file 2014 (v2) for this analysis. The RAND data files are a clean and easy-to-use version of the HRS data, containing information from the Core and Exit Interviews of the HRS. It includes derived variables constructed and named consistently across waves. My sample focuses on respondents aged 51 to 75. The sample selection for this study uses the same criteria and restrictions as Bonsang et al. (2012). Following Bonsang et al. (2012), I exclude individuals with missing information for labour force status, the year they left the labour force and the cognitive tests. I also remove individuals who never worked, those who return to the labour force after retirement, and those who left their job before age 50. While Bonsang et al. (2012) use six waves (1998 - 2008) of HRS data, I use three additional waves of data to extend the sample to 2014. The final analytic sample corresponds to an unbalanced panel including, 85,354 observations for 22,453 individuals (1998 - 2014).[2]

[2] The sample size is larger than the 54,377 observations for 14,710 individuals in Bonsang et al. (2012).

4.4.2 Definition of retirement

In this study, I define an individual as retired if their labour force status indicates they are retired and do not plan to return to work (Lazear (1986)). This definition excludes individuals who report being partly retired, disabled, or not in the labour force. Figure 4.2 shows the proportion of the total sample retired by age. About 30% of the sample is retired by age 61. This proportion increases by $10 - 12\%$ by age 62, the current early retirement age for benefit claiming in the United States. The proportion retired further increases by approximately 20% between ages 62 and 66. This increase is in line with expectations that reaching the early and regular retirement age for benefit claiming increases the probability that the individual will retire.

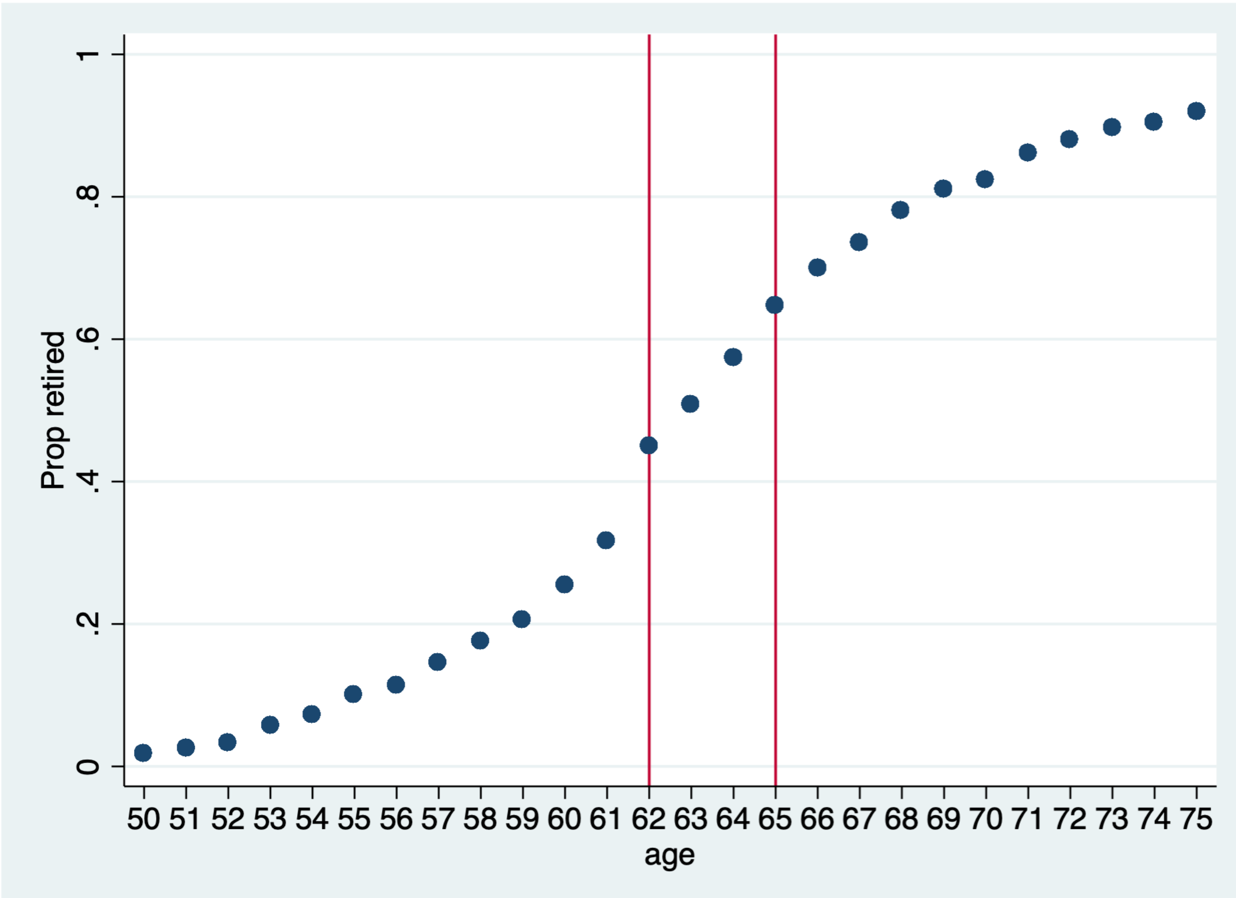
I use a similar strategy to (Bonsang et al. (2012)) for the construction of the retirement variable. I construct a dummy variable using HRS information on the year and month of the individual's last job. To estimate the impact of time spent in retirement on health behaviours and cognition, the retirement variable takes a value of "1" if the individual is retired for at least one year. On average, 47% of the sample is retired, and 41% is retired for at least one year. Using retired for at least one year as a proxy for retirement duration provides a clearer picture of the negative impact of retirement on cognitive function, as it eliminates any reduction in the effect that might occur during the "honeymoon phase" as individuals settle into retirement.[3]

4.4.3 Measure of cognitive function

The HRS cognitive measures are the standard tests used in the psychology of ageing and geriatric literature on cognitive impairment and dementia. The measurements of cognitive function in the HRS are the outcomes of tests of memory, verbal fluency and numeracy. Since the cognitive process is multidimensional, each test measures a different component. This analysis focuses on episodic memory, assessed through a verbal learning and recall test

[3] I also did the analysis using retired as the dependent variable and obtained similar results.

Figure 4.2: Proportion Retired by Age (Full Sample)



Notes: Data from the Health and Retirement Study (1998 - 2014). Vertical lines at ages 62 and 65 represent the retirement benefit claiming thresholds in the United States, respectively.

available in the HRS. In the HRS, surveyors give the respondents a list of 10 words and ask them to recall as many words as possible. The surveyors record the number of words (between 0 and 10) that the respondents identify immediately. After a delay of 5 minutes, the surveyors again ask the respondents to recall the same list of 10 words. Similar to Celidoni et al. (2017); Bonsang et al. (2012); Rohwedder and Willis (2010), the final memory score is the sum of the words recalled at the immediate recall and delayed recall phase (ranging from 0 to 20).

Episodic memory is the ability to recall experiences and is among the first cognitive functions affected by ageing. A reduction in memory (rather than verbal literacy or numeracy) is one of the earliest signs of dementia in the elderly (Anderson and Craik (2000); Prull et al. (2000)). Therefore, researchers prefer episodic memory to other measures of cognitive functioning available in the HRS data. In addition, the score on a verbal learning and recall test provides a more significant variability in scores as there is no cluster of maximum or minimum outcomes (Bonsang et al. (2012)). The major limitation of tests of verbal learning and recall is the possibility of re-testing effects, which involves an improvement in the memory tests of individuals across waves. To account for the re-testing effect, the HRS, since 1996, has used four lists of ten words that do not overlap. Respondents are randomly assigned the initial list, and the surveyors give a different list of words in four successive waves. The list assigned also differs from the one received by the respondents' spouses in the current or previous wave (Ofstedal et al. (2005)). These precautions reduce the possibility of respondents remembering the words from earlier lists.

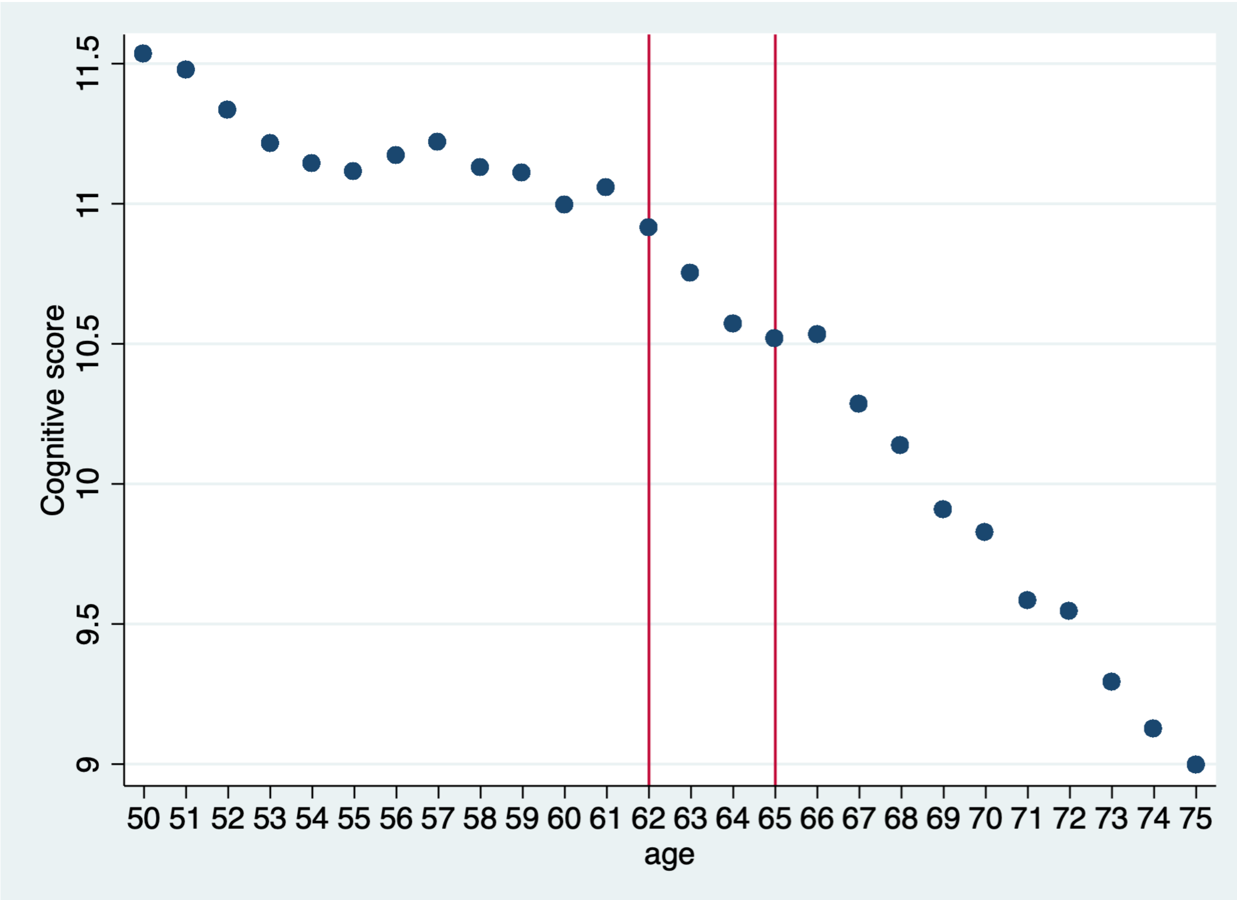
Figure 4.3 shows a negative association between cognitive abilities and age. The total number of words recalled decreases linearly, and there is a reduction in the scores around the retirement benefit claiming ages. The memory score has a sample mean of 10.6 and a standard deviation of 3.3 (Table 4.1).[4] The average memory score for the retired sub-sample is 9.89 compared to 11.14 for the non-retired.

4.4.4 Measure of physical activity

The HRS provides information on the intensity and frequency of physical activity, but does not provide information on the duration. To determine the level of physical activity, respondents respond to three questions on how often they participate in vigorous, moderate or light physical exercise. The possible responses are every day, more than once per week, once per week, one to three times per month, or never. Vigorous physical activity involves "running or jogging, swimming, cycling, aerobics or gym workout, tennis, or digging with a spade or shovel". Moderate physical activity includes "gardening, cleaning the car, walking at a

[4] The average cognitive score and the standard deviation are similar to 10.6 and 3.4, respectively, from Bonsang et al. (2012).

Figure 4.3: Cognitive Score by Age



Notes: Data from the Health and Retirement Study (1998 - 2014). Vertical lines at ages 62 and 65 represent the retirement benefit claiming thresholds in the United States, respectively.

moderate pace, dancing, floor or stretching exercises", and light physical activity involves "sports or activities that are mildly energetic, such as vacuuming, laundry, home repairs".

Following Kämpfen and Maurer (2016), I use the US physical activity guidelines as a benchmark to measure an individual's level of physical activity. I create a dummy variable with a value of "1" if the individual satisfies the physical activity guidelines and "0" otherwise. The 2018 guidelines indicate that to achieve positive health benefits, the elderly should engage in 150 minutes to 300 minutes a week of moderate activity or 75 minutes to 150 minutes a week of vigorous activity, or an equivalent combination of moderate and vigorous activity (of Health and Human Services (2018)). The respondent meets the guidelines if they engage in (a) vigorous or moderate activity every day or do both every day; (b) vigorous activity more than once a week and moderate activity at least 1–3 times per month; (c) vigorous activity once a week and moderate activity at least more than once a week (Kämpfen and Maurer (2016)). Approximately 40.6% of the sample meets the 2018 physical activity guidelines, and about 37% of those meeting the guidelines are retired (Table 4.1).

Figure 4.4 shows an overall decrease in the proportion of individuals meeting the 2018 Physical Activity Guidelines with age. However, the proportion of persons meeting the Guidelines increases between ages 63 and 64 as well as 65 to 67. This descriptive evidence suggests that there is a delay in the declining age trend around the early and normal retirement benefit claiming thresholds. This is in line with the literature on the positive effects of retirement on engaging in physical activity.

4.4.5 Control Variables

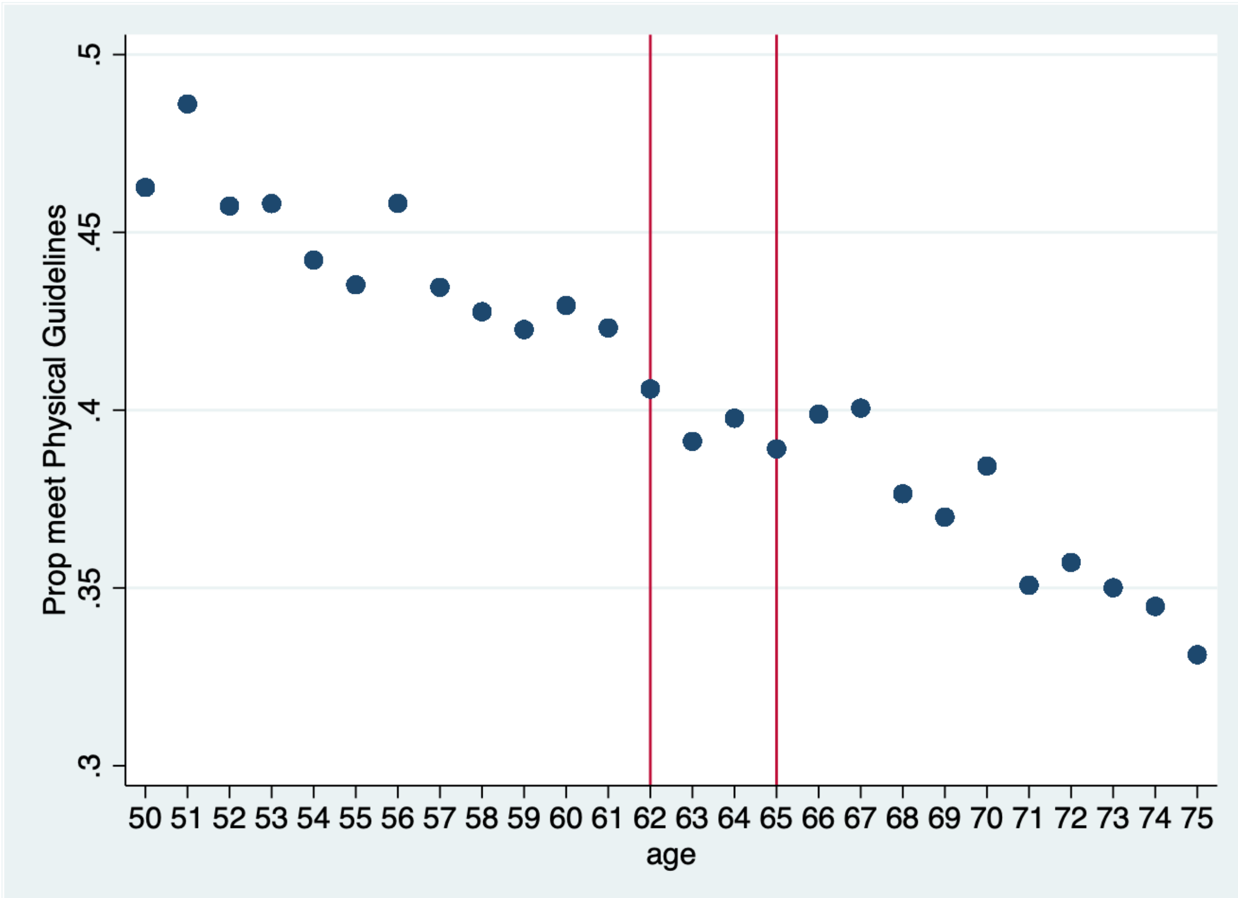
I include age and a quadratic function of age as control variables to differentiate the impact of health behaviour and retirement on cognition separately from the age effect. According to Heller-Sahlgren (2017), since the instrumental variable analysis used in the study provides random variation in retirement behaviour, controls for the respondent's age are the only covariates necessary to ensure a causal interpretation of the relationship between retirement, physical activity and the cognitive score.

4.4.6 Heterogeneity

To capture the impact of individual heterogeneity, I include four additional covariates in the analysis. I include gender, educational attainment, whether living with a partner and the partner's retirement status to investigate whether the impact of retirement and physical activity on the cognitive score differs based on observed individual characteristics.

Individuals with a higher level of education have a higher cognitive score (Alley et al. (2007); Wilson et al. (2009)) and retire later (Stenberg and Westerlund (2013)) than in-

Figure 4.4: Meeting the 2018 Physical Activity Guidelines by Age



Notes: Data from the Health and Retirement Study (1998 - 2014). Vertical lines at ages 62 and 65 represent the retirement benefit claiming thresholds in the United States, respectively.

dividuals with a lower level of education. However, there is no clear evidence of whether higher education significantly influences age-related decline in cognition. If educational attainment matters for the impact of retirement on cognitive scores, the coefficient on the retirement variable would be substantially different for those with high and low educational attainment. On average, in the sample, 84% of respondents have at least a High School or GED and 24% at least a Bachelor's degree or higher (Table 4.1). I capture the individual's educational attainment using two dummy variables representing (1) attaining at least a high school diploma or GED and (2) attaining a bachelor's degree or higher.

Regarding gender, research shows that women live longer and have better health outcomes than men as they age. In addition, women also retire earlier from the labour force than men (Barford et al. (2006); Baum et al. (2021)). In the sample, 44.3% is male, and 43.5% of the males are retired (Table 4.1). Additionally, Atalay et al. (2019) find that women increase their time spent on mental and household activities during retirement. Hence, they experience a smaller decline in cognitive scores during retirement than their male counterparts. The descriptive statistics support this finding, as females have a cognitive score approximately 14% higher than males. I use a dummy variable to capture the individual's gender, where "1" means that the respondent is male.

I include a dummy variable to reflect whether the respondent is married or living with a partner to determine whether living with a partner significantly influences the impact of retirement on cognitive scores. According to Wilson et al. (2009), and Håkansson et al. (2009), living with a partner positively affects an individual's mental health and health outcomes. In the sample, on average, 79% of respondents live with a partner and 67.6% of those are retired. Finally, I examine if the retirement status of the respondent's partner significantly influences the impact of retirement on cognitive scores. Previous literature found that spouses typically retire close to each other (Gustman and Steinmeier (2000, 2004)). One explanation is that spouses will obtain greater pleasure from retirement if they retire together (leisure complementarity). Schirle (2008) finds that an increase in the labour force participation of wives explains one-fourth, one-half, and one-third of the increase in the participation of husbands in the United States, Canada, and the United Kingdom, respectively. To capture leisure complementarity, I include a dummy variable to represent the retirement status of the respondent's partner, where "1" means the partner is retired. On average, 25.9% of the sample has a retired partner (Table 4.1).

To determine if the individual covariates (educational attainment, gender, living with a partner, and the retirement status of the partner) significantly influence the impact of retirement and physical activity on cognitive function, I examine separate regression equations for each covariate. If these covariates matter for the impact of retirement and physical activity on the cognitive scores, the coefficients on the interaction terms would be significantly different.

Table 4.1 shows descriptive statistics for the set of variables used in the analysis. The table provides the overall sample mean and the means by the respondents' retirement status. Retired respondents are older, have a lower cognitive score and a lower probability of meeting the physical activity guidelines (36.5%) compared to Non-Retired respondents. 67.6% of the retired respondents live with a partner, and 41% have a partner that is also retired. Breaking down the sample by gender and retirement status showed that, on average, females retire at an earlier age and have a higher cognitive score than males. An examination of the level of education reveals that in the sample, males tend to have a higher probability of having a high school diploma or at least a Bachelor's degree.

Table 4.1: Summary Statistics - Retired versus Non-Retired

Variable	Mean	Not-Retired	Retired	Not-Retired Gender		Retired Gender	
				M	F	M	F
A. Outcome variable							
Cognitive score	10.556	11.137	9.892	10.545	11.621	9.150	10.463
	[3.329]	[3.17]	[3.382]	[3.121]	[3.123]	[3.234]	[3.382]
B. Dependent variable							
Retired (Yes or No)	0.467	0	1				
	[0.499]						
Retired at least 1 year (Yes or No)	0.409	0	0.876				
	[0.492]		[0.329]				
Meet physical activity guidelines (Yes or No)	0.406	0.442	0.365	0.502	0.392	0.412	0.329
	[0.492]	[0.497]	[0.481]	[0.500]	[0.488]	[0.492]	[0.470]
D. Covariates							
Age	62.753	58.627	67.471	59.079	58.258	67.676	67.314
	[6.968]	[5.476]	[5.296]	[5.565]	[5.374]	[5.201]	[5.362]
Gender (Male = 1)	0.443	0.450	0.435				
	[0.497]	[0.497]	[0.496]				
Living with partner	0.700	0.721	0.676	0.817	0.643	0.789	0.589
	[0.458]	[0.449]	[0.468]	[0.387]	[0.479]	[0.408]	[0.492]
Partner is retired	0.259	0.128	0.410	0.107	0.145	0.413	0.407
	[0.438]	[0.334]	[0.492]	[0.309]	[0.352]	[0.492]	[0.491]
At least Bachelor or higher	0.238	0.283	0.187	0.324	0.249	0.209	0.170
	[0.426]	[0.450]	[0.390]	[0.468]	[0.432]	[0.407]	[0.375]
At least High School or GED	0.835	0.873	0.793	0.864	0.880	0.407	0.801
	[0.371]	[0.333]	[0.405]	[0.342]	[0.325]	[0.413]	[0.399]
No. of Observations (N)	86593	46195	40398	20785	25410	17561	22837

Source: Health and Retirement Study 1998 - 2014.
All respondents were aged between 51 and 75. Standard deviations are in brackets.

4.5 Empirical Results

4.5.1 Retirement and Cognitive Scores

Table 4.2 provides evidence of the association between retirement and cognitive scores. Column (i) estimates a simple OLS regression that assumes the exogeneity of retirement duration and controls for individual time-invariant omitted variables. The OLS fixed effect (OLS-FE) estimate shows a slightly negative but statistically significant relationship. Retirement duration reduces cognitive scores by 0.1 points, an approximately 1% reduction compared to the sample average of 10.6 (Table 4.1). However, given the possible endogeneity between retirement duration and cognitive scores, the OLS-FE estimation would yield biased results. Possible instruments for retirement duration are the retirement benefit claiming age thresholds. Early represents reaching age 62, the earliest age for retirement benefit claiming in the United States. While normal means reaching age 65, the full age for retirement benefit claiming.

In column (iv), I estimate a reduced form regression, including age, quadratic age, and the instruments (early and normal) as control variables. The reduced form regression expresses the main endogenous variable cognitive score as a function of the exogenous variables, the retirement benefit claiming age thresholds. The reduced form regression suggests a reduction in the cognitive scores at the retirement age thresholds. The OLS-FE and the reduced form estimation support previous findings in the literature regarding the negative correlation between retirement and cognitive function (Bonsang et al. (2012); Coe and Lindeboom (2008); Celidoni and Rebba (2017); Atalay et al. (2019)).

To estimate the causal link between retirement duration and cognitive scores, I use 2SLS regressions to account for the endogeneity of retirement. Table 4.2, column (ii) shows the coefficients of the first stage equation (Equation 4.3), which represents the probability of being retired for at least one year. The first stage result indicates that the instruments (early and normal) have a positive and significant impact on the probability of being retired

Table 4.2: Cognitive Functioning and Retirement (1998 - 2014)

	FE Cognitive (i)	First Stage: Retired at least one year (ii)	Second Stage: Cognitive (iii)	Reduced Form Cognitive (iv)
Retired at least one year	-0.100***		-0.5875**	
	(0.031)		(0.2982)	
Instruments				
Early (>62 years old)		0.0845****		-0.064*
		(0.004)		(0.036)
Normal (>65 years old)		0.1306****		-0.068
		(0.005)		(0.042)
Controls				
Age (years)	0.4691****	-0.0262****	0.4519****	0.473****
	(0.040)	(0.005)	(0.041)	(0.041)
Age^2	-0.4194*****	0.004	-0.411****	-0.418****
	(0.023)	(0.003)	(0.024)	(0.025)
Test of over identification restriction (p-value)			0.5677	
Durbin-Wu-Hausman (p-value)			0.0284	
Cragg-Donald Wald F-statistic		356.54		
Year FE	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
N	86593	86593	86593	86593

Source: Health and Retirement Study 1998 - 2014. Age^2= $Age^2/100$.
All respondents were aged between 51 and 75. Robust standard errors are in parentheses.
Significance: $^{****}p < 0.001$, $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

for at least one year. The first stage result also shows that the pension eligibility thresholds strongly predict the change in retirement behaviour in the United States. Exceeding the early and full pension eligibility thresholds increases the probability of being retired for at least one year by 8.5 and 13.1 percentage points, respectively.[5] The first stage Cragg-Donald Wald F-statistic for the instruments is 356.54 (Table 4.2 (column ii)). The F-statistic is well above the commonly used rule of thumb of 10 (Staiger and Stock (1997)) and critical values provided by Stock and Yogo (2005) for testing for weak identification. In addition,

[5]Similar to the probability of 10.8 and 6.8 percentage points obtained by Bonsang et al. (2012) for sample period 1998-2008. The difference in the construction of the instrument for early accounts for the difference in the estimates.

the Sargan-Hansen test of over-identification restriction provides a p-value of 0.5677 which rejects the null hypothesis at the 5% level and concludes that there is no correlation between the instrumental variables and the error term. The Sargan-Hansen test provides evidence that the two instrumental variables, exceeding the early and full retirement ages for benefit claiming, used in the analysis are valid.

Using 2SLS, I estimate the second stage equation of the effect of retirement on cognition. I estimate (Equation 4.1), omitting the variable for physical activity. Table 4.2 (column iii) indicates a negative and statistically significant causal effect of being retired for at least one year on the cognitive scores. Column (iii) shows a reduction in the cognitive score of 0.59 points. This reduction corresponds to a 5.6% decrease in cognitive scores compared to the sample average.[6] The IV-FE estimate measures the Local Average Treatment Effects. It captures the effect of retirement on cognitive scores only for the individuals who retire at the early and full eligibility ages. This excludes individuals who retire prior to reaching the early eligibility age and those works past the full retirement age. In contrast, the OLS-FE regression estimates the average effect of being retired for all individuals who retire during the sample period. The Durbin-Wu-Hausman test, with a p-value of 0.0284, rejects the null hypothesis of the exogeneity of retirement at the 5% level. Rejecting the null hypothesis of exogeneity of retirement implies that being retired for at least one year is interrelated with our dependent variable, so the estimates of the simple OLS-FE regression are inconsistent, and the 2SLS model is preferred.

Comparing the IV-FE with the OLS-FE estimation in column (i) shows that not accounting for the endogeneity of retirement causes the OLS-FE regression to underestimate the negative effect of retirement on cognitive scores. There are three possible sources of endogeneity in the model: (1) measurement error in the retirement and cognitive scores obtained from the HRS, (2) reverse causality between retirement duration and cognitive function, and (3) an omitted variable correlated with the retirement and/or the cognitive

[6] average cognitive score of 10.6 for the full sample shown in Table 4.1.

variable. Measurement error, if classical, has an attenuation bias, so the OLS estimator is a lower bound for the true effect. Correcting for the attenuation bias would improve the impact of retirement on cognitive function (i.e. the IV-FE estimator would increase). Since correcting the bias increases the negative effect of retirement on cognitive function, it might be reverse causality rather than measurement error that dominates. However, given that the association between retirement and cognitive function is negative in both directions, it is difficult to determine the direction of the bias that would occur due to reverse causality.[7]

Additionally, the difference between the OLS-FE and the IV-FE estimator could be due to an omitted variable bias. As seen in Figure 4.1, a positive association exists between retirement and engaging in physical activity and between physical activity and cognitive scores. So even though I use instrumental variables to correct for the possible reverse causality between retirement and cognition, the omission of physical activity could introduce a bias.

4.5.2 Retirement and Physical Activity

Table 4.3 provides evidence of the association between retirement and physical activity. The OLS-FE estimation in Table 4.3 column (i) assumes the exogeneity of retirement duration and controls for individual time-invariant omitted variables. The OLS-FE estimate shows a positive relationship between retirement and physical activity. Retirement duration increases the probability of meeting the physical activity guidelines by 2.5 percentage points. This finding supports the view that when individuals retire, they are no longer burdened with work-related stress and have more time available. The increase in leisure time might encourage them to engage in more healthy behaviours such as physical activity (Kämpfen and Maurer (2016); Eibich (2015); Kesavayuth et al. (2018)). The reduced form regression column (iv) expresses the primary endogenous variable, physical activity, as a function of the exogenous variables, the retirement benefit claiming age thresholds. The reduced form

[7] Basu (2015) notes that if the coefficient on the bi-directional estimators are the same the sign of the bias depends on whether the product of the coefficients is greater than or less than 1.

regression suggests an increase in the likelihood of meeting the physical activity guidelines at the retirement age thresholds. The OLS-FE and the reduced form estimation support the findings by Kämpfen and Maurer (2016); Eibich (2015); Kesavayuth et al. (2018) of a positive correlation between retirement and engaging in physical activity.

To establish the causal effect of retirement duration on engaging in physical activity, Table 4.3 (column ii) shows the coefficients of the first stage equation (Equation 4.3).[8] Using 2SLS, I estimate the causal effect of being retired for at least one year on meeting the physical activity guideline (Equation 4.4). In Table 4.3, column (iii) I find that in the IV-FE regression, being retired for at least one year increases the probability that individuals will meet the Physical Activity Guidelines by 12.6 percentage points.[9] The Durbin-Wu-Hausman test, with a p-value of 0.0447, rejects the null hypothesis of the exogeneity of retirement at the 5% level. The IV-FE result is larger than the OLS-FE model, which shows that individuals retired for at least one year are 2.5 percentage points more likely to meet the guidelines (Table 4.3, column (i)). This result suggests that ignoring the possible endogeneity of retirement would underestimate the positive effect of retirement on meeting the physical activity guidelines.

The bias in the estimation could be due to measurement error in the construction of the physical activity variable or reverse causality between retirement duration and meeting the physical activity guidelines. While measurement error, if classical, creates an attenuation bias in OLS, it is difficult to determine the direction or magnitude of the bias that occurs due to reverse causality. Figure 4.1, illustrates a positive association between retirement and physical activity and a negative association between engaging in physical activity and retirement. Given the bidirectional relationship between the two variables has opposite signs the bias resulting from the reserve causality may be negative (Basu (2015)). Correcting for the negative bias from measurement error and reverse causality accounts for the increase in

[8]The first stage regression coefficients are the same as Table 2 Column (ii).

[9]This estimate is lower in magnitude than the increase of 27.6 percentage points obtained by Kämpfen and Maurer (2016) using FE-2SLS. The data set and sample size used in the analysis is a reason for this difference.

the IV-FE estimator compared to OLS-FE.

Table 4.3: Physical Activity and Retirement (1998 - 2014)

	FE Physical (i)	First Stage: Retired at least one year (ii)	Second Stage: Physical (iii)	Reduced Form Physical (iv)
Retired at least one year	0.025****		0.1262**	
	(0.005)		(0.051)	
Instruments				
Early (>62 years old)		0.0845****		0.003
		(0.004)		(0.006)
Normal (>65 years old)		0.1306****		0.0021***
		(0.005)		(0.007)
Controls				
Age (years)	0.0015	-0.0262****	0.005	0.005
	(0.007)	(0.005)	(0.007)	(0.007)
Age^2	-0.010***	0.004	-0.0120***	-0.014***
	(0.004)	(0.003)	(0.004)	(0.004)
Test of over identification restriction (p-value)			0.0665	
Durbin-Wu-Hausman (p-value)			0.0447	
F-statistic		357.76		
Year FE	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
N	86593	86593	86593	86593

Source: Health and Retirement Study 1998 - 2014. Age^2= $Age^2/100$.
All respondents were aged between 51 and 75. Robust standard errors are in parentheses.
Significance: $^{****}p < 0.001$, $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

4.5.3 Physical Activity as a Mechanism

Table 4.4 provides evidence of the association between meeting the physical activity guidelines and cognitive scores. The OLS-FE estimate in Table 4.4 column (i) shows that meeting the physical activity guidelines positively affects the cognitive score. This finding supports the literature's positive association between engaging in physical activity and cognitive function (Weuve et al. (2004); Frith and Loprinzi (2017); Cadar (2018)). In Table 4.4, column (ii), I include being retired for at least a year, meeting the physical activity guidelines and an interaction term between retirement duration and physical activity in the OLS-FE estimation. The OLS-FE analysis in column (ii) assumes that retirement and engaging in physical activity are both exogenous. The result shows that the inclusion of physical activity (column (ii)) results in a 1.3% reduction in cognitive scores for individuals retired for at least one year.[10] This reduction is slightly more substantial than the effect of being retired for at least one year obtained for the OLS-FE regression without physical activity as a control (Table 4.2 column (i)). The coefficient on the interaction term indicates that engaging in physical activity during retirement increases cognitive scores by 0.09 points, approximately 0.8 percentage points, compared to the sample average of 10.6. The interaction coefficient suggests that changes in the probability of meeting the physical activity guidelines during retirement affect the retirement variable's impact on cognitive scores.

To establish the paper's main result, I estimate Equation 4.1 to examine whether changes in physical activity during retirement contribute to the impact of being retired on cognitive scores.[11] I correct for the endogeneity between retirement and cognitive function using early and normal eligibility ages for pension claiming as instruments. Reverse causality may exist between the cognitive score and health behaviours, but I did not correct this in the model. The cognitive measure I use in this analysis depends on how many words respondents can recall in a given period. Given the absence of literature on a negative association between cognitive function and physical activity, I believe that a low memory recall score in the short term has little impact on whether the individual meets the physical activity. However, examining the long-run effect on cognitive scores might require considering the possible endogeneity between meeting the physical activity guidelines and cognitive scores.

Table 4.4 (column iii) shows the coefficients of the first stage equation (Equation 4.3), which represents the probability of being retired for at least one year using the retirement

[10] a reduction in average cognitive score of 10.6 for the full sample in Table 4.1 by 0.14 points.

[11] Similar to Eibich (2015), who includes physical activity as a control variable to estimate the cross-sectional models for one year of data.

benefit claiming age thresholds as instruments.[12] Using 2SLS, I analyse the second stage equation of the effect of retirement on cognition, including meeting the physical activity guidelines and an interaction term between retirement duration and physical activity. In Table 4.4, column (iii), I find that in the IV-FE regression, being retired for at least one year reduces the cognitive score by about 7.5% (a 0.8 point reduction). Including the interaction between retirement duration and meeting the Physical Activity Guidelines in Table 4.4, column (iv) results in a 4.5% increase in the cognitive score in the IV-FE model. This coefficient should not be interpreted as a causal effect since the model does not address the possible endogeneity of physical activity.

The reduction in the negative effect of retirement in the IV-FE estimation in Table 4.4, column (iv) compared to Table 4.2, column (iii) suggests that the omission of the physical activity variable results in a biased and inconsistent estimator. If there is reverse causality between physical activity and cognitive function, the sign of the bias would likely be negative (Basu (2015)). This negative bias means that the results from column(iii) would underestimate the positive effect of engaging in physical activity during retirement on cognitive function.

In summary, the results demonstrate that meeting the Physical Activity Guidelines could improve cognitive scores in retirement and slow cognitive functioning decline. Hence, engaging in physical activity is a possible mechanism through which being retired for at least one year impacts cognitive function.

4.5.4 Heterogeneity

Tables 4.5 and 4.6 explore the impact of heterogeneity on the causal effect of retirement duration on cognitive scores. Given the statistically significant causal impact of being retired for at least one year on cognitive scores, I investigate if the effects differ due to individual characteristics such as gender, educational attainment, living with a partner and the

[12] The first stage regression coefficients are the same as Table 4.2 column (ii).

Table 4.4: Physical Activity, Retirement and Cognitive Function (1998 - 2014)

	OLS-FE (i)	OLS-FE Cognitive (ii)	First Stage: Retired (iii)	Second Stage Cognitive (iii)
Retired at least one year		-0.138****		-0.824**
		(0.036)		(0.406)
Physical activity	0.095****	0.060**		-0.089
	(0.023)	(0.029)		(0.093)
Retired*Physical		0.091**		0.483**
		(0.043)		(0.235)
Instruments				
Early (>62 years old)			0.085****	
			(0.004)	
Normal (>65 years old)			0.131****	
			(0.005)	
Test of over identification restriction (p-value)				0.5989
Durbin-Wu-Hausman (p-value)				0.0257
F-statistic			356.54	
Year FE	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes
N	86593	86593	86593	86593

Note: Health and Retirement Study 1998 - 2014.
All models include age (years) and quadratic age as controls. $Age^2 = Age^2/100$.
All respondents were aged between 51 and 75. Robust standard errors are in parentheses.
Significance: $^{****}p < 0.001$, $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

retirement status of the partner. I estimate separate models for each covariate.

Table 4.5, columns (ii) and (iii) shows that there is effect heterogeneity concerning gender. There is a statistically significant difference in the impact of retirement duration on the cognitive score for men. Retirement duration reduces the cognitive scores of men by 9.4% compared to the sample average of 10.6. The decrease in the cognitive scores for men is 25% higher than the IV-FE results for the combined group in column (i). Generally, retirement provides relief from work-related stress, which can be detrimental to cognitive function. In addition, women typically face tremendous stress working and handling family demands simultaneously. So retirement might provide better protection of the cognitive scores for women than men. The coefficient on the interaction term between retired for at least one year and meeting the physical activity guidelines is positive and statistically significant for men. Engaging in physical activity during retirement increases the cognitive scores of men by 6.4%. Of the retired respondents, 41.2% of the men are likely to meet the physical activity guidelines compared to 32.9% of women. This result supports the hypothesis that engaging in physical activity during retirement has a protective impact on cognitive scores.

Table 4.5, columns (iv) and (v), examine the effect of educational attainment. Table 4.5 (column v) indicates that retirement duration significantly impact on the cognitive scores of respondents without a university degree. The reduction is similar to the 7.5% reduction observed in the full sample IV-FE estimation (Table 4.5 (column i)). This finding aligns with Mazzonna and Peracchi (2012), who finds that cognitive scores' decline varies with educational attainment and is more significant for less educated women than highly educated women. However, the interaction between retired for at least one year and meeting the physical activity guidelines has no significant effect.

Individuals living with a partner, Table 4.6 columns (ii) and (iii), experience a more considerable decline in cognitive scores of 10.4% compared to the 7.5% reduction in the full sample (column i). Engaging in physical activity during retirement increases the cognitive scores of men by 6.1%. There is little evidence of heterogeneity for individuals whose partner is also retired (Table 4.6 (columns (iv) and (v)).

4.6 Conclusion

This paper uses instrumental variable analysis to investigate the impact of heterogeneity and physical activity on the causal relationship between being retired for at least one year and cognitive scores. The results indicate that retirement harms cognitive scores, reducing the cognitive score by 5.6% compared to the sample average (10.6) (Table 4.2). In contrast, retirement positively impacts the probability of engaging in physical activity. Retirees in-

Table 4.5: Heterogeneous Effect
Gender and Educational Attainment (1998 - 2014)

	IV-FE (i)	Male (ii)	Female (iii)	University (iv)	No University (v)
Retired at least one year	-0.824**	-1.041*	-0.708	-0.997	-0.815*
	(0.406)	(0.606)	(0.547)	(1.228)	(0.426)
Physical Activity	-0.089	-0.191	-0.021	-0.071	-0.108
	(0.093)	(0.148)	(0.118)	(0.229)	(0.102)
Retired*Physical	0.483**	0.677*	0.356	0.557	0.491
	(0.235)	(0.360)	(0.311)	(0.751)	(0.243)
N	86593	38346	48247	20610	65983

Note: Health and Retirement Study 1998 - 2014.
All models include age (years) and quadratic age as controls. $Age^2 = Age^2/100$.
All respondents were aged between 51 and 75. Robust standard errors are in parentheses.
Significance: $^{****}p < 0.001$, $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

Table 4.6: Heterogeneous Effect
Living with Partner and Partner Retired (1998 - 2014)

	IV-FE (i)	Partner (ii)	No Partner (iii)	Partner Retired (iv)	Partner Not Retired (v)
Retired at least one year	-0.824**	-1.091**	0.093	-0.815	-0.701
	(0.406)	(0.502)	(0.859)	(1.744)	(0.534)
Physical Activity	-0.089	-0.158	0.130	-0.294	-0.010
	(0.093)	(0.112)	(0.197)	(0.715)	(0.094)
Retired*Physical	0.483**	0.653**	-0.043	0.549	0.379
	(0.235)	(0.295)	(0.469)	(1.094)	(0.310)
N	86593	60627	25966	22463	64130

Note: Health and Retirement Study 1998 - 2014.
All models include age (years) and quadratic age as controls. $Age^2 = Age^2/100$.
All respondents were aged between 51 and 75. Robust standard errors are in parentheses.
Significance: $^{****}p < 0.001$, $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$

creasing their time spent on physical activity significantly increases the likelihood of meeting the Physical Activities Guidelines by 12.6 percentage points (Table 4.3, column (iii)).

The inclusion of the physical activity variable in the OLS-FE positively affects cognitive scores. The OLS-FE provides an approximately 0.1-point association between meeting the physical activity guidelines and cognitive scores (Table 4.4, column (ii)). Meeting the physical activity guidelines increases the cognitive scores of older workers by 1% compared to the sample average. The inclusion of meeting the physical activity guidelines in the IV-FE estimation suggests that engaging in physical activity is a possible mechanism through which being retired for at least one year impacts cognitive function. Table 4.4, column (iii) shows that the interaction between retirement and engaging in physical activity reduces the negative impact of retirement duration on cognitive scores in comparison to the results from Table 4.2, column (iii). This implies that it is not the act of retirement that impacts cognitive scores, but how individuals utilize the time they previously allocated to work.

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Summary and Concluding Remarks

The second chapter of my thesis investigates the effect of ageing on the labour market attachment of workers aged 50 to 69 in Canada and the United States. I show that as workers age, there is a greater inclination towards part-time employment driven by personal preference, retirement and/or social security earnings limit. Further, an examination of the unemployed reveals that the main reasons are business conditions, layoffs, and the end of short-term employment. The main factors contributing to the non-attachment of workers classified as NILF are retirement and ending short-term jobs.

Policymakers can use the descriptive evidence I provide in this chapter to inform public policy to increase the labour force participation of older workers. Public policy should encourage firms to create more jobs with flexible work arrangements to keep older workers attached to the labour market. Moreover, though pension age eligibility thresholds provide incentives for older workers to exit the labour market, the focus of public pension reform should be to make it more attractive for workers who are physically able to continue working while collecting pension benefits rather than increasing the age threshold for pension eligibility.

In the third chapter of my thesis, I investigate how workers 50 to 69 years in Canada react to an involuntary job separation. In this chapter, I extend the findings on the labour market attachment from chapter two to test two hypotheses: (1) The likelihood of re-employment decreases with prior job loss and the individual's age. (2) The likelihood of exiting the labour market increases with prior job loss and the individual's age. My results show that the probability of re-employment decreases for workers who experience an involuntary job separation compared to workers separated for other reasons. Moreover, the likelihood of labour market exit increases for workers who experience an involuntary job separation compared to those who experience no separation or those separated for other reasons.

The main limitation of this study is the assumption that involuntary job separation is exogenous. However, even if the involuntary separation occurs due to the business closing down, self-selection into unemployment is still possible. Workers might choose to leave their job in anticipation of the company shutting down. Also, I cannot determine if workers retire following an involuntary separation. Instead, I define a labour market exit as a permanent move from employment or unemployment to NILF. The outlined limitations mean that my analysis provides an association between involuntary job separation and labour market outcomes: re-employment and exit. Access to richer Canadian data, similar to the US Health and Retirement Survey that allows me to follow the separated older workers over time, would provide a more robust analysis. However, this study represents an essential first step in the empirical analysis of the impact of involuntary separation on the labour market exit of older workers in Canada.

My final chapter investigates whether engaging in physical activity during retirement will improve cognitive function. I examine three outcomes, the impact of (1) retirement on cognitive function, (2) retirement on physical activity, and (3) the combined impact of retirement and physical activity on cognitive function. My results support the findings in the literature that retirement negatively impacts the cognitive function of older workers (Bonsang et al. (2012); Atalay et al. (2019)). Additionally, my results show that physical activity protects against cognitive ageing during retirement.

The findings of this chapter suggest that policies aimed at increasing the level of physical activity of retirees could help to slow the decline of cognitive function among older adults. However, lifestyle behaviours are not always independent and may have an additive effect on cognitive function (Cadar (2018)). Another limitation is the possibility of reverse causality between physical activity and cognitive function, which I do not correct in this analysis. In light of this, if the data exists, a worthwhile addition would be to include other health behaviours (e.g. smoking, alcohol consumption) in the analysis.

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